Backspin Entry of POLLEN into the HIVE CELL: Model Equations and Derivations

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Abstract

We derive, step by step, the equations of a model that asks whether a single powered 98 mm[U] flywheel, with a fixed curved hood or an optional passive upper roller, can put enough backspin on POLLEN (a perforated polyethylene ball of diameter about 71 mm[C] and mass about 25 g[A] from the vendor listing [3]) to make it enter the upward-facing CELL of the pivoting HIVE from the unfavourable side, where the CELL roof stands between the robot and the mouth. The model has five parts: the HIVE cross-section and an exact clean-entry test, a flight model with drag, a capped Magnus lift and spin decay, a launcher contact model, a rigid-body rim and interior collision law, and a robustness layer that turns entry windows into a shooter specification. Every equation matches what the code in `src/` computes.

Every result in this paper is a model prediction. No aerodynamic or contact property of POLLEN has been measured: drag, spin lift, spin decay, restitution and friction are all bounded from pickleball and smooth-sphere literature, which are not POLLEN measurements. Within those bounds the model predicts that a clean far-side entry is physically possible, but only with a tight shot-to-shot repeatability (about ± 0.075 m/s[P] in exit speed and $\pm 0.15°$[P] in elevation) after the wheel speed has been tuned on the field. Backspin shifts the clean window to other speeds and angles rather than being required for it: no-spin shots have clean windows of their own, and more spin is not always better, because the lift coefficient saturates in this model and backspin can lower retention after a contact.

Contents

1 Introduction **3**
Notation 4

2 Problem setup, HIVE geometry and the clean-entry criterion **9**
2.1 The question in two dimensions 9
2.2 Frames 10
2.3 The cross-section and its key points 11
2.4 Poses, the mouth normal and the two sides 12
2.5 Tilt and its bounds 12
2.6 Configuration space: inflating the obstacles by the ball radius 13
2.7 The mouth gate and the inside region 13
2.8 The clean-entry criterion 14
2.9 Steepness on the unfavourable side: the minimum descent angle 15
2.10 Release point and the robot height limit 18
2.11 Assumptions 18

3 Flight dynamics: drag, Magnus lift and spin decay **19**
3.1 Frame, state and parameters . . . 19
3.2 Equations of motion . . . 20
3.3 Drag . . . 20
3.4 Reynolds number and spin parameter . . . 21
3.5 The Magnus force . . . 22
3.6 The lift law and its saturation . . . 23
3.7 Spin decay . . . 25
3.8 The backspin axis convention . . . 26
3.9 The 2D reduction . . . 27
3.10 How large are drag and Magnus compared with gravity? . . . 27
3.11 Numerical integration . . . 29
3.12 Summary of modelling assumptions . . . 30

4 Single-flywheel launcher: exit speed and spin **31**
4.1 Frames, sign conventions and the wheel surface speed . . . 31
4.2 Contact kinematics: where the ball touches and how fast that point moves . . . 32
4.3 The ideal hood and where "$v = u/2$" comes from . . . 33
4.4 Equations of motion during contact . . . 33
4.5 Why a fixed hood gives backspin, and the band of S . . . 36
4.6 Rolling and grip-limited regimes . . . 37
4.7 The ratio v/u, and why $v = u/2$ is not universal . . . 38
4.8 Flywheel slow-down and energy per shot . . . 39
4.9 The passive upper roller . . . 40
4.10 Inverse problem: wheel speed for a target exit speed . . . 41
4.11 Assumptions . . . 42

5 Contact mechanics: rim and interior bounces **42**
5.1 Frame, contact normal and contact-point velocity . . . 43
5.2 Ball inertia: the hollow-shell factor α . . . 44
5.3 The impact law, step by step . . . 44
5.4 Energy can only be lost . . . 46
5.5 Which way does backspin push? The kick direction . . . 47
5.6 From one impact to a trajectory . . . 48
5.7 Outcome classes and the trapping test . . . 49
5.8 Why backspin can reduce retention on this side . . . 49
5.9 What is assumed . . . 51

6 From windows to requirements: robustness, Monte Carlo and the shooter specification **51**
6.1 The clean window on a grid . . . 53
6.2 The largest clean square and the $\pm$ tolerance . . . 54
6.3 Many conditions: the robust intersection and the field-tuned hood . . . 56
6.4 Release errors as a Gaussian blur . . . 58
6.5 Azimuth: why the in-plane distance scales with $\cos\psi$ . . . 59
6.6 Three-dimensional Monte Carlo . . . 61
6.7 How sure is a success rate? The Wilson interval . . . 63
6.8 From exit speed to wheel and ball rpm . . . 64

6.9 The shooter specification card 65

7 Whole-field model, live solver and missing effects 66
7.1 Whole-field oblique model 66
7.2 The 3D cell: side walls and pentagon roof 66
7.3 Live solver: knife-edge and tolerant minima 67
7.4 Robot and turret view 67
7.5 Venue air, drag options, drafts and the grip limit 67
7.6 Shooting on the move, hit chance and lift options 68
7.7 Hit-chance guidance and the measurement protocol 70
7.8 Missing effects and how to measure them 71

8 Limitations and what to measure 71

1 Introduction

The game question. In the FTC BIOBUZZ game the centre HIVE is a rigid beam with a box-shaped CELL at each end. It rotates about a horizontal pivot axis between two stable tipped positions, and only one CELL faces upward at a time. Teams LAUNCH POLLEN into that upward-facing CELL to make the HIVE tip [1, 2]. From one side of the field the CELL mouth faces the robot (the *favourable* side). From the other side the CELL roof is between the robot and the mouth (the *unfavourable* side), and an ordinary lob has to fall very steeply to get in.

The team has seen backspinning POLLEN behave in a way that seemed to help on the unfavourable side. The proposed mechanism is a high arc with strong backspin whose path is reshaped by the Magnus force. One point needs care from the start: at the apex of the arc only the vertical velocity is zero, $v_z = 0$; the ball keeps its horizontal speed, and spin gives it no sudden new push there. The Magnus force acts during the whole flight and depends on the spin and the air-relative velocity together. Whether it helps is therefore a trajectory-and-clearance calculation, which is what this paper sets up.

Two hypotheses. The team reported two effects, and we keep them apart:

(A) **Air-only clean entry.** A backspinning ball follows a Magnus-shaped flight that passes over the roof and enters the CELL without touching any part of the HIVE.

(B) **Magnus flight plus a rim contact.** The ball flies a Magnus-shaped path, touches the CELL rim or interior, and the spin-dependent collision redirects it into the CELL (or out of it).

Neither hypothesis is assumed to be true. Sections 2–3 and 6 test (A); Section 5 tests (B); Section 4 asks which exit states a single powered wheel can actually produce.

How to read the numbers. Every number carries one grey tag that says where it comes from:

Tag	Macro	Meaning
[C]	`\confirmed`	official CAD or competition manual
[A]	`\assumed`	assumed, chosen or swept value
[D]	`\derived`	computed from other values by arithmetic or geometry
[P]	`\predicted`	output of the model (a prediction, not a measurement)
[U]	`\userval`	value given by the team

The aerodynamic and contact bounds come from pickleball and smooth-sphere work [5, 6, 7], collected in the project literature synthesis [8]; the launcher model starts from the flywheel analysis of [4]. Every worked example was computed by running the repository code. The code is the source of truth: where a project document disagrees with the code, the text follows the code and the source file carries a comment recording the difference. Aerodynamic parameter ranges come from `research/params_lit.json`, the current default; the older first-pass set `data/aero_params.json` is named only where a result was produced with it.

What the paper does not claim. It does not claim that more backspin is always better, that a dual-powered-wheel shooter is needed, or that a single successful shot shows repeatability. The literature coefficients for pickleballs and smooth spheres are used only as bounds.

Notation

Tables 1–4 list each symbol with the meaning it has where it first appears. Units are SI unless stated. Some letters are reused later with a local meaning; those reuses are listed below and are always defined again at their first use in the section concerned.

Reused letters. The alphabet is small, so a few letters change meaning between sections. Each is redefined at its first use there: c_1 is also the initial slope of the linear-capped lift law (Section 3); κ is also the drag constant $\rho A/(2m)$ (1/m) of the 2D flight equations; α is also the moment-of-inertia shape factor in $I_b = \alpha m R^2$ (Sections 3 and 5); τ is also the spin-down torque of the literature model (N m) and, in Section 6, the hood-angle tolerance (deg); h is also the integration step size (s); θ is also the local flight-path angle in the curvature example of Section 3; in Section 4, x is the arc length travelled by the ball centre along the contact path, v the ball-centre speed along it, s the ball's absolute spin rate (positive = backspin), u the wheel surface speed and $D = 2R_w$ the wheel diameter; in Section 5, s is the coordinate along the gate ($s = 0$ at the far rim); in Section 6, β is the HIVE tilt and ε_v a relative speed error (dimensionless).

Table 1: Notation, part 1 of 4: HIVE geometry and the clean-entry test (Section 2).

Symbol	Meaning	Units
x	HIVE-frame horizontal coordinate in the tipping plane, $+x$ toward the audience; origin on the tile below the pivot axis	m
z	HIVE-frame vertical coordinate above the tiles	m
y	coordinate along the pivot axis	m
$\boldsymbol{p} = (x, z)$	point or ball-centre position in the tipping plane	m
z_p	pivot-axis height above the tiles ($1.11633^{[\mathsf{C}]}$)	m
u	cell-local coordinate along the cell axis, toward the up-CELL mouth	m
w	cell-local coordinate perpendicular to u, toward the CELL roof	m
s	lateral coordinate across the CELL opening (zero on the centre plane)	m
τ	HIVE tilt: angle of the cell axis above horizontal (nominal $30^{\circ\,[\mathsf{C}]}$)	rad (quoted in deg)
σ	pose sign, $+1$ for `pose_pos`, -1 for `pose_neg`	–
$\boldsymbol{p}_r = (-d, h)$	release point of the ball centre in the shot frame	m
d	horizontal distance from the release point to the pivot plane $x = 0$	m
h	ball-centre height above the tiles at release	m
u_M	cell-local mouth plane ($0.54508^{[\mathsf{C}]}$)	m
u_B	cell-local back wall ($0.23952^{[\mathsf{C}]}$)	m
w_F	cell-local floor, lower lip ($-0.03734^{[\mathsf{C}]}$)	m
w_C	cell-local ceiling, upper lip; roof underside at the peak ($0.32004^{[\mathsf{C}]}$)	m
w_T	cell-local roof-top envelope ($0.33325^{[\mathsf{C}]}$)	m
L	mouth length $w_C - w_F$ ($0.35738^{[\mathsf{D}]}$)	m
$\boldsymbol{a}_j, \boldsymbol{b}_j$	end points of obstacle segment j	m
$\boldsymbol{n}$	outward unit normal of the mouth, $(\sigma \cos\tau,\ \sin\tau)$	–
$\boldsymbol{m}$	unit direction of the mouth line from the lower to the upper lip	–
$\boldsymbol{v} = (v_x, v_z)$	ball velocity	m/s
$z_{\mathrm{lip}}(\tau)$	lower-lip height as a function of tilt	m
$\Delta z_{\mathrm{lip}},\ \Delta\tau$	checklist lip-height tolerance and the resulting tilt tolerance	in; deg
R	POLLEN radius ($0.0355^{[\mathsf{C}]}$)	m
λ_j^*	clipped fraction along segment j of the nearest point to $\boldsymbol{p}$	–
$\boldsymbol{q}_j$	nearest point on segment j to $\boldsymbol{p}$	m
$\delta_j(\boldsymbol{p})$	distance from $\boldsymbol{p}$ to segment j	m
$c(\boldsymbol{p})$	clearance: minimum distance to the obstacles minus R; contact if and only if $c < 0$	m
Δ	C-space raster cell size ($1\,\mathrm{mm}^{[\mathsf{A}]}$)	m
$\mathcal{I}$	inside region of the CELL (a quadrilateral in u, w)	m (region)
$\boldsymbol{A}, \boldsymbol{B}$	low and high ends of the gate (mouth-plane edge of the inside region)	m
L_g	gate length $L - 2R$ ($0.28638^{[\mathsf{D}]}$)	m

Table 2: Notation, part 2 of 4: Section 2, continued: paths, crossing and descent angle.

Symbol	Meaning	Units
v	launch speed	m/s
θ	launch elevation angle	deg
t	time since release	s
$\boldsymbol{\Pi}(t) = (\Pi_x, \Pi_z)$	path shape relative to the release point (translation-invariant without wind)	m
t_k	time of solver sample k	s
s_g	fraction along the gate from $\boldsymbol{A}$ (0) to $\boldsymbol{B}$ (1)	–
x_g	x of the gate point at the sample height	m
g	standard gravity ($9.80665^{[\mathsf{C}]}$)	m/s^2
v_{z0}	initial vertical launch speed	m/s
c_2, c_1, c_0	coefficients of the mouth-line crossing quadratic	m/s^2, m/s, m
t_1, t_2	roots of the mouth-line crossing quadratic	s
φ	descent angle of the velocity below horizontal, atan2$(-v_z, v_x)$	deg
φ_{kin}	kinematic minimum descent angle $90^\circ - \tau$	deg
α	excess descent angle $\varphi + \tau - 90^\circ$	deg
$\boldsymbol{T} = (x_T, z_T)$	roof top at the mouth, the highest HIVE point	m
κ	path curvature under gravity, $g\cos\varphi/\lvert\boldsymbol{v}\rvert^2$	1/m
$\hat{\boldsymbol{e}}_{uw}$	unit direction of motion in the cell-local frame, $(-\sin\alpha, -\cos\alpha)$	–
w_c	cell-local w at which the centre crosses the mouth plane	m
λ	distance travelled back along the path from the crossing	m
$\boldsymbol{C}, \boldsymbol{D}$	upper-lip and lower-lip inner corners (u_M, w_C), (u_M, w_F)	m
δ_C, δ_D	distance of the straight path line from the upper and lower lip corner	m
φ_{gate}	minimum descent angle for a clean straight path under the code's gate criterion ($66.33^{\circ[\mathsf{D}]}$ at $30^{\circ[\mathsf{C}]}$ tilt)	deg
φ_{fit}	stricter both-lips minimum descent angle ($71.46^{\circ[\mathsf{D}]}$ at $30^{\circ[\mathsf{C}]}$ tilt)	deg
$W_\perp(\alpha)$	width of the clean straight-path window, measured perpendicular to the path	m
$H_{\max}$	R105 robot height limit ($0.7366^{[\mathsf{C}]}$)	m
$\Delta w(s)$	roof drop for lateral offset s (2D proxy of the pentagon roof)	m
$h_0, h_w, s_{\text{half}}$	pentagon peak height, side-wall height and half-width used in the roof drop	m

Table 3: Notation, part 3 of 4: Flight dynamics (Section 3).

Symbol	Meaning	Units
x, y, z; $\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}, \hat{\boldsymbol{z}}$	world frame: x horizontal toward the target, y sideways, z up; unit vectors	m; unit vectors –
$\boldsymbol{r}$	ball-centre position	m
$\boldsymbol{\omega}$	spin (angular velocity) vector	rad/s
$\boldsymbol{\omega}_0$, ω_0	launch spin vector; signed launch spin rate (positive = backspin) used by `backspin_omega`	rad/s
ω_y	y component of spin in the 2D reduction (negative = backspin toward $+x$)	rad/s
m	ball mass	kg
$D = 2R$	ball diameter	m
ρ	air density	kg/m^3
ν	kinematic viscosity of air	m^2/s
$A = \pi R^2$	frontal (cross-sectional) area	m^2
$\boldsymbol{u}$	wind velocity	m/s
$\boldsymbol{v}_{\mathrm{rel}} = \boldsymbol{v} - \boldsymbol{u}$	air-relative velocity	m/s
$\boldsymbol{F}_D$	drag force	N
$\boldsymbol{F}_M$	Magnus force	N
q	dynamic pressure $\frac{1}{2}\rho\lvert\boldsymbol{v}_{\mathrm{rel}}\rvert^2$	Pa
C_d	drag coefficient (constant per run)	–
C_L	lift (Magnus) coefficient, a function of S	–
$C_{L,\max}$	lift-coefficient ceiling	–
S	spin parameter $\lvert\boldsymbol{\omega}\rvert R/\lvert\boldsymbol{v}_{\mathrm{rel}}\rvert$	–
S_0	spin parameter at launch (signed in `feasibility.py`, $> 0 =$ backspin)	–
$S_{\mathrm{sat}} = C_{L,\max}/c_1$	saturation spin parameter above which C_L is capped	–
ε_v	regularisation speed (10^{-6}[A]) in S and in the Magnus direction	m/s
Re	Reynolds number $\lvert\boldsymbol{v}_{\mathrm{rel}}\rvert\, 2R/\nu$ (diagnostic only)	–
$\hat{\boldsymbol{n}}_M$	Magnus direction, the unit vector of $\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}}$	–
k	spin-decay rate	1/s
$t_{1/2}$	spin half-life $\ln 2/k$	s
v_0	launch speed	m/s
$C_M = \beta S$	spin-down moment coefficient (literature model, not in the code)	–
β	spin-down moment slope (literature model)	–
$I_b = \alpha m R^2$	ball moment of inertia	kg m^2
k_{torque}	speed-dependent decay rate from the torque model (not implemented)	1/s

Table 4: Notation, part 4 of 4: Section 3, continued, and Sections 4–6.

Symbol	Meaning	Units
s_b	backspin sign, $-\operatorname{sgn}(\omega_y)$, $+1$ for backspin toward $+x$	–
v_t	terminal speed $\sqrt{2mg/(\rho C_d A)}$	m/s
$a_D,\ a_M$	drag and Magnus accelerations	m/s^2
$a_\perp$	acceleration normal to the velocity	m/s^2
r_c	local radius of curvature of the path	m
$\Delta z_{\text{apex}},\ \Delta x_{\text{land}}$	apex-height gain and landing-point shift caused by spin, compared with no spin	m (tabulated in cm)
$\boldsymbol{y},\ \boldsymbol{f},\ \boldsymbol{k}_1, \ldots, \boldsymbol{k}_4$	generic ODE state, right-hand side, RK4 stages	state units; stages per s
$\boldsymbol{a}_1, \boldsymbol{a}_2, \boldsymbol{a}_3$	acceleration stages in the position form of RK4	m/s^2
$r_{\text{tol}},\ a_{\text{tol}}$	relative and absolute tolerances of `solve_ivp`	–; state units
e_i	local error estimate of state component i	state units
C	RK4 global-error constant	m/s^4
n	flywheel speed (Section 4)	rev/min
Ω	flywheel angular velocity, positive when the wheel surface at the contact moves forward	rad/s
Ω_t	passive-roller angular velocity, positive when its surface at the contact moves forward	rad/s
R_w	flywheel radius	m
e	coefficient of restitution of a contact (Section 5)	–
e_t	tangential restitution coefficient	–
μ	Coulomb friction coefficient of a contact	–
ψ	shot azimuth away from the tipping plane (Section 6)	deg or rad
$\hat{p}$	estimated success rate k/n from k successes in n shots	–

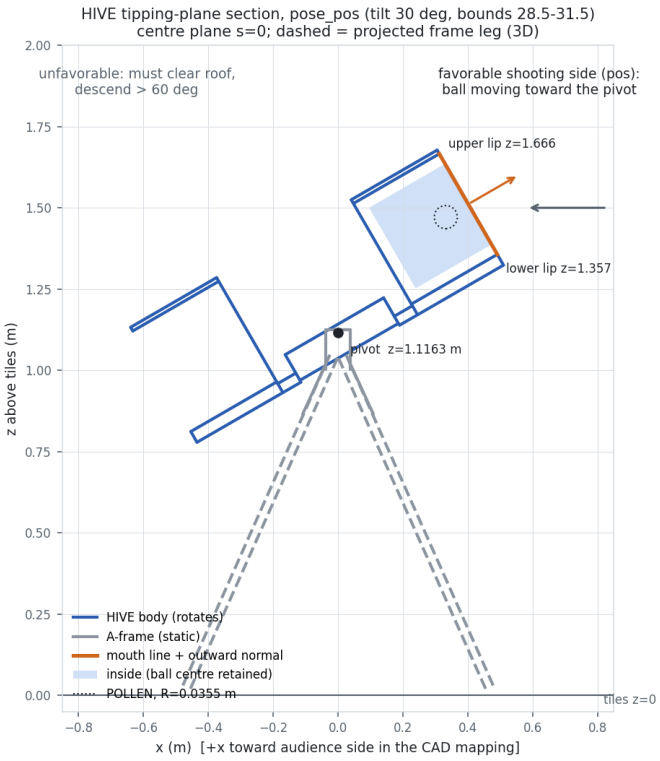


Figure 1: HIVE tipping-plane section in `pose_pos` at the $30°^{[C]}$ tilt (bounds $28.5–31.5°^{[D]}$), centre plane $s = 0$. Blue: HIVE body (rotates about the pivot at $z_p = 1.1163\,\mathrm{m}^{[C]}$); grey: static A-frame (dashed legs are projected from 3D); orange: mouth line and its outward normal $\boldsymbol{n}$; shaded: inside region $\mathcal{I}$ for the ball centre; dotted circle: POLLEN with $R = 0.0355\,\mathrm{m}^{[C]}$. The geometry is derived from the official CAD (Section 2); it is not a model prediction. In this pose the favourable side is on the right; an unfavourable-side shot comes from the left and must clear the roof.

2 Problem setup, HIVE geometry and the clean-entry criterion

Before any physics of spin can be judged, we need a precise answer to a purely geometric question: *what does a ball path have to look like to count as "in"?* This section sets up the coordinate frames, describes the HIVE cross-section that the model uses, defines the *clean-entry* test exactly as the code applies it, and derives from the geometry alone how steeply the ball must be falling when it arrives from the unfavourable side. Everything later in the paper (drag, Magnus lift, the launcher) only changes *which paths* the robot can make; this section decides *which paths score*.

2.1 The question in two dimensions

The target is the upward-facing CELL of the pivoting HIVE at the centre of the field. The HIVE is one rigid beam with a box-shaped CELL at each end. It rests on a damper stop, so one CELL points up and outward and the other points down and outward. A shot from the side the up CELL faces is the *favourable* side. A shot from the other side is the *unfavourable* side: the up CELL's roof is between the robot and the mouth. Our question is whether backspin from a single powered wheel can bend a POLLEN path enough to score from the unfavourable side.

The model works in the vertical *tipping plane*, the plane in which the CELLS rotate. It uses the

cross-section of the HIVE at its centre plane. A pure backspin shot fired in that plane stays in it, so a two-dimensional (2D) model is enough for centred shots. Off-centre shots and the CELL side walls are three-dimensional effects that this section treats only as a stated limitation (Section 2.11).

2.2 Frames

HIVE frame. The origin is on the tile surface directly below the HIVE pivot axis, at the HIVE centre plane. The axis x (in m) is horizontal and lies in the tipping plane, with $+x$ pointing toward the audience side of the field. The axis z (in m) points up, and y (in m) runs along the pivot axis. The pivot axis is at height

$$z_p = 1.11633 \text{ m}^{[\mathsf{C}]}, \tag{1}$$

which is $43.95\,\text{in}^{[\mathsf{C}]}$ above the tiles (Competition Manual 9.6.1 and the official field CAD). A point in the tipping plane is written $\boldsymbol{p} = (x, z)$.

Cell-local frame. The HIVE body is stored once, in a frame that rotates with it. The coordinate u (in m) runs along the cell axis, from the pivot outward toward the mouth of the CELL that is up; $u > 0$ is the up CELL and $u < 0$ the down CELL. The coordinate w (in m) is perpendicular to u and points toward the CELL roof. Both CELLS have their roofs on the $+w$ side, so the body is mirror-symmetric about $u = 0$. The lateral coordinate s (in m) runs across the CELL opening; $s = 0$ is the centre plane, the plane used in this paper.

From cell-local to HIVE frame. Let τ (in rad, quoted in degrees) be the *tilt*, the angle of the cell axis above the horizontal. Let $\sigma \in \{+1, -1\}$ be the pose sign: $\sigma = +1$ when the up CELL is on the $+x$ side and $\sigma = -1$ when it is on the $-x$ side. The code (`Hive.from_json`, inner function `world`) places a cell-local point (u, w) in the HIVE frame in two steps. First it rotates by τ:

$$\begin{pmatrix} x' \\ z' - z_p \end{pmatrix} = \begin{pmatrix} \cos\tau & -\sin\tau \\ \sin\tau & \cos\tau \end{pmatrix} \begin{pmatrix} u \\ w \end{pmatrix}. \tag{2}$$

Then it mirrors the horizontal coordinate when the up CELL is on the $-x$ side, $x = \sigma x'$. Combining the two steps gives

$$x = \sigma\,(u\cos\tau - w\sin\tau), \tag{3}$$

$$z = z_p + u\sin\tau + w\cos\tau. \tag{4}$$

Equations (3)–(4) are used whenever the tilt or the pivot height differ from their nominal values. At the nominal pose the code reads pre-computed HIVE-frame segments from `data/hive_geometry.json`; these agree with (3)–(4) to the five decimals stored.

Shot frame. The flight solver always fires toward $+x$. A shooter standing on the $x < 0$ side therefore uses the HIVE frame unchanged. A shooter on the $x > 0$ side fires toward $-x$, so the code mirrors the *geometry* ($x \to -x$, `Hive.mirrored`) instead of the trajectory. In the shot frame the robot releases the ball at

$$\boldsymbol{p}_r = (x_r, z_r) = (-d,\ h), \tag{5}$$

where d (in m) is the horizontal distance from the release point to the pivot plane $x = 0$, and h (in m) is the height of the ball *centre* above the tiles at release.

2.3 The cross-section and its key points

Table 5 lists the cell-local coordinates that set the mouth. All of them come from the official field CAD (STEP file, parsed by the project), at the centre plane $s = 0$.

Table 5: Cell-local coordinates of the up CELL at $s = 0$ (from `body_local` in `data/hive_geometry.json`).

Symbol	Meaning	Value
u_M	mouth plane (outer rib face)	$0.54508\,\mathrm{m}^{[C]}$
u_B	back wall (back skin)	$0.23952\,\mathrm{m}^{[C]}$
w_F	floor (top of the bottom skin)	$-0.03734\,\mathrm{m}^{[C]}$
w_C	ceiling (roof underside at the pentagon peak)	$0.32004\,\mathrm{m}^{[C]}$
w_T	roof top (envelope of the rib tops)	$0.33325\,\mathrm{m}^{[C]}$

Mouth length. The mouth runs along the line $u = u_M$ from the lower lip (u_M, w_F) to the upper lip (u_M, w_C). Its length is

$$L = w_C - w_F = 0.32004 - (-0.03734) = 0.35738\ \mathrm{m}^{[D]}. \tag{6}$$

The CELL depth is $u_M - u_B = 0.30556\,\mathrm{m}^{[D]}$ (manual: about $12\,\mathrm{in}^{[C]}$).

Worked example: the lips in the HIVE frame. Take $\sigma = +1$ and $\tau = 30^{\circ[C]}$, so $\cos\tau = 0.86603^{[D]}$ and $\sin\tau = 0.5^{[D]}$. For the lower lip, $(u, w) = (0.54508, -0.03734)$, equations (3)–(4) give

$$\begin{aligned} x &= 0.54508 \times 0.86603 - (-0.03734) \times 0.5 \\ &= 0.47205 + 0.01867 = 0.49072\ \mathrm{m}^{[D]}, \\ z &= 1.11633 + 0.54508 \times 0.5 + (-0.03734) \times 0.86603 \\ &= 1.11633 + 0.27254 - 0.03234 = 1.35653\ \mathrm{m}^{[D]}. \end{aligned}$$

That is $53.41\,\mathrm{in}^{[D]}$, inside the Field Acceptance Checklist value of $53.5 \pm 0.5\,\mathrm{in}^{[C]}$. Running the same formula in python for the other key points gives Table 6.

Table 6: Key points of the up CELL, pose `pose_pos`, $\tau = 30^{\circ}$ (python, `hive.py`).

Point	x (m)	z (m)	Official check
Lower lip (u_M, w_F)	$0.49072^{[D]}$	$1.35653^{[D]}$	$53.5\,\mathrm{in}^{[C]}$ (checklist)
Upper lip (u_M, w_C)	$0.31203^{[D]}$	$1.66603^{[D]}$	$65.6\,\mathrm{in}^{[C]}$ (Fig. 9-10)
Roof top at mouth (u_M, w_T)	$0.30543^{[D]}$	$1.67747^{[D]}$	highest point of the HIVE
Roof top at back (u_B, w_T)	$0.04081^{[D]}$	$1.52469^{[D]}$	–

Obstacles. The code treats the HIVE as a set of solid line segments in the (x, z) plane. Segment j runs from $\boldsymbol{a}_j$ to $\boldsymbol{b}_j$ (both in m). Each pose has $16^{[D]}$ CELL segments (8 per CELL: back wall, roof outer, upper lip, roof inner, floor, lower lip, underside, tube top), $4^{[D]}$ pivot-plate segments, and $9^{[D]}$ static A-frame segments. Four of the A-frame segments are the legs. They slant across the

field, so at $s = 0$ only their top exists; the file flags them `projected`, and the code drops them by default (`include_projected=False`). The default obstacle set therefore has $25^{[D]}$ segments ($29^{[D]}$ with the legs). The A-frame does not rotate with the HIVE.

2.4 Poses, the mouth normal and the two sides

Poses. The HIVE has two resting poses. In `pose_pos` the up CELL is on the $+x$ (audience) side, $\sigma = +1$. In `pose_neg` it is on the $-x$ side, $\sigma = -1$. `pose_neg` is the exact mirror image ($x \to -x$) of `pose_pos`. Every TIP flips the pose.

Outward normal of the mouth. The mouth lies on $u = u_M$, so the direction pointing out of the CELL through the mouth is the cell-local unit vector $(1, 0)$. Rotating it with (2) gives $(\cos\tau, \sin\tau)$, and the mirror multiplies the x part by σ:

$$\boldsymbol{n} = (\sigma \cos\tau,\ \sin\tau). \tag{7}$$

This is exactly the code's `normal = np.array([sp * ct, st])`. At $\tau = 30^{\circ[C]}$ and $\sigma = +1$, $\boldsymbol{n} = (0.86603, 0.5)^{[D]}$: the mouth faces outward and $30^{\circ[D]}$ above the horizontal. The mouth *line* is perpendicular to $\boldsymbol{n}$, and in `pose_pos` its direction from lower to upper lip is $\boldsymbol{m} = (-\sin\tau, \cos\tau) = (-0.5, 0.86603)^{[D]}$.

Which way is "in"? A ball with velocity $\boldsymbol{v} = (v_x, v_z)$ (in m/s) is moving into the CELL through the mouth only if its velocity points against the outward normal:

$$\boldsymbol{v} \cdot \boldsymbol{n} < 0. \tag{8}$$

Favourable and unfavourable sides. The *favourable* side is the side the up CELL is on (`favorable_side`: `"pos"` for `pose_pos`, `"neg"` for `pose_neg`). A ball fired from there travels toward the pivot, which is roughly $-\boldsymbol{n}$, so (8) is easy to satisfy. The *unfavourable* side is the other one: the ball travels away from the shooter in the same direction as $\boldsymbol{n}_x$, so it can only satisfy (8) by falling steeply (Section 2.9). In the code (`feasibility.Geometry.make`) the unfavourable case is `pose_pos` used as it is, with the shooter at $x = -d$ firing toward $+x$; the favourable case is the same HIVE mirrored. Results for `pose_neg` are the exact mirror image and are not computed separately.

2.5 Tilt and its bounds

The nominal tilt is $\tau_0 = 30^{\circ[C]}$: Fig. 9-10 of the manual labels it, and the CAD base tube measures $30.0^{\circ[C]}$. Real fields differ, so we bound it using the one tilt-sensitive number the field checklist controls, the lower-lip height.

Step 1: lip height as a function of tilt. From (4) with $(u, w) = (u_M, w_F)$:

$$z_{\text{lip}}(\tau) = z_p + u_M \sin\tau + w_F \cos\tau. \tag{9}$$

Step 2: sensitivity. Differentiating (9) with z_p held fixed,

$$\frac{\mathrm{d}z_{\text{lip}}}{\mathrm{d}\tau} = u_M \cos\tau - w_F \sin\tau. \tag{10}$$

Comparing with (3) shows that this is exactly x_{lip} for $\sigma = +1$: the lip moves up at a rate equal to its horizontal distance from the pivot, as expected for a rotation about the pivot. At τ_0 it is $0.49072\,\text{m/rad}^{[D]} = 19.32\,\text{in/rad}^{[D]} = 0.3372\,\text{in/deg}^{[D]}$.

Step 3: invert for the tolerance. The checklist allows $\Delta z_{\text{lip}} = \pm 0.5\,\text{in}^{[C]}$. With the pivot at its nominal height,

$$\Delta\tau \approx \frac{\Delta z_{\text{lip}}}{\mathrm{d}z_{\text{lip}}/\mathrm{d}\tau} = \frac{0.5}{0.3372} = 1.48^{\circ\,[D]} \;\rightarrow\; \tau \in [28.5^\circ, 31.5^\circ]^{[D]}. \tag{11}$$

If the pivot height is also off by the manual's general $\pm 1\,\text{in}^{[C]}$ tolerance, the lip can move $1.5\,\text{in}^{[D]}$ for the same checklist reading, giving $1.5/0.3372 = 4.45^{\circ\,[D]}$, rounded to the wide bounds $[25.5^\circ, 34.5^\circ]^{[D]}$ (`low_wide`, `high_wide`). The code re-poses the body at any tilt with (3)–(4); Table 7 in Section 2.9 shows how the entry requirement moves with τ.

2.6 Configuration space: inflating the obstacles by the ball radius

A real ball touches a wall when its *surface* reaches the wall. It is easier to track one point, the ball centre, and grow every obstacle by the ball radius instead. This is the *configuration-space* idea, and it is how `hive.py` works. The POLLEN radius is

$$R = 0.0355\ \text{m}^{[C]} \tag{12}$$

(manual 9.8: diameter "approximately" $7.1\,\text{cm}^{[C]}$, so about $\pm 1\,\text{mm}^{[A]}$ is uncertain).

Distance to one segment. For a ball centre $\boldsymbol{p}$ and segment j, the code (`seg_point_dist`) first finds the nearest point's position along the segment and clips it to the segment:

$$\lambda_j^* = \text{clip}\left(\frac{(\boldsymbol{p}-\boldsymbol{a}_j)\cdot(\boldsymbol{b}_j-\boldsymbol{a}_j)}{\|\boldsymbol{b}_j-\boldsymbol{a}_j\|^2},\ 0,\ 1\right), \qquad \boldsymbol{q}_j = \boldsymbol{a}_j + \lambda_j^*(\boldsymbol{b}_j-\boldsymbol{a}_j). \tag{13}$$

The clip is what gives the inflated segment its rounded end caps: beyond an end, the nearest point is the end itself. The distance is $\delta_j(\boldsymbol{p}) = \|\boldsymbol{p}-\boldsymbol{q}_j\|$ (in m).

Clearance. The clearance (in m) is the smallest gap between the ball surface and any obstacle:

$$c(\boldsymbol{p}) = \min_j \delta_j(\boldsymbol{p}) - R. \tag{14}$$

The ball is in *contact* when $c(\boldsymbol{p}) < 0$ (strictly; $c = 0$ is a graze and not a contact).

Along a path. A path is a list of centre positions $\boldsymbol{p}_0, \boldsymbol{p}_1, \ldots$ at the solver's time steps. The exact classifier (`Hive.classify_path`) tests every *chord* $\boldsymbol{p}_k\boldsymbol{p}_{k+1}$ against every obstacle with the exact segment-to-segment distance (`seg_seg_dist`), so contact between samples is not missed. The sweep (`evaluate_gate`) uses a precomputed clearance grid instead (`CSpaceRaster`), with cells of side $\Delta = 1\,\text{mm}^{[A]}$. It looks up the cell containing each sample. Because the distance function changes by at most 1 m per metre moved, the lookup error is at most $\Delta/\sqrt{2} = 0.71\,\text{mm}^{[D]}$.

2.7 The mouth gate and the inside region

Inside region. The code calls a ball centre *inside* the CELL when it lies in the quadrilateral $\mathcal{I}$ whose cell-local corners are

$$\mathcal{I}: \quad u \in [\,u_B + R,\ u_M\,], \qquad w \in [\,w_F + R,\ w_C - R\,]. \tag{15}$$

In words: behind the mouth plane, and at least one radius away from the back wall, the floor and the ceiling. It is mapped to the HIVE frame with (3)–(4). Note that the edge on the mouth plane is *not* moved in by R: the ball counts as inside as soon as its centre is behind the mouth plane.

Gate. The *gate* is the edge of $\mathcal{I}$ that lies on the mouth plane (`Hive.gate`), ordered low end first:

$$\boldsymbol{A} = \text{world}(u_M,\ w_F + R), \qquad \boldsymbol{B} = \text{world}(u_M,\ w_C - R), \tag{16}$$

where world$(\cdot)$ is (3)–(4). Its length is

$$L_g = (w_C - R) - (w_F + R) = L - 2R = 0.35738 - 0.071 = 0.28638\ \text{m}^{[\text{D}]}. \tag{17}$$

For `pose_pos` at $\tau = 30°^{[\text{C}]}$ the code gives $\boldsymbol{A} = (0.47298, 1.38728)\,\text{m}^{[\text{D}]}$ and $\boldsymbol{B} = (0.32979, 1.63529)\,\text{m}^{[\text{D}]}$. The gate is the window the ball *centre* must pass through; the ball surface then just clears the lips.

2.8 The clean-entry criterion

Definition. A path is *clean* when the ball centre crosses the gate $\overline{\boldsymbol{AB}}$ moving inward, (8), and the clearance (14) stayed non-negative at every earlier point of the path. Contact *after* the crossing (for example with the floor) does not change the label; whether such a ball stays in is the job of the contact model. The exact classifier returns one of three labels: `clean`; `hit:<name>`, the first obstacle touched before entry; or `miss`, when the path reaches neither (for example it falls short onto the field).

How the sweep finds gate crossings. Without wind, drag and Magnus forces do not depend on position. So a launch with speed v (in m/s), elevation θ (in degrees) and spin gives one path *shape* $\boldsymbol{\Pi}(t) = (\Pi_x(t), \Pi_z(t))$ (in m, starting at $(0, 0)$ at time $t = 0\,\text{s}$), and the release point only shifts it:

$$\boldsymbol{p}(t) = \boldsymbol{p}_r + \boldsymbol{\Pi}(t) = \big(-d + \Pi_x(t),\ h + \Pi_z(t)\big). \tag{18}$$

`evaluate_gate` uses this to avoid re-integrating for every release point. For a fixed h, it keeps every sample k of every path that satisfies two conditions:

$$A_z \le h + \Pi_z(t_k) \le B_z, \tag{19}$$

$$\dot{\boldsymbol{\Pi}}(t_k) \cdot \boldsymbol{n} < 0. \tag{20}$$

The first condition says the sample is at a height where the gate is; the second is (8). Because the gate is a straight, non-horizontal segment, a height fixes a unique point on it. Let s_g be the fraction along the gate (0 at $\boldsymbol{A}$, 1 at $\boldsymbol{B}$):

$$s_g = \frac{h + \Pi_z(t_k) - A_z}{B_z - A_z}, \qquad x_g = A_x + s_g\,(B_x - A_x). \tag{21}$$

Requiring the shifted sample to sit exactly on that gate point, $-d + \Pi_x(t_k) = x_g$, gives the one release distance at which this sample enters:

$$d = \Pi_x(t_k) - x_g. \tag{22}$$

Candidates outside the swept range widened by $0.06\,\text{m}^{[\text{A}]}$ at each end, $d \in [0.49, 1.81]\,\text{m}^{[\text{D}]}$ (grid 0.55 to $1.75\,\text{m}^{[\text{A}]}$), are dropped. Each remaining candidate is clean if the raster clearance is ≥ 0 at every sample from the first one whose shifted x has reached the left edge of the raster box up to and including sample k (samples outside the box count as free). (For a path whose v_x reverses, the check starts at the first sample.) Two consecutive clean samples $k, k+1$ of the same path bound a continuous interval of clean release distances, and `grid_from_candidates` marks every grid value of d inside it as clean. The exact classifier reaches the same label in a different way: the first sample inside $\mathcal{I}$ before any chord with negative clearance. The two agree because the only way into $\mathcal{I}$ from outside without contact is through the gate.

Worked example: one shot through the gate. To check the formulas by hand we use a vacuum path (no air at all), so that $\mathbf{\Pi}(t)$ is a parabola. Air is added in later sections. Take $v = 7.0\,\mathrm{m/s}^{[A]}$, $\theta = 81.0^{\circ[A]}$, $h = 0.737\,\mathrm{m}^{[A]}$ and $d = 1.00\,\mathrm{m}^{[A]}$, with $g = 9.80665\,\mathrm{m/s}^{2[C]}$ (standard gravity, `flight.G_STANDARD`). Then $v_x = v\cos\theta = 1.0950\,\mathrm{m/s}^{[D]}$ and $v_{z0} = v\sin\theta = 6.9138\,\mathrm{m/s}^{[D]}$.

Step 1: crossings of the mouth line. The centre is on the mouth line when $(\boldsymbol{p}(t) - \boldsymbol{A}) \cdot \boldsymbol{n} = 0$. With $\boldsymbol{p}(t) = (-d + v_x t,\ h + v_{z0}t - \frac{1}{2}gt^2)$ this is a quadratic in t:

$$\underbrace{\left(-\tfrac{1}{2}g\,n_z\right)}_{c_2} t^2 + \underbrace{\left(n_x v_x + n_z v_{z0}\right)}_{c_1} t + \underbrace{\left(n_x(-d - A_x) + n_z(h - A_z)\right)}_{c_0} = 0. \tag{23}$$

Its roots are $t_1 = 0.5057\,\mathrm{s}^{[P]}$ and $t_2 = 1.2911\,\mathrm{s}^{[P]}$.

Step 2: test each crossing. At t_1 the ball is still rising ($v_z = +1.954\,\mathrm{m/s}^{[P]}$), $\boldsymbol{v}\cdot\boldsymbol{n} = +1.926\,\mathrm{m/s}^{[P]}$ (outward), and $s_g = 6.42^{[P]}$, far outside $[0, 1]$: the ball passes the extended mouth line high above the HIVE. It fails both (19) and (20). At t_2 the centre is at $(0.4138, 1.4897)\,\mathrm{m}^{[P]}$, $s_g = 0.413^{[P]}$ (inside the gate), $v_z = -5.748\,\mathrm{m/s}^{[P]}$ and $\boldsymbol{v}\cdot\boldsymbol{n} = -1.926\,\mathrm{m/s}^{[P]}$ (inward). This is the entry.

Step 3: no earlier contact. `hive.classify_detail` on the exact parabola returns `clean` with a minimum clearance of $30.5\,\mathrm{mm}^{[P]}$ before entry, reached as the ball passes the upper lip. The raster sweep (`evaluate_gate` with `grid_from_candidates`) gives $28.3\,\mathrm{mm}^{[P]}$ for the same cell. The difference is not raster error: the grid cell at $d = 1.00\,\mathrm{m}^{[A]}$ takes the smaller margin of the two bracketing gate samples, which sit at $d = 1.0025\,\mathrm{m}^{[P]}$ and $0.9981\,\mathrm{m}^{[P]}$ (one solver step of $\Delta t = 2\,\mathrm{ms}^{[A]}$, so $v_x\,\Delta t \approx 2.2\,\mathrm{mm}^{[D]}$ apart); the exact classifier at $d = 1.0025\,\mathrm{m}^{[P]}$ gives $28.0\,\mathrm{mm}^{[P]}$.

Step 4: the apex. The highest point is at $t = v_{z0}/g = 0.705\,\mathrm{s}^{[P]}$, $z = h + v_{z0}^2/(2g) = 3.174\,\mathrm{m}^{[P]}$. There $v_z = 0$ but the ball is still moving sideways at $v_x = 1.095\,\mathrm{m/s}^{[P]}$; it is not stationary.

Step 5: the neighbours. Keeping $v = 7.0\,\mathrm{m/s}^{[A]}$ and changing only θ: $78.0^{\circ[A]}$ is a `miss`, $80.5^{\circ[A]}$ is `hit:up_floor` (clearance on the first touching chord $-1.6\,\mathrm{mm}^{[P]}$), $81.5^{\circ[A]}$ is `hit:up_roof_outer` ($-3.9\,\mathrm{mm}^{[P]}$) and $84.0^{\circ[A]}$ is `hit:pivot_plate_top`. The overlap reported at first contact depends on the time step of the path samples, so only its sign is meaningful. In the sweep (`evaluate_gate` with `grid_from_candidates`), on a sub-grid (v from $5.0\,\mathrm{m/s}^{[A]}$ to $9.0\,\mathrm{m/s}^{[A]}$ in $0.1\,\mathrm{m/s}^{[A]}$ steps, θ from $60^{\circ[A]}$ to $89.75^{\circ[A]}$ in $0.25^{\circ[A]}$ steps) at this (h, d), $82^{[P]}$ launches reach the gate and $47^{[P]}$ of them are clean; the other $35^{[P]}$ first touch `up_roof_outer` ($22^{[P]}$) or `up_upper_lip` ($13^{[P]}$). The clean launches form a thin diagonal band in the (v, θ) plane. This is a statement about the model in vacuum, not about how often a real launcher would score.

2.9 Steepness on the unfavourable side: the minimum descent angle

We now derive, from the geometry alone, how steeply the ball must fall to enter from the unfavourable side. Work in the unfavourable shot frame: `pose_pos`, $\sigma = +1$, shooter at $x = -d$, ball moving toward $+x$.

Descent angle. Define the descent angle φ (in degrees) of the velocity, measured below the horizontal and positive when falling, as the code does:

$$\varphi = \operatorname{atan2}(-v_z,\ v_x), \qquad \boldsymbol{v} = \|\boldsymbol{v}\|\,(\cos\varphi,\ -\sin\varphi). \tag{24}$$

Step 1: the kinematic condition. Insert (24) and (7) into (8):

$$\boldsymbol{v}\cdot\boldsymbol{n} = \|\boldsymbol{v}\|\,(\cos\varphi\cos\tau - \sin\varphi\sin\tau) = \|\boldsymbol{v}\|\cos(\varphi + \tau) < 0. \tag{25}$$

For a ball still moving toward $+x$ ($-90^\circ < \varphi < 90^\circ$) this needs $\varphi + \tau > 90^\circ$, so

$$\varphi > \varphi_{\text{kin}} = 90^\circ - \tau = 60^\circ\,^{[\mathsf{D}]}. \tag{26}$$

This is the "steeper than 60°" rule. It only says the ball is moving into the plane of the mouth; it says nothing yet about fitting past the lips. It is convenient to measure the excess over this angle,

$$\alpha = \varphi + \tau - 90^\circ, \tag{27}$$

so that (26) reads $\alpha > 0$.

Step 2: the roof. The highest point of the HIVE is the roof top at the mouth, $\boldsymbol{T} = (0.30543, 1.67747)\,\text{m}^{[\mathsf{D}]}$ (Table 6). Every gate point has $x > 0.3298\,\text{m}^{[\mathsf{D}]}$ and the release has $x = -d < 0$, so the centre must cross the vertical line $x = x_T$. At that moment the centre is directly above $\boldsymbol{T}$ (below it is inside the HIVE), so the clearance (14) is non-negative only if

$$z(x_T) \geq z_T + R = 1.67747 + 0.0355 = 1.7130\ \text{m}^{[\mathsf{D}]}. \tag{28}$$

So the path must rise at least $1.7130 - h$ above the release point. In vacuum this needs a vertical launch speed of at least $\sqrt{2g(1.7130 - h)}$, which is $4.375\,\text{m/s}^{[\mathsf{D}]}$ at $h = 0.737\,\text{m}^{[\mathsf{A}]}$. Drag raises this bound; backspin lift can lower it. Both are treated in later sections.

Step 3: a straight line near the mouth. Close to the mouth the path is almost straight. In the worked example the ball reaches the gate at $5.85\,\text{m/s}^{[\mathsf{P}]}$ and $\varphi = 79.2^\circ\,^{[\mathsf{P}]}$. Under gravity alone its curvature is $\kappa = g\cos\varphi/\|\boldsymbol{v}\|^2 = 0.054\,\text{m}^{-1\,[\mathsf{D}]}$, a radius of $18.7\,\text{m}^{[\mathsf{D}]}$. Over a $0.2\,\text{m}^{[\mathsf{A}]}$ chord near the mouth the path departs from a straight line by about $0.3\,\text{mm}^{[\mathsf{D}]}$. We therefore treat the last part of the path as a straight line with descent angle φ.

Step 4: go to the cell-local frame. Rotate the direction of motion into the cell-local frame with the inverse of (2). Using $\varphi + \tau = 90^\circ + \alpha$,

$$\hat{\boldsymbol{e}}_{uw} = \big(\cos(\varphi + \tau),\ -\sin(\varphi + \tau)\big) = (-\sin\alpha,\ -\cos\alpha). \tag{29}$$

In the CELL's own frame the ball is falling almost parallel to the mouth plane, from the roof side toward the floor side, while moving inward at the small rate $\sin\alpha$. Let the centre cross the mouth plane at (u_M, w_c), with $w_F + R \leq w_c \leq w_C - R$ (on the gate). Before the crossing the centre was at

$$\big(u_M + \lambda\sin\alpha,\ w_c + \lambda\cos\alpha\big), \qquad \lambda \geq 0, \tag{30}$$

with λ (in m) the distance travelled back along the path. It is outside the mouth ($u > u_M$) the whole time.

Step 5: clear the upper lip. The obstacle nearest this ray is the inner corner of the upper lip, $\boldsymbol{C} = (u_M, w_C)$; the rest of the lip and the roof are further away from it. The distance from $\boldsymbol{C}$ to the line through (u_M, w_c) in direction (29) is the length of the cross product of $\boldsymbol{C} - (u_M, w_c) = (0,\ w_C - w_c)$ with the unit direction:

$$\delta_C = \big|\,0\cdot(-\cos\alpha) - (w_C - w_c)(-\sin\alpha)\,\big| = (w_C - w_c)\sin\alpha. \tag{31}$$

The nearest point lies at $\lambda = (w_C - w_c)\cos\alpha \geq 0$, so it is *before* the crossing and counts. The clean test needs $\delta_C \geq R$:

$$w_c \leq w_C - \frac{R}{\sin\alpha}. \tag{32}$$

Step 6: the lower lip does not bind before entry. For the lower-lip corner $\boldsymbol{D} = (u_M, w_F)$ the same calculation gives the nearest point at $\lambda = (w_F - w_c)\cos\alpha < 0$, which is *after* the crossing. Before the crossing the closest the ball gets to $\boldsymbol{D}$ is at the crossing itself, $w_c - w_F \geq R$, which the gate already guarantees. A ball crossing near the low end may touch the lower lip or floor just after it enters, but by the definition in Section 2.8 it is already in.

Step 7: the minimum descent angle. A clean straight path exists when (32) leaves room above the gate's low end, $w_C - R/\sin\alpha \geq w_F + R$, that is $\sin\alpha \geq R/(w_C - w_F - R) = R/(L - R)$. Hence

$$\boxed{\varphi > \varphi_{\text{gate}} = 90^\circ - \tau + \arcsin\frac{R}{L - R}} \tag{33}$$

With $R/(L-R) = 0.0355/0.32188 = 0.1103^{[D]}$, $\arcsin(0.1103) = 6.33^{\circ[D]}$ and $\varphi_{\text{gate}} = 66.33^{\circ[D]}$ at $\tau = 30^{\circ[C]}$.

A stricter version: passing both lips. If we also ask that the ball touch nothing until it is completely past the mouth plane, the line must also stay R from $\boldsymbol{D}$: $\delta_D = (w_c - w_F)\sin\alpha \geq R$. Combined with (32) this needs $L\sin\alpha \geq 2R$:

$$\varphi > \varphi_{\text{fit}} = 90^\circ - \tau + \arcsin\frac{2R}{L} = 60^\circ + 11.46^\circ = 71.46^{\circ[D]}. \tag{34}$$

The quantity $L\sin\alpha$ is the width of the mouth as seen along the path. This is the "about 71.5°" figure in the project documents. The code's clean test is (33), not (34).

Window width. The set of clean straight paths at a given α is $w_c \in [\,w_F + R,\ w_C - R/\sin\alpha\,]$. Measured perpendicular to the path, its width (in m) is

$$W_\perp(\alpha) = \Big(w_C - \frac{R}{\sin\alpha} - w_F - R\Big)\sin\alpha = (L - R)\sin\alpha - R. \tag{35}$$

For a vertical drop, $\varphi = 90^\circ$, so $\alpha = \tau$ and the window is horizontal: $W_\perp = 0.32188 \times 0.5 - 0.0355 = 0.1254\,\text{m}^{[D]}$. Under the stricter both-lips rule the same drop leaves $L\sin\tau - 2R = 0.1077\,\text{m}^{[D]}$. At $\varphi = 70^{\circ[A]}$ ($\alpha = 10^\circ$) the clean window is only $W_\perp = 0.0204\,\text{m}^{[D]}$, about $0.29^{[D]}$ ball diameters ($0.57^{[D]}$ radii).

Worked check against the code. We fired families of straight lines at the gate with `Hive.classify_path` (`pose_pos`, HIVE frame, $400^{[A]}$ entry points along the gate per angle) and bisected on φ for the smallest angle with any clean line. Table 7 compares the result with (33). The perpendicular widths found at $\varphi = 70^\circ$, 72° and 90° were $0.0202^{[P]}$, $0.0312^{[P]}$ and $0.1249\,\text{m}^{[P]}$, against $0.0204^{[D]}$, $0.0314^{[D]}$ and $0.1254\,\text{m}^{[D]}$ from (35); the small shortfall comes from the discrete entry points, spaced $0.72\,\text{mm}^{[D]}$ along the gate, so up to one spacing is lost at each end of the window. The vertical drop is clean for $x \in [0.348, 0.4726]\,\text{m}^{[P]}$, from one radius right of the upper lip ($0.31203 + 0.0355 = 0.3475\,\text{m}^{[D]}$) to the gate's low end ($A_x = 0.47298\,\text{m}^{[D]}$).

A steeper HIVE (larger τ) turns the mouth more toward the sky and lowers every limit one degree per degree of tilt. The arcsin term does not depend on tilt because the CELL itself does not change shape. The worked shot of Section 2.8 entered at $\varphi = 79.21^{\circ[P]}$, above all three limits at $\tau = 30^{\circ[C]}$.

Table 7: Descent-angle limits on the unfavourable side versus tilt. φ_{kin}: (26); φ_{gate}: (33) (the code's criterion); φ_{fit}: (34); "code" is the bisection with `Hive.classify_path`; z_{lip} from the re-posed `Hive`.

τ	z_{lip} (m)	φ_{kin}	φ_{gate}	code	φ_{fit}
$25.5°^{[D]}$	$1.3173^{[D]}$	$64.5°^{[D]}$	$70.83°^{[D]}$	–	$75.96°^{[D]}$
$28.5°^{[D]}$	$1.3436^{[D]}$	$61.5°^{[D]}$	$67.83°^{[D]}$	$67.85°^{[P]}$	$72.96°^{[D]}$
$30.0°^{[C]}$	$1.3565^{[D]}$	$60.0°^{[D]}$	$66.33°^{[D]}$	$66.35°^{[P]}$	$71.46°^{[D]}$
$31.5°^{[D]}$	$1.3693^{[D]}$	$58.5°^{[D]}$	$64.83°^{[D]}$	$64.85°^{[P]}$	$69.96°^{[D]}$
$34.5°^{[D]}$	$1.3943^{[D]}$	$55.5°^{[D]}$	$61.83°^{[D]}$	–	$66.96°^{[D]}$

2.10 Release point and the robot height limit

The release point (5) has two free numbers. The distance d is measured from the release point to the pivot plane $x = 0$ (not to the mouth). The field's inner perimeter wall is at $|x| = 1.8009\,\text{m}^{[C]}$, so d cannot exceed that. The sweep grid uses d from $0.55\,\text{m}^{[A]}$ to $1.75\,\text{m}^{[A]}$ in $0.05\,\text{m}^{[A]}$ steps.

The height h is limited by the robot size rule. Manual rule R105 caps the expanded robot at $H_{\max} = 29\,\text{in}^{[C]} = 0.7366\,\text{m}^{[C]}$ measured from the field surface. The project assumes that the ball centre at release is no higher than this:

$$h \le H_{\max} = 0.7366\ \text{m}^{[A]}, \tag{36}$$

and the sweep uses $h \in \{0.20, 0.35, 0.50, 0.65, 0.737\}\,\text{m}^{[A]}$. If instead the whole ball had to stay inside the sizing volume, the centre limit would be $H_{\max} - R = 0.701\,\text{m}^{[D]}$; the code uses the former. The lowest point of the HIVE (the down CELL's lower lip) is at $0.778\,\text{m}^{[D]}$, above the release, so every shot must rise to pass over the roof, condition (28).

2.11 Assumptions

- The model is 2D, in the HIVE centre plane $s = 0$, where the mouth is tallest. Off-centre the pentagonal roof is lower. For a lateral offset $|s|$, `hive.py` lowers the roof segments and the top of the mouth by
$$\Delta w(s) = (h_0 - h_w)\,\frac{\min(|s|,\, s_{\text{half}})}{s_{\text{half}}}, \tag{37}$$
with peak height $h_0 = 0.3556\,\text{m}^{[C]}$, side-wall height $h_w = 0.19329\,\text{m}^{[C]}$ and half-width $s_{\text{half}} = 0.254\,\text{m}^{[C]}$. This is a 2D proxy; the side walls themselves are not modelled.

- The obstacles are solid line segments with no thickness of their own. The roof top and the underside are envelopes of the ribs, accurate to about $11\,\text{mm}^{[D]}$. The ballast screw near each CELL opening and the three preloaded NECTAR are not modelled.

- The four A-frame legs are left out by default, because at $s = 0$ only their tops exist.

- The ball is a rigid sphere of radius R; its centre is the tracked point.

- "Clean" means the centre crosses the gate inward with no earlier contact (an assumed criterion). Contact after the crossing, and whether the ball then stays in, belongs to the contact model. A planned bank shot off the roof is outside this channel (and is a G417 risk).

- The tilt is a fixed $30°^{[C]}$ rest angle, with $\pm 1.5°^{[D]}$ and $\pm 4.5°^{[D]}$ bounds; the HIVE does not move during the shot.

- The descent-angle limits (33) and (34) treat the last few centimetres of the path as straight.
- The release height limit (36) applies to the ball centre.
- The worked shot is in vacuum and illustrates the method only; its numbers are not a prediction for POLLEN in air.

3 Flight dynamics: drag, Magnus lift and spin decay

Once POLLEN leaves the shooter, nothing on the robot can change its path any more. Three things act on the ball in the air: gravity, air drag and the Magnus force from spin. This section derives, step by step, exactly what the flight solver `src/flight.py` computes. The goal is to answer two practical questions. How large is the spin force compared with gravity and drag? And does adding more backspin keep helping? In this model the answer to the second question is no: above a certain spin the lift coefficient stops growing, and extra spin changes nothing.

Every aerodynamic coefficient in this section is a literature bound, not a POLLEN measurement. The ranges come from `research/params_lit.json`. Pickleball and smooth-sphere data set the bounds, but no one has yet measured drag, lift or spin decay on a real POLLEN ball.

3.1 Frame, state and parameters

Motivation. Before writing any force we need a coordinate frame and a list of what the solver tracks. Sign errors in the Magnus force are the most common bug in spin models, so the frame must be fixed first.

Frame. The world frame is right-handed. The axis x is horizontal and points toward the target, y is horizontal and sideways, and z points up. The unit vectors are $\hat{\boldsymbol{x}}$, $\hat{\boldsymbol{y}}$ and $\hat{\boldsymbol{z}}$. All quantities are in SI units (m, s, kg, rad).

State. The solver integrates a nine-component state: the position of the ball centre $\boldsymbol{r}$ (m), its velocity $\boldsymbol{v}$ (m/s) and its spin (angular velocity) vector $\boldsymbol{\omega}$ (rad/s). The ball is treated as a rigid sphere of mass m (kg) and outer radius R (m).

Parameters. Table 8 lists the physical constants read by `load_params`. The acceleration of gravity is g (m/s^2), the air density is ρ (kg/m^3) and the kinematic viscosity of air is ν (m^2/s). The frontal (cross-sectional) area of the ball is

$$A = \pi R^2, \tag{38}$$

in m^2. With the nominal radius, $A = 3.959 \times 10^{-3}\,\mathrm{m}^{2\,[\mathrm{D}]}$.

Assumptions.

- The world frame is inertial (no Coriolis or centrifugal terms) and the floor is flat.
- The ball is a rigid sphere of fixed m and R; it does not deform in flight.
- ρ, ν and g are constant during a shot.

Table 8: Physical constants used by the flight solver (`research/params_lit.json`). Only the nominal values of ρ, ν, m and R are read by `load_params`; the bounds are used by the robustness sweeps.

Symbol	Meaning	Unit	Nominal	Bounds
g	standard gravity	$\mathrm{m/s^2}$	$9.80665^{[C]}$	–
ρ	air density	$\mathrm{kg/m^3}$	$1.2^{[A]}$	$1.0^{[A]}$ to $1.25^{[A]}$
ν	kinematic viscosity	$\mathrm{m^2/s}$	$1.5 \times 10^{-5\,[A]}$	$1.45 \times 10^{-5\,[A]}$ to $1.6 \times 10^{-5\,[A]}$
m	ball mass	kg	$0.02495^{[A]}$	$0.0235^{[A]}$ to $0.0275^{[A]}$
R	outer radius	m	$0.0355^{[C]}$	$0.035^{[A]}$ to $0.036^{[A]}$

3.2 Equations of motion

Motivation. Newton's second law gives the translation; a simple decay law gives the spin. The only subtlety is wind: aerodynamic forces depend on how the ball moves *through the air*, not over the floor.

Air-relative velocity. Let $\boldsymbol{u}$ (m/s) be the wind velocity. The air-relative velocity is

$$\boldsymbol{v}_{\mathrm{rel}} = \boldsymbol{v} - \boldsymbol{u}. \tag{39}$$

Both aerodynamic forces below use $\boldsymbol{v}_{\mathrm{rel}}$; gravity does not. The code accepts a constant $\boldsymbol{u}$ or a function $\boldsymbol{u}(t, \boldsymbol{r})$. The default is still air, $\boldsymbol{u} = \boldsymbol{0}$, so then $\boldsymbol{v}_{\mathrm{rel}} = \boldsymbol{v}$.

Newton's law and the spin law. Gravity is $m\boldsymbol{g}$ with $\boldsymbol{g} = -g\,\hat{\boldsymbol{z}}$. Adding the drag force $\boldsymbol{F}_D$ and the Magnus force $\boldsymbol{F}_M$ (both in N) gives

$$\frac{d\boldsymbol{r}}{dt} = \boldsymbol{v}, \tag{40}$$

$$m\,\frac{d\boldsymbol{v}}{dt} = -mg\,\hat{\boldsymbol{z}} + \boldsymbol{F}_D(\boldsymbol{v}_{\mathrm{rel}}) + \boldsymbol{F}_M(\boldsymbol{v}_{\mathrm{rel}}, \boldsymbol{\omega}), \tag{41}$$

$$\frac{d\boldsymbol{\omega}}{dt} = -k\,\boldsymbol{\omega}. \tag{42}$$

Here k (1/s) is the spin-decay rate (Section 3.7). In `_make_rhs` the right-hand side is exactly this: the acceleration is $(\boldsymbol{F}_D + \boldsymbol{F}_M)/m$ with g subtracted from the z component, and each spin component is multiplied by $-k$. No other torque acts on the ball, so the spin axis never turns.

Assumptions.

- The ball is a point mass for translation; only its spin vector is tracked for rotation.
- Buoyancy and the added mass of the surrounding air are neglected. The air displaced by the outer envelope has a mass of $2.25 \times 10^{-4}\,\mathrm{kg}^{[D]}$, which is $0.9\,\%^{[D]}$ of m.
- The wind is prescribed; the ball's own wake and the robot's airflow are ignored.

3.3 Drag

Motivation. Drag is the largest aerodynamic force on POLLEN for almost every parameter set in this model (the one exception, high lift with low drag, is in Section 3.10). It slows the ball and shortens the shot.

Derivation. The dynamic pressure q (Pa) is the kinetic energy of the oncoming air per unit volume. The drag magnitude is the dynamic pressure times the frontal area times a dimensionless drag coefficient C_d. Drag points straight against the air-relative velocity, so we multiply by the unit vector $-\boldsymbol{v}_{\mathrm{rel}}/|\boldsymbol{v}_{\mathrm{rel}}|$:

$$q = \tfrac{1}{2}\,\rho\,|\boldsymbol{v}_{\mathrm{rel}}|^2, \tag{43}$$

$$|\boldsymbol{F}_D| = q\,C_d\,A, \tag{44}$$

$$\boldsymbol{F}_D = -|\boldsymbol{F}_D|\,\frac{\boldsymbol{v}_{\mathrm{rel}}}{|\boldsymbol{v}_{\mathrm{rel}}|} = -\tfrac{1}{2}\,\rho\,C_d\,A\,|\boldsymbol{v}_{\mathrm{rel}}|\,\boldsymbol{v}_{\mathrm{rel}}. \tag{45}$$

In the last step one factor of $|\boldsymbol{v}_{\mathrm{rel}}|$ cancels. The form $|\boldsymbol{v}_{\mathrm{rel}}|\,\boldsymbol{v}_{\mathrm{rel}}$ is what `_aero` computes (`kd = -q*cd*vmag`, then `kd*vx`, ...; note that the code's variable `q` is $\frac{1}{2}\rho A$, not the dynamic pressure q of Eq. (43)). It has no division, so drag goes smoothly to zero as $|\boldsymbol{v}_{\mathrm{rel}}| \to 0$.

Values. C_d is constant during a run. The bounds are $C_d = 0.33^{[A]}$ (low), $0.5^{[A]}$ (nominal) and $0.9^{[A]}$ (high). The low end comes from outdoor pickleball free flight; the high end covers a spinning perforated ball, because smooth-sphere data show that spin raises drag. The solver also accepts a function $C_d(\mathrm{Re})$ or $C_d(\mathrm{Re}, S)$, but no sweep uses one.

Worked example. Take a launch at $|\boldsymbol{v}| = 7\,\mathrm{m/s}^{[A]}$ and elevation $60^{\circ[A]}$ in still air, so $\boldsymbol{v} = (3.5^{[A]}, 0, 6.062^{[D]})\,\mathrm{m/s}$. With the nominal values, $\frac{1}{2}\rho A = 2.3755 \times 10^{-3}\,\mathrm{kg/m}^{[D]}$, so $|\boldsymbol{F}_D| = 2.3755 \times 10^{-3[D]} \times 0.5^{[A]} \times 49^{[D]} = 0.0582\,\mathrm{N}^{[D]}$. `aero_forces` returns $\boldsymbol{F}_D = (-0.02910, 0, -0.05040)\,\mathrm{N}^{[P]}$, which is the same magnitude pointing exactly against $\boldsymbol{v}$. The weight is $mg = 0.2447\,\mathrm{N}^{[D]}$, so drag is $0.238^{[D]}$ of the weight at this speed.

Assumptions.

- C_d is constant along the path; no Reynolds-number or spin dependence is active.
- No "knuckle" side force from the holes is modelled. `params_lit.json` lists a non-Magnus hole force as a scatter source, but the solver does not include it.

3.4 Reynolds number and spin parameter

Motivation. Two dimensionless numbers describe the flow around a spinning ball. The Reynolds number says which flow regime we are in; the spin parameter says how fast the surface moves compared with the air.

Reynolds number. With the ball diameter $D = 2R$ (m),

$$\mathrm{Re} = \frac{|\boldsymbol{v}_{\mathrm{rel}}|\,2R}{\nu}. \tag{46}$$

For speeds of $2^{[A]}$, $5^{[A]}$, $7^{[A]}$ and $10\,\mathrm{m/s}^{[A]}$ this gives $\mathrm{Re} = 9.47 \times 10^{3[D]}$, $2.37 \times 10^{4[D]}$, $3.31 \times 10^{4[D]}$ and $4.73 \times 10^{4[D]}$. This is below the smooth-sphere drag crisis, but it overlaps the band where smooth-sphere lift data show a dip (the "lift crisis" recorded in `params_lit.json`). In the default model Re is only a diagnostic: C_d is constant and C_L depends on S alone. It is reported so that a run can be checked against the range of the literature data.

Spin parameter. A point on the equator of the ball moves at speed $|\boldsymbol{\omega}|R$ relative to the centre. The spin parameter is the ratio of that surface speed to the airspeed:

$$S = \frac{|\boldsymbol{\omega}|\, R}{|\boldsymbol{v}_{\mathrm{rel}}|}. \tag{47}$$

S has no units. $S = 1$ means the surface speed equals the airspeed, which is what an ideal no-slip hood gives at the exit. Note that the code uses the full spin magnitude $|\boldsymbol{\omega}|$, not only the part of $\boldsymbol{\omega}$ perpendicular to $\boldsymbol{v}_{\mathrm{rel}}$.

Regularisation. Equation (47) divides by zero when the ball is at rest relative to the air. The code therefore uses

$$S = \frac{|\boldsymbol{\omega}|\, R}{\sqrt{|\boldsymbol{v}_{\mathrm{rel}}|^2 + \varepsilon_v^2}}, \tag{48}$$

with the regularisation speed $\varepsilon_v = 10^{-6}\,\mathrm{m/s}^{[A]}$. For any airspeed above about $10^{-3}\,\mathrm{m/s}^{[A]}$ the difference is far below measurement accuracy.

Worked example. For the launch above, an exit spin parameter $S_0 = 1.06^{[A]}$ (the nominal hood value in `params_lit.json`) needs $|\boldsymbol{\omega}_0| = S_0|\boldsymbol{v}|/R = 1.06^{[A]} \times 7^{[A]}/0.0355^{[C]} = 209.0\,\mathrm{rad/s}^{[D]}$, which is $1996\,\mathrm{rpm}^{[D]}$. The Reynolds number is $3.31 \times 10^{4\,[D]}$.

3.5 The Magnus force

Motivation. A spinning ball drags the air near its surface around with it. The flow is deflected to one side, and the ball is pushed to the other side. This sideways push is the Magnus force. It is the only way spin changes the flight path in this model.

Magnitude and direction. The Magnus force has the same dynamic-pressure form as drag, with a lift coefficient C_L (no units) in place of C_d. Its direction is perpendicular to both the spin axis and the airflow, given by the cross product $\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}}$:

$$|\boldsymbol{F}_M| = \tfrac{1}{2}\,\rho\, C_L\, A\, |\boldsymbol{v}_{\mathrm{rel}}|^2, \tag{49}$$

$$\hat{\boldsymbol{n}}_M = \frac{\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}}}{|\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}}|}, \tag{50}$$

$$\boldsymbol{F}_M = |\boldsymbol{F}_M|\, \hat{\boldsymbol{n}}_M = \tfrac{1}{2}\,\rho\, C_L(S)\, A\, |\boldsymbol{v}_{\mathrm{rel}}|^2\, \frac{\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}}}{|\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}}|}. \tag{51}$$

The unit vector $\hat{\boldsymbol{n}}_M$ carries only the direction, so the spin rate enters the force only through $C_L(S)$. This fact matters a great deal in Section 3.6.

Regularised form used in the code. To keep the direction finite when $\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}} \to \boldsymbol{0}$, `_aero` computes

$$\boldsymbol{F}_M = \tfrac{1}{2}\,\rho\, C_L(S)\, A\, |\boldsymbol{v}_{\mathrm{rel}}|^2\, \frac{\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}}}{\sqrt{|\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}}|^2 + \varepsilon_v^2 |\boldsymbol{\omega}|^2}}, \tag{52}$$

and sets $\boldsymbol{F}_M = \boldsymbol{0}$ when $\boldsymbol{\omega} = \boldsymbol{0}$, when $\boldsymbol{v}_{\mathrm{rel}} = \boldsymbol{0}$ or when no lift law is given. Because of the factor $|\boldsymbol{v}_{\mathrm{rel}}|^2$ and the bounded C_L, the force vanishes smoothly at zero airspeed; there is no fictitious force from the $1/|\boldsymbol{v}_{\mathrm{rel}}|$ in S.

Magnus does no work in still air. A useful property follows from the cross product. For any vectors, $(\boldsymbol{\omega} \times \boldsymbol{v}_{\text{rel}}) \cdot \boldsymbol{v}_{\text{rel}} = 0$, so

$$\boldsymbol{F}_M \cdot \boldsymbol{v}_{\text{rel}} = 0. \tag{53}$$

In still air $\boldsymbol{v}_{\text{rel}} = \boldsymbol{v}$, so the power $\boldsymbol{F}_M \cdot \boldsymbol{v}$ is zero. The Magnus force bends the path but cannot speed the ball up or slow it down. Only gravity and drag change the speed.

Worked example. For the launch above ($S_0 = 1.06^{[A]}$, nominal set, so $C_L = 0.25^{[D]}$; see below), `aero_forces` returns $\boldsymbol{F}_M = (-0.02520, 0, 0.01455)\,\text{N}^{[P]}$ with $|\boldsymbol{F}_M| = 0.0291\,\text{N}^{[P]}$. Check: $2.3755 \times 10^{-3\,[D]} \times 0.25^{[D]} \times 49^{[D]} = 0.0291\,\text{N}^{[D]}$. The dot product with $\boldsymbol{v} = (3.5, 0, 6.062)$ is $-0.02520 \times 3.5 + 0.01455 \times 6.062 = 0.0000^{[D]}$ (to four decimals), as Eq. (53) requires.

Assumptions.

- C_L depends on S only, and S uses $|\boldsymbol{\omega}|$, not its component perpendicular to $\boldsymbol{v}_{\text{rel}}$. For pure backspin in a vertical plane the two are equal.
- The direction rule $\hat{\boldsymbol{n}}_M = \text{unit}(\boldsymbol{\omega} \times \boldsymbol{v}_{\text{rel}})$ holds for every S: the model has no reverse (negative) Magnus effect. The literature judges reverse Magnus unlikely at POLLEN's Reynolds numbers (the documented negative-lift windows are at $\text{Re} \geq 4.6 \times 10^{4\,[A]}$ and $S < 0.6^{[A]}$), but does not exclude it, because no perforated ball has been tested above $S = 0.56^{[A]}$.

3.6 The lift law and its saturation

Motivation. How C_L grows with spin decides whether more spin helps. For POLLEN the shape of $C_L(S)$ is the largest unknown in the whole project.

Linear-capped law. `cl_linear_capped` implements

$$C_L(S) = \min\big(c_1 S,\ C_{L,\max}\big), \qquad S \geq 0, \tag{54}$$

where c_1 (no units) is the initial slope and $C_{L,\max}$ (no units) is the ceiling. The law rises linearly from zero and then stays flat. It has no offset, so $C_L = 0$ when $S = 0$.

Saturation spin. The two branches meet where $c_1 S = C_{L,\max}$. Solving for S gives the saturation spin parameter

$$S_{\text{sat}} = \frac{C_{L,\max}}{c_1}. \tag{55}$$

For $S \geq S_{\text{sat}}$ the code returns $C_{L,\max}$ exactly (`return v if v < cl_max else cl_max`). Table 9 gives the four sets in `params_lit.json`.

How S changes along the path. With constant spin decay (Eq. (58) below) the spin parameter along the path is

$$S(t) = \frac{|\boldsymbol{\omega}_0|\, e^{-kt}\, R}{|\boldsymbol{v}_{\text{rel}}(t)|} = S_0\, e^{-kt}\, \frac{|\boldsymbol{v}_{\text{rel}}(0)|}{|\boldsymbol{v}_{\text{rel}}(t)|}. \tag{56}$$

The ball slows on the way up, so S *rises* toward the apex even though the spin itself decays. In the worked shot below, S grows from $1.06^{[A]}$ at launch to $2.53^{[P]}$ at the apex and falls back to $1.08^{[P]}$ at the floor.

Table 9: Lift sets in `research/params_lit.json`. All are literature bounds, not POLLEN measurements. S_{sat} is from Eq. (55).

Set	c_1	$C_{L,\max}$	S_{sat}	Basis (summary)
floor	$0.02^{[A]}$	$0.04^{[A]}$	$2.00^{[D]}$	pickleball fit, Magnus-only reading
low	$0.2^{[A]}$	$0.15^{[A]}$	$0.75^{[D]}$	pickleball slope; smooth-sphere lift dip
nominal	$0.35^{[A]}$	$0.25^{[A]}$	$0.714^{[D]}$	between pickleball and smooth sphere
high	$0.8^{[A]}$	$0.4^{[A]}$	$0.50^{[D]}$	smooth-sphere slope and plateau

Why spin above S_{sat} changes nothing in this model. Look at where $\boldsymbol{\omega}$ enters the equations of motion. The translational equation (41) depends on $\boldsymbol{\omega}$ only through (i) $C_L(S)$ and (ii) the unit vector $\hat{\boldsymbol{n}}_M$. The spin law (42) scales $\boldsymbol{\omega}$ but never turns it, so $\hat{\boldsymbol{n}}_M$ depends on the spin *direction*, which is fixed at launch, and not on the spin rate. Now suppose $S(t) \geq S_{\text{sat}}$ for the whole flight. Then $C_L = C_{L,\max}$ at every instant, and the Magnus force becomes

$$\boldsymbol{F}_M = \tfrac{1}{2}\,\rho\, C_{L,\max}\, A\, |\boldsymbol{v}_{\text{rel}}|^2\, \hat{\boldsymbol{n}}_M \qquad (S \geq S_{\text{sat}}), \tag{57}$$

which contains no spin rate at all. Two launches with the same $\boldsymbol{v}_0$ and spin axis but different spin rates, both above saturation for the whole flight, therefore obey identical equations from identical initial positions and velocities. They follow the same path. For the same reason the decay rate k has no effect on the path while S stays above S_{sat}.

This is a model assumption, not a measured property of POLLEN. The cap was chosen because smooth-sphere and baseball data flatten out and because no data exist for a perforated ball at $S \approx 1$ to 2. A real ball might keep gaining lift, might show a dip (the `dip_variant` table in `params_lit.json`, which `src/flight.py` does not read), and might gain drag with spin, which the constant-C_d model leaves out. So the model cannot support the claim that more spin always helps, and it does not make that claim.

Worked example. Release from $(0, 0, 0.4)\,\text{m}^{[A]}$ at $7\,\text{m/s}^{[A]}$ and $60^{\circ[A]}$, nominal set, $k = 0.03\,1/\text{s}^{[A]}$. Running `simulate` with $S_0 = 1.0^{[A]}$, $1.06^{[A]}$, $1.5^{[A]}$ and $2.0^{[A]}$ gives the same apex at $x = 1.846\,\text{m}^{[P]}$, $z = 2.155\,\text{m}^{[P]}$ and the same landing point $x = 3.764\,\text{m}^{[P]}$. The largest position difference between the $S_0 = 1.0$ and $S_0 = 2.0$ paths is $0.0\,\text{m}^{[P]}$ in double precision, because the smallest S on the $S_0 = 1.0$ path is $1.0^{[P]}$, already above $S_{\text{sat}} = 0.714^{[D]}$. Changing k to $0.005^{[A]}$ or $0.15\,1/\text{s}^{[A]}$ also leaves the path unchanged. Table 10 compares the sets against the same launch without spin. Only the floor set responds to the extra spin: its $S_{\text{sat}} = 2.0^{[D]}$ lies above $S_0 = 1.06$, so with $S_0 = 1.06$ it is below its cap for most of the path (it saturates only near the apex, where S peaks at $2.53^{[P]}$), while with $S_0 = 2.0$ it is capped for the whole flight.

The landing shift is negative: on this steep shot backspin makes the ball land slightly *shorter*. Section 3.8 shows why: while the ball rises, the Magnus force points up *and back*.

Assumptions.

- C_L follows Eq. (54) with one of the four sets; no Reynolds-number dependence, no lift dip, no offset.
- The cap $C_{L,\max}$ holds for all S up to the largest value reached in flight (`params_lit.json` gives a valid range of 0 to $2^{[A]}$; apex values above that, as in the example, are an extrapolation of the flat cap).

Table 10: Change caused by backspin relative to the same launch with no spin (7 m/s[A], 60°[A], release height 0.4 m[A], $C_d = 0.5$[A], $k = 0.03\,1/\mathrm{s}$[A]). Δz_{apex} is the apex height gain and Δx_{land} the landing-point shift, in cm. One model run per cell; these are model sensitivities, not repeatable test results.

	Δz_{apex} (cm)		Δx_{land} (cm)	
Set	S_0=1.06	S_0=2.0	S_0=1.06	S_0=2.0
floor	+0.74[P]	+1.06[P]	−0.18[P]	−0.67[P]
low	+3.98[P]	+3.98[P]	−2.86[P]	−2.86[P]
nominal	+6.60[P]	+6.60[P]	−5.29[P]	−5.29[P]
high	+10.47[P]	+10.47[P]	−9.72[P]	−9.72[P]

3.7 Spin decay

Motivation. Air friction on the surface slows the spin during flight. The model uses the simplest law that captures this.

Derivation. Equation (42) acts on each component separately: $d\omega_i/dt = -k\,\omega_i$. Separating variables, $d\omega_i/\omega_i = -k\,dt$, and integrating from 0 to t gives $\ln(\omega_i(t)/\omega_{i,0}) = -kt$, so

$$\boldsymbol{\omega}(t) = \boldsymbol{\omega}_0\, e^{-kt}. \tag{58}$$

All components shrink by the same factor, so the spin axis stays fixed in space. The time for the spin to halve is

$$t_{1/2} = \frac{\ln 2}{k}. \tag{59}$$

`flight.simulate` and `contact.free_flight` integrate Eq. (42) as part of the state; `feasibility.integrate_batch` and `robustness.integrate3d` use the closed form (58) directly.

Table 11: Spin-decay bounds from `params_lit.json` and their effect over a 1.2 s[A] flight.

Bound	k (1/s)	Spin lost in 1.2 s[A]	$t_{1/2}$ (s)
low	0.005[A]	0.6 %[D]	139[D]
nominal	0.03[A]	3.5 %[D]	23.1[D]
high	0.15[A]	16.5 %[D]	4.62[D]

Values.

Where k comes from (literature model, not in the code). `params_lit.json` records a torque model in which the spin-down torque grows with speed. With a moment coefficient $C_M = \beta S$ (β has no units), the torque is $\tau = -R\,\rho\,A\,C_M\,|\boldsymbol{v}|^2$ (N m). The ball's moment of inertia is $I_b = \alpha m R^2$ (kg m^2). The shape factor α has no units. Substituting $S = |\boldsymbol{\omega}|R/|\boldsymbol{v}|$ and $A = \pi R^2$

into $I_b\, d|\boldsymbol{\omega}|/dt = \tau$:

$$\alpha m R^2 \frac{d|\boldsymbol{\omega}|}{dt} = -R\,\rho\,\pi R^2\,\beta\,\frac{|\boldsymbol{\omega}|R}{|\boldsymbol{v}|}\,|\boldsymbol{v}|^2 = -\pi\rho\beta R^4\,|\boldsymbol{v}|\,|\boldsymbol{\omega}|,$$
$$k_{\text{torque}} = \frac{\pi\,\rho\,\beta\,R^2\,|\boldsymbol{v}|}{\alpha\,m}. \tag{60}$$

With $\rho = 1.2$[A], $R = 0.0355$[C], $\alpha = 0.63$[D] and $m = 0.02495$[A] this is $k_{\text{torque}} = 0.302\,\beta\,|\boldsymbol{v}|$[D]. The nominal $\beta = 0.02$[A] at $5\,\text{m/s}$[A] gives $0.030\,1/\text{s}$[D], which is where the nominal constant k comes from. The solver uses a *constant* k; the speed-dependent form is not implemented.

Assumptions.

- k is constant during a flight, equal to its value near $5\,\text{m/s}$[A].
- No torque turns the spin axis (no gyroscopic or wobble effects).

3.8 The backspin axis convention

Motivation. "Backspin" must be turned into a definite vector before the solver can use it. `backspin_omega` does this, and a unit test checks that backspin on a ball moving forward gives an upward Magnus force.

Definition. For a launch velocity $\boldsymbol{v}_0$ and a spin rate ω_0 (rad/s; positive for backspin, negative for topspin),

$$\boldsymbol{\omega}_0 = \omega_0\,\frac{\boldsymbol{v}_0 \times \hat{\boldsymbol{z}}}{|\boldsymbol{v}_0 \times \hat{\boldsymbol{z}}|}. \tag{61}$$

The axis is horizontal and perpendicular to the vertical plane of motion. It is undefined for a vertical launch; the code raises an error then.

Derivation for a shot along $+x$. Write $\boldsymbol{v}_0 = (v_0\cos\theta,\, 0,\, v_0\sin\theta)$ with elevation θ (rad). Using $\boldsymbol{a} \times \hat{\boldsymbol{z}} = (a_y,\, -a_x,\, 0)$,

$$\boldsymbol{v}_0 \times \hat{\boldsymbol{z}} = (0,\, -v_0\cos\theta,\, 0) \implies \boldsymbol{\omega}_0 = -\,\omega_0\,\hat{\boldsymbol{y}} \quad (\cos\theta > 0). \tag{62}$$

So backspin on a ball moving toward $+x$ points along $-y$. Now take the cross product with the velocity (still air, so $\boldsymbol{v}_{\text{rel}} = \boldsymbol{v}$). Using $\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}} = -\hat{\boldsymbol{z}}$ and $\hat{\boldsymbol{y}} \times \hat{\boldsymbol{z}} = \hat{\boldsymbol{x}}$,

$$\boldsymbol{\omega}_0 \times \boldsymbol{v}_0 = -\omega_0\left(v_0\cos\theta\;\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}} + v_0\sin\theta\;\hat{\boldsymbol{y}} \times \hat{\boldsymbol{z}}\right) = \omega_0 v_0\,(-\sin\theta,\, 0,\, \cos\theta),$$
$$\hat{\boldsymbol{n}}_M = (-\sin\theta,\, 0,\, \cos\theta) \qquad (\omega_0 > 0). \tag{63}$$

This is the flight direction $(\cos\theta, 0, \sin\theta)$ turned by $+90^\circ$ in the x–z plane. For a level shot ($\theta = 0$) it points straight up, which is the unit-tested convention. For a rising ball ($\theta > 0$) it points up *and backward*; for a falling ball it points up and forward. That is why backspin shortened the steep shot in Table 10.

Worked example. For $\theta = 60^\circ$[A] and $\omega_0 = 209.0\,\text{rad/s}$[D], the code gives $\boldsymbol{\omega}_0 = (0, -209.0, 0)\,\text{rad/s}$[D] and $\hat{\boldsymbol{n}}_M = (-0.866, 0, 0.500)$[D]. This matches $(-\sin 60^\circ, 0, \cos 60^\circ)$.

Assumptions.

- The axis is set once, from $\boldsymbol{v}_0$, and stays fixed. If the horizontal velocity ever reversed, the same $\boldsymbol{\omega}$ would act as topspin; the code flags such paths with `horizontal_reversal`.
- No sidespin or axis tilt is used in the 2D baseline.

3.9 The 2D reduction

Motivation. Most sweeps run in the vertical x–z plane, because 2D runs are far cheaper than 3D runs. We now show that pure backspin with no sideways wind keeps the ball in that plane, and derive the scalar equations the fast sweeps use.

Derivation. Take $y_0 = 0$, $v_{y,0} = 0$, $u_y = 0$ and $\boldsymbol{\omega} = \omega_y\,\hat{\boldsymbol{y}}$ (for backspin toward $+x$, $\omega_y < 0$). Write $\boldsymbol{v}_{\mathrm{rel}} = (v_x, 0, v_z)$ for its components. Then

$$\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}} = (\omega_y v_z - 0,\ 0 - 0,\ 0 - \omega_y v_x) = \omega_y\,(v_z,\ 0,\ -v_x), \tag{64}$$

so $|\boldsymbol{\omega} \times \boldsymbol{v}_{\mathrm{rel}}| = |\omega_y|\,|\boldsymbol{v}_{\mathrm{rel}}|$ and

$$\boldsymbol{F}_M = \tfrac{1}{2}\,\rho\,C_L\,A\,|\boldsymbol{v}_{\mathrm{rel}}|\ \mathrm{sgn}(\omega_y)\,(v_z,\ 0,\ -v_x). \tag{65}$$

Neither force has a y component, so v_y stays zero and the ball stays in the plane. Define the drag constant $\kappa = \rho A/(2m)$ (1/m) and the backspin sign $s_b = -\,\mathrm{sgn}(\omega_y)$ ($s_b = +1$ for backspin toward $+x$). Dividing Eq. (41) by m and inserting Eqs. (45) and (65):

$$\frac{dv_x}{dt} = -\kappa\,C_d\,|\boldsymbol{v}_{\mathrm{rel}}|\,v_x - s_b\,\kappa\,C_L(S)\,|\boldsymbol{v}_{\mathrm{rel}}|\,v_z, \tag{66}$$

$$\frac{dv_z}{dt} = -g - \kappa\,C_d\,|\boldsymbol{v}_{\mathrm{rel}}|\,v_z + s_b\,\kappa\,C_L(S)\,|\boldsymbol{v}_{\mathrm{rel}}|\,v_x, \tag{67}$$

$$\omega_y(t) = \omega_{y,0}\,e^{-kt}, \qquad S = \frac{|\omega_y|\,R}{|\boldsymbol{v}_{\mathrm{rel}}|}. \tag{68}$$

These are the equations in `feasibility.integrate_batch` (with s_b given by the sign of the signed exit spin parameter S_0, positive for backspin) and in `contact._acc` (written with ω_y directly, backspin negative). The nominal value is $\kappa = 0.0952\,1/\mathrm{m}^{[\mathsf{D}]}$.

Worked example. At launch ($v_x = 3.5^{[\mathsf{A}]}$, $v_z = 6.062\,\mathrm{m/s}^{[\mathsf{D}]}$, $|\boldsymbol{v}| = 7\,\mathrm{m/s}^{[\mathsf{A}]}$, $C_L = 0.25^{[\mathsf{D}]}$), the Magnus terms of Eqs. (66)–(67) are $-0.0952 \times 0.25 \times 7 \times 6.062 = -1.010\,\mathrm{m/s}^{2\,[\mathsf{D}]}$ in x and $+0.0952 \times 0.25 \times 7 \times 3.5 = +0.583\,\mathrm{m/s}^{2\,[\mathsf{D}]}$ in z. Multiplying by m gives $(-0.0252, 0.0146)\,\mathrm{N}^{[\mathsf{D}]}$, the same as the 3D `aero_forces` result in Section 3.5.

Assumptions.

- Pure backspin about a horizontal axis perpendicular to the launch plane.
- No sideways wind and no sideways launch component.

3.10 How large are drag and Magnus compared with gravity?

Motivation. Before trusting any trajectory map it helps to know the size of each force. Ratios to g make that easy to read.

Terminal speed. For a ball falling straight down without spin, the speed stops growing when drag balances weight. Setting $\frac{1}{2}\rho C_d A v_t^2 = mg$ and solving for the terminal speed v_t (m/s):

$$v_t = \sqrt{\frac{2\,m\,g}{\rho\,C_d\,A}}. \tag{69}$$

For $C_d = 0.33^{[A]}$, $0.5^{[A]}$ and $0.9^{[A]}$ this gives $v_t = 17.67^{[D]}$, $14.35^{[D]}$ and $10.70\,\mathrm{m/s}^{[D]}$.

Drag and Magnus over g. Divide each force magnitude by mg and use Eq. (69) to replace $\rho A/(2m)$ by $g/(C_d v_t^2)$:

$$\frac{a_D}{g} = \frac{\rho\,C_d\,A\,|\boldsymbol{v}_{\mathrm{rel}}|^2}{2\,m\,g} = \left(\frac{|\boldsymbol{v}_{\mathrm{rel}}|}{v_t}\right)^2, \tag{70}$$

$$\frac{a_M}{g} = \frac{\rho\,C_L\,A\,|\boldsymbol{v}_{\mathrm{rel}}|^2}{2\,m\,g} = \frac{C_L}{C_d}\left(\frac{|\boldsymbol{v}_{\mathrm{rel}}|}{v_t}\right)^2, \tag{71}$$

$$\frac{a_M}{a_D} = \frac{C_L}{C_d}. \tag{72}$$

Here a_D and a_M (m/s^2) are the drag and Magnus accelerations. Both grow with the square of the airspeed, so both are small near the apex and largest at launch and at the end of a long fall. Because $C_L \le C_{L,\max} \le 0.4^{[A]}$ and $C_d \ge 0.33^{[A]}$, Magnus is smaller than drag for every set except the high-lift, low-drag corner ($0.4/0.33 = 1.2^{[D]}$). At the nominal values the ratio is at most $0.25/0.5 = 0.5^{[D]}$.

Table 12: Accelerations as a fraction of g (nominal ρ, m, R). Drag for the three C_d bounds; Magnus at $S = 1^{[A]}$ for the four lift sets (only the floor set is below its cap there).

	a_D/g			a_M/g at $S = 1$			
$\lvert\boldsymbol{v}_{\mathrm{rel}}\rvert$ (m/s)	C_d 0.33	0.5	0.9	floor	low	nominal	high
$3^{[A]}$	$0.029^{[D]}$	$0.044^{[D]}$	$0.079^{[D]}$	$0.002^{[D]}$	$0.013^{[D]}$	$0.022^{[D]}$	$0.035^{[D]}$
$5^{[A]}$	$0.080^{[D]}$	$0.121^{[D]}$	$0.218^{[D]}$	$0.005^{[D]}$	$0.036^{[D]}$	$0.061^{[D]}$	$0.097^{[D]}$
$7^{[A]}$	$0.157^{[D]}$	$0.238^{[D]}$	$0.428^{[D]}$	$0.010^{[D]}$	$0.071^{[D]}$	$0.119^{[D]}$	$0.190^{[D]}$
$9^{[A]}$	$0.260^{[D]}$	$0.393^{[D]}$	$0.708^{[D]}$	$0.016^{[D]}$	$0.118^{[D]}$	$0.197^{[D]}$	$0.315^{[D]}$

Path bending. What changes the shape of the path is the force component perpendicular to the velocity. For a path with local radius of curvature r_c (m), the normal acceleration is

$$a_\perp = \frac{|\boldsymbol{v}|^2}{r_c} \quad\Longrightarrow\quad r_c = \frac{|\boldsymbol{v}|^2}{a_\perp}. \tag{73}$$

Drag has no normal component. Gravity contributes $-g\cos\theta$, and in still air the Magnus force is entirely normal (Eq. (53)).

Worked example. At launch ($7\,\mathrm{m/s}^{[A]}$, $60^{\circ[A]}$, nominal, $S_0 = 1.06^{[A]}$): $a_D/g = 0.238^{[D]}$ and $a_M/g = 0.119^{[D]}$. The normal accelerations are $-4.903\,\mathrm{m/s}^{2[D]}$ from gravity and $+1.166\,\mathrm{m/s}^{2[D]}$ from Magnus, so backspin cancels $24\,\%^{[D]}$ of the downward bending at that moment. The radius of curvature grows from $9.99^{[D]}$ to $13.11\,\mathrm{m}^{[D]}$. At the apex of the same shot ($t = 0.594\,\mathrm{s}^{[P]}$), $v_z = 0$

but the ball is still moving horizontally at $v_x = 2.885\,\mathrm{m/s}^{[P]}$; it is not stationary. There $S = 2.53^{[P]}$ but C_L is still capped at $0.25^{[D]}$, and the Magnus acceleration is only $0.020\,g^{[P]}$, pointing straight up. Spin has its largest effect on the fast parts of the path, not at the apex.

Wind example. A $1\,\mathrm{m/s}^{[A]}$ headwind, $\boldsymbol{u} = (-1, 0, 0)\,\mathrm{m/s}^{[A]}$, makes $\boldsymbol{v}_{\mathrm{rel}} = (4.5, 0, 6.062)\,\mathrm{m/s}^{[D]}$ with $|\boldsymbol{v}_{\mathrm{rel}}| = 7.55\,\mathrm{m/s}^{[D]}$. The spin parameter falls to $S = 0.983^{[D]}$ (still above S_{sat}), and `aero_forces` gives $|\boldsymbol{F}_D| = 0.0677\,\mathrm{N}^{[P]}$ and $|\boldsymbol{F}_M| = 0.0339\,\mathrm{N}^{[P]}$, both $16\,\%^{[D]}$ larger than in still air because $|\boldsymbol{v}_{\mathrm{rel}}|^2$ grew from $49^{[D]}$ to $57\,\mathrm{m}^2/\mathrm{s}^{2[D]}$.

Assumptions.

- The table uses the nominal ρ, m and R; the bounds of Table 8 scale every entry by $\rho R^2/m$, which moves it by $-27\,\%^{[D]}$ to $+14\,\%^{[D]}$.
- The terminal speed is for a non-spinning ball falling vertically; the shots in this project stay well below it.

3.11 Numerical integration

Motivation. Equations (40)–(42) have no closed-form solution once drag and lift are both present, so they are integrated numerically. The project uses two kinds of integrator, and we need to know that they agree.

Adaptive Runge–Kutta in `flight.simulate`. `simulate` calls SciPy's `solve_ivp` with the Dormand–Prince 5(4) method (`RK45`; `DOP853` is an option). At every step the method estimates its own local error e_i for each state component y_i and accepts the step only if

$$\sqrt{\frac{1}{9}\sum_{i=1}^{9}\left(\frac{e_i}{a_{\mathrm{tol}} + r_{\mathrm{tol}}\,\max(|y_{n,i}|, |y_{n+1,i}|)}\right)^2} \le 1, \tag{74}$$

otherwise it retries with a smaller step. The defaults are $r_{\mathrm{tol}} = 10^{-9[A]}$, $a_{\mathrm{tol}} = 10^{-11[A]}$ and a maximum step of $10^{-3}\,\mathrm{s}^{[A]}$. Dense output gives the state at any time. Event functions find the apex (first crossing of v_z from positive to negative, non-terminal), the floor (z falling through `z_ground`, terminal) and optional planes. The apex event records $v_z = 0$ together with the nonzero v_x.

Fixed-step RK4 in the sweeps. The sweeps integrate many shots at once with the classical fourth-order Runge–Kutta method. For $\dot{\boldsymbol{y}} = \boldsymbol{f}(t, \boldsymbol{y})$ and step h (s):

$$\begin{aligned} \boldsymbol{k}_1 &= \boldsymbol{f}(t_n, \boldsymbol{y}_n), \qquad \boldsymbol{k}_2 = \boldsymbol{f}\big(t_n + \tfrac{h}{2}, \boldsymbol{y}_n + \tfrac{h}{2}\boldsymbol{k}_1\big), \\ \boldsymbol{k}_3 &= \boldsymbol{f}\big(t_n + \tfrac{h}{2}, \boldsymbol{y}_n + \tfrac{h}{2}\boldsymbol{k}_2\big), \qquad \boldsymbol{k}_4 = \boldsymbol{f}(t_n + h, \boldsymbol{y}_n + h\boldsymbol{k}_3), \\ \boldsymbol{y}_{n+1} &= \boldsymbol{y}_n + \tfrac{h}{6}\big(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4\big). \end{aligned} \tag{75}$$

`feasibility.integrate_batch` and `robustness.integrate3d` write the position update in a shorter form. Since $d\boldsymbol{r}/dt = \boldsymbol{v}$, the four position slopes are $\boldsymbol{v}_n$, $\boldsymbol{v}_n + \frac{h}{2}\boldsymbol{a}_1$, $\boldsymbol{v}_n + \frac{h}{2}\boldsymbol{a}_2$ and $\boldsymbol{v}_n + h\boldsymbol{a}_3$, where $\boldsymbol{a}_j$ are the acceleration stages. Inserting them into Eq. (75):

$$\boldsymbol{r}_{n+1} = \boldsymbol{r}_n + \tfrac{h}{6}\big(6\boldsymbol{v}_n + h\boldsymbol{a}_1 + h\boldsymbol{a}_2 + h\boldsymbol{a}_3\big) = \boldsymbol{r}_n + h\Big(\boldsymbol{v}_n + \tfrac{h}{6}(\boldsymbol{a}_1 + \boldsymbol{a}_2 + \boldsymbol{a}_3)\Big), \tag{76}$$

which is exactly the code's `x + dt*(vx + dt/6*(a1x + a2x + a3x))`. The global error of RK4 scales as

$$\left|\boldsymbol{r}_h(t) - \boldsymbol{r}(t)\right| \approx C\,h^4, \tag{77}$$

so halving h should cut the error by $2^4 = 16$. The sweeps use $h = 2\,\mathrm{ms}^{[A]}$ (`integrate_batch`, `integrate3d`) and $h = 1\,\mathrm{ms}^{[A]}$ (`contact.free_flight`). The two batch integrators store positions in single precision (`float32`).

Worked check: do they agree? Take the worked shot ($7\,\mathrm{m/s}^{[A]}$, $60^{\circ[A]}$, $S_0 = 1.06^{[A]}$, nominal set, $1.2\,\mathrm{s}^{[A]}$ of flight) and use `DOP853` with $r_{\mathrm{tol}} = 10^{-12\,[A]}$ as the reference.

- `simulate` (RK45, default tolerances): largest position difference $1.1 \times 10^{-15}\,\mathrm{m}^{[P]}$.
- `contact.free_flight` (RK4, double precision), error at $t = 1.2\,\mathrm{s}$: $9.3\times10^{-11\,[P]}$, $5.8\times10^{-12\,[P]}$, $3.5 \times 10^{-13\,[P]}$ and $2.2 \times 10^{-14}\,\mathrm{m}^{[P]}$ for $h = 8^{[A]}$, $4^{[A]}$, $2^{[A]}$ and $1\,\mathrm{ms}^{[A]}$. Successive ratios are $16.1^{[D]}$, $16.6^{[D]}$ and $16.0^{[D]}$, as Eq. (77) predicts.
- `integrate_batch` (2D) and `integrate3d` (3D), largest position difference: $1.2 \times 10^{-7}\,\mathrm{m}^{[P]}$ for every step from $8^{[A]}$ to $1\,\mathrm{ms}^{[A]}$. This floor is the `float32` rounding of positions between $2^{[D]}$ and $4\,\mathrm{m}^{[D]}$ (spacing $2.4 \times 10^{-7}\,\mathrm{m}^{[D]}$, so a rounding error of at most $1.2 \times 10^{-7}\,\mathrm{m}^{[D]}$), not truncation error.

The unit tests demand agreement better than $10^{-5}\,\mathrm{m}^{[A]}$ (`tests/test_hive.py`) and $2 \times 10^{-6}\,\mathrm{m}^{[A]}$ (`tests/test_contact.py`). Integration error is therefore at least four orders of magnitude smaller than the millimetre- to centimetre-scale differences between lift sets in Table 10. The parameters, not the numerics, limit what the model can say.

Assumptions.

- The check is for one smooth free-flight path; contact events are handled separately (Section on contact) and have their own step control.
- Agreement between integrators shows the equations are solved correctly; it says nothing about whether the equations describe a real POLLEN ball.

3.12 Summary of modelling assumptions

1. Rigid sphere, point-mass translation, inertial frame, flat floor; buoyancy and added mass neglected.
2. Forces use the air-relative velocity $\boldsymbol{v} - \boldsymbol{u}$; the wind is prescribed.
3. Drag: $\boldsymbol{F}_D = -\frac{1}{2}\rho C_d A|\boldsymbol{v}_{\mathrm{rel}}|\boldsymbol{v}_{\mathrm{rel}}$ with C_d constant per run (bounds $0.33^{[A]}$ to $0.9^{[A]}$); no spin-dependent drag.
4. Magnus: direction $\mathrm{unit}(\boldsymbol{\omega}\times\boldsymbol{v}_{\mathrm{rel}})$, magnitude $\frac{1}{2}\rho C_L A|\boldsymbol{v}_{\mathrm{rel}}|^2$; no reverse Magnus, no non-Magnus hole forces.
5. Lift law $C_L = \min(c_1 S, C_{L,\max})$, independent of Re. Above $S_{\mathrm{sat}} = C_{L,\max}/c_1$ extra spin changes nothing; this is a model assumption.
6. Spin decay $\boldsymbol{\omega} = \boldsymbol{\omega}_0 e^{-kt}$ with constant k; the spin axis is fixed.

7. Backspin axis unit($\boldsymbol{v}_0 \times \hat{\boldsymbol{z}}$), so pure backspin keeps the ball in the vertical launch plane.

8. All aerodynamic coefficients are literature bounds (pickleball and smooth-sphere data), not POLLEN measurements. The worked numbers above are single model runs and say nothing about shot-to-shot repeatability.

4 Single-flywheel launcher: exit speed and spin

The flight sections take the exit state of the ball as an input: an exit speed v and a spin parameter S. This section asks the opposite question. Given *one* powered 98 mm[U] flywheel with either a fixed curved hood or one passive (unpowered) upper roller, which pairs (v, S) can the launcher actually produce, how fast must the wheel turn, and what does one shot cost the wheel? Everything here follows `src/launcher.py`. The model is uncalibrated: no POLLEN has been measured on a launcher yet, so every simulated output is a prediction.

4.1 Frames, sign conventions and the wheel surface speed

The model is planar and one-dimensional along the contact path. Let x be the arc length travelled by the ball centre along the path (m), positive in the direction of travel, and let z point away from the wheel. The state is

- v: speed of the ball centre along the path (m/s), positive forward;
- s: absolute angular velocity of the ball (rad/s), positive for *backspin*, i.e. the wheel-side point of the ball moves forward relative to the centre and the hood-side point moves backward;
- Ω: flywheel angular velocity (rad/s), positive when the wheel surface at the contact moves forward;
- Ω_t: passive-roller angular velocity (rad/s), positive when the roller surface at the contact moves forward (a "counter-rotating" roller).

Wheel speeds enter and leave the code in revolutions per minute, n (rev/min), and are converted with

$$\Omega = \frac{2\pi}{60}\, n. \tag{78}$$

The wheel has radius R_w (m) and diameter $D = 2R_w$. A point on its rim moves at

$$u = R_w\, \Omega = \frac{D}{2} \cdot \frac{2\pi n}{60} = \frac{\pi D\, n}{60}, \tag{79}$$

which is the *wheel surface speed* (m/s). The spin parameter used by every other section is defined with the *nominal* ball radius R_b (m),

$$S = \frac{s\, R_b}{v}, \tag{80}$$

so $S = 1$ means the ball's nominal surface turns as fast as its centre moves.

Worked example. With $D = 0.098\,\mathrm{m}$[U] and $n = 3000\,\mathrm{rev/min}$[A], (78) gives $\Omega = 314.16\,\mathrm{rad/s}$[D] and (79) gives $u = \pi(0.098)(3000)/60 = 15.39\,\mathrm{m/s}$[D].

Symbol	Meaning	Default	Range used
m	ball mass	$0.025\,\mathrm{kg}$[C]	0.0245–$0.0255\,\mathrm{kg}$[A]
R_b	nominal ball radius	$0.0355\,\mathrm{m}$[C]	–
k_I	ball inertia factor, $I_b = k_I m R_b^2$	0.63[A]	0.55–0.667[A]
R_w	flywheel radius	$0.049\,\mathrm{m}$[U]	–
I_w	flywheel + steel flywheel inertia	$2.65 \times 10^{-4}\,\mathrm{kg\,m^2}$[A]	0.5–$5 \times 10^{-4}\,\mathrm{kg\,m^2}$[A]
μ_w	friction, wheel tread on ball	0.8[A]	0.5–1.2[A]
μ_t	friction, hood or roller on ball	0.3[A] (hood), 0.8[A] (roller)	0.05–1.0[A]
δ	diametral compression at the gap	$0.004\,\mathrm{m}$[U]	0.002–$0.008\,\mathrm{m}$[A]
k_n	linear diametral stiffness, $N = k_n \delta$	$4000\,\mathrm{N/m}$[A]	1500–$10000\,\mathrm{N/m}$[A]
λ	share of δ that shrinks the rolling radius	1[A]	0–1[A]
c_{rr}	rolling-hysteresis lever arm / r_c	0.02[A]	0–0.06[A]
θ	hood wrap angle	$90°$[A]	45–$150°$[A]
v_{in}	feed speed into the contact (no spin)	$0.5\,\mathrm{m/s}$[A]	0.2–$1.5\,\mathrm{m/s}$[A]
R_t	roller radius	$0.024\,\mathrm{m}$[A]	0.0125–$0.050\,\mathrm{m}$[A]
I_t	roller inertia	$1 \times 10^{-5}\,\mathrm{kg\,m^2}$[A]	10^{-6}–$2 \times 10^{-4}\,\mathrm{kg\,m^2}$[A]
τ_c	roller bearing Coulomb torque	$5 \times 10^{-4}\,\mathrm{N\,m}$[A]	0–$3 \times 10^{-3}\,\mathrm{N\,m}$[A]
c_v	roller viscous drag coefficient	$1 \times 10^{-6}\,\mathrm{N\,m\,s}$[A]	0–$10^{-5}\,\mathrm{N\,m\,s}$[A]
Δt_{shot}	time between shots	$0.5\,\mathrm{s}$[A]	0.25–$1.5\,\mathrm{s}$[A]
n_{free}, τ_s	goBILDA 5203 1:1 free speed, stall torque	$6000\,\mathrm{rev/min}$[C], $0.144\,\mathrm{N\,m}$[C]	–

Table 13: Launcher inputs and their provenance (`src/launcher.py`, `PARAMS`).

Parameters. Table 13 lists the defaults of `PARAMS` and of the dataclasses `Ball`, `Wheel`, `Contact`, `Hood` and `Roller`.

4.2 Contact kinematics: where the ball touches and how fast that point moves

Motivation. Friction acts on the *surface* of the ball, so everything depends on how fast the two touching points move compared with the wheel and the hood.

Ball inertia. The ball is a hollow perforated shell. The code writes its moment of inertia about a diameter as

$$I_b = k_I\, m\, R_b^2, \tag{81}$$

with k_I dimensionless ($k_I = 2/3$ for a thin shell, $2/5$ for a solid sphere) and I_b in $\mathrm{kg\,m^2}$.

Effective rolling radius. The ball is squeezed by δ (diametral, m) between the wheel and the hood. The code assumes a fraction λ of that squeeze shortens the lever arm from the ball centre to each contact, shared equally between the two contacts:

$$r_c = R_b - \lambda\,\frac{\delta}{2}. \tag{82}$$

$\lambda = 1$ puts all of the squeeze in the ball; $\lambda = 0$ is rigid-radius rolling on a flattened patch.

Velocity of the two contact points. Use a right-handed frame $(\hat{x}, \hat{y}, \hat{z})$ with $\hat{x}$ forward and $\hat{z}$ away from the wheel. With the sign convention above, backspin $s > 0$ is the angular-velocity

vector $\boldsymbol{\omega} = -s\,\hat{y}$. The wheel contact sits at $\boldsymbol{r}_w = -r_c\hat{z}$ from the ball centre and the hood (or roller) contact at $\boldsymbol{r}_t = +r_c\hat{z}$. Rigid-body kinematics, $\boldsymbol{v}_P = \boldsymbol{v}_c + \boldsymbol{\omega} \times \boldsymbol{r}_P$, gives

$$\boldsymbol{v}_{P,w} = v\hat{x} + (-s\hat{y}) \times (-r_c\hat{z}) = (v + s\,r_c)\,\hat{x}, \tag{83}$$

$$\boldsymbol{v}_{P,t} = v\hat{x} + (-s\hat{y}) \times (+r_c\hat{z}) = (v - s\,r_c)\,\hat{x}, \tag{84}$$

using $\hat{y} \times \hat{z} = \hat{x}$. For a hood concentric with the wheel both contact normals pass through the ball centre and the wheel axis, so the same two expressions hold along the curved path, with x measured along the arc.

Slip velocities. The code defines slip as *surface speed minus ball-point speed*:

$$\sigma_w = R_w\Omega - (v + s\,r_c), \tag{85}$$

$$\sigma_t = R_t\Omega_t - (v - s\,r_c), \qquad \text{(hood: } R_t\Omega_t \equiv 0\text{)}. \tag{86}$$

$\sigma > 0$ means the surface runs ahead of the ball, so its friction pushes the ball *forward*.

Worked example. With $R_b = 0.0355\,\mathrm{m}$[C], $\delta = 0.004\,\mathrm{m}$[U] and $\lambda = 1$[A], (82) gives $r_c = 0.0355 - 0.002 = 0.0335\,\mathrm{m}$[D]. With $m = 0.025\,\mathrm{kg}$[C] and $k_I = 0.63$[A], (81) gives $I_b = 0.63 \times 0.025 \times 0.0355^2 = 1.985 \times 10^{-5}\,\mathrm{kg\,m^2}$[D].

4.3 The ideal hood and where "$v = u/2$" comes from

Motivation. The most quoted rule of thumb for a hooded flywheel is that the ball leaves at half the wheel surface speed. This subsection derives it, so that the conditions it needs are visible.

Assume rolling without slip at *both* contacts at the moment of exit, a fixed hood, and a wheel speed Ω_f at exit. Setting $\sigma_t = 0$ and $\sigma_w = 0$ in (85)–(86):

$$0 = v - s\,r_c, \tag{87}$$

$$R_w\Omega_f = v + s\,r_c. \tag{88}$$

Adding the two equations eliminates s: $R_w\Omega_f = 2v$. Subtracting them eliminates v: $R_w\Omega_f = 2sr_c$. Hence

$$v = \frac{R_w\,\Omega_f}{2}, \qquad s = \frac{v}{r_c}, \qquad S = \frac{sR_b}{v} = \frac{R_b}{r_c}. \tag{89}$$

Three things are hidden in (89). (i) Ω_f is the wheel speed *at exit*, which is lower than the speed before the shot (Section 4.8). (ii) Both contacts must have stopped slipping, which needs enough friction and enough contact length (Section 4.6). (iii) The compression does not change v but does raise S above 1. The code function `hood_ideal` uses the rigid case $r_c = R_b$ (so $S = 1$).

Worked example. At $n = 3000\,\mathrm{rev/min}$[A] with no wheel slow-down ($\Omega_f = 314.16\,\mathrm{rad/s}$[D]), $v = 0.049 \times 314.16/2 = 7.70\,\mathrm{m/s}$[D] and $s = v/R_b = 216.8\,\mathrm{rad/s}$[D] ($2070\,\mathrm{rev/min}$[D]). With $\delta = 4\,\mathrm{mm}$[U], $S = 0.0355/0.0335 = 1.060$[D].

4.4 Equations of motion during contact

Motivation. Real contacts slip, the wheel slows, and the hood pushes back. The code integrates these effects through the whole contact rather than assuming the ideal state.

Momentum and angular momentum. Let F_w and F_t (N) be the friction forces *on the ball* at the wheel and the top contact, positive forward. M_w and M_t (N m) are the rolling-hysteresis moments on the ball (backspin-positive), τ_m the motor torque at the wheel shaft and τ_b the roller bearing drag (both N m). The code integrates

$$m\,\dot{v} = F_w + F_t, \tag{90}$$

$$I_b\,\dot{s} = r_c\,(F_w - F_t) + M_w + M_t, \tag{91}$$

$$I_w\,\dot{\Omega} = -R_w F_w + \tau_m(\Omega) + M_w, \tag{92}$$

$$I_t\,\dot{\Omega}_t = -R_t F_t - \tau_b(\Omega_t) - M_t \quad \text{(roller only)}. \tag{93}$$

Each line is Newton's law for one body. In (91) a forward force at the wheel contact (below the centre) adds backspin and a forward force at the top contact removes it, each with lever arm r_c. In (92) the reaction $-F_w$ acts on the wheel rim at radius R_w. In the code the hysteresis reaction on the wheel is `-Mw_wheel` with `Mw_wheel = -Mw_ball`, which is the $+M_w$ written here; in (93), `+Mt_roll` $= -M_t$.

Normal loads on a concentric hood. The ball is pressed by the same squeeze at both contacts, $N_0 = k_n\delta$. Its centre moves on a circle of radius

$$\rho = R_w + r_c \tag{94}$$

about the wheel axis. The wheel pushes the ball outward with N_w and the hood pushes it inward with N_t, so the net inward force must supply the centripetal acceleration and the sum stays at $2k_n\delta$:

$$N_t - N_w = \frac{mv^2}{\rho}, \qquad N_t + N_w = 2k_n\delta \implies N_w = k_n\delta - \frac{mv^2}{2\rho}, \quad N_t = k_n\delta + \frac{mv^2}{2\rho}. \tag{95}$$

If N_w would become negative the code sets $N_w = 0$, puts the whole centripetal load on the hood, $N_t = mv^2/\rho$, and raises the flag `lost_wheel_contact`. The wheel loses contact when $v^2 > 2\rho k_n\delta/m$. The normal loads use v at the start of each step.

Normal load in a roller nip. The ball centre travels straight through the gap between the wheel and the roller. At offset x from the narrowest point, the distance from the wheel axis to the ball centre is $\sqrt{(R_w+R_b)^2+x^2} \approx (R_w+R_b) + x^2/[2(R_w+R_b)]$, so the overlap with the wheel shrinks by $x^2/[2(R_w+R_b)]$; the same holds for the roller with R_t. The remaining squeeze and load are

$$\delta(x) = \delta - \frac{\kappa}{2}x^2, \qquad \kappa = \frac{1}{R_w+R_b} + \frac{1}{R_t+R_b}, \qquad N_w = N_t = k_n \max\big(\delta(x), 0\big), \tag{96}$$

with κ in 1/m. The load vanishes at $\delta(x) = 0$, so the contact runs over

$$-x_h \le x \le x_h, \qquad x_h = \sqrt{2\delta/\kappa}. \tag{97}$$

The roller path has no centripetal term. The hood path runs from $x = 0$ to $x_{end} = \theta\rho$ (with θ in rad).

Hysteresis, motor and bearing. The ball is not a perfect spring. The code applies a rolling resistance moment $e = c_{rr}\, r_c\, N$ at each contact that opposes the relative rotation of the ball and the surface, smoothed with $\varepsilon = 1\,\mathrm{rad/s}$[A]:

$$M_w = -c_{rr} r_c N_w \tanh\frac{s+\Omega}{\varepsilon}, \qquad M_t = -c_{rr} r_c N_t \tanh\frac{s-\Omega_t}{\varepsilon} \quad (\Omega_t \equiv 0 \text{ for the hood}). \tag{98}$$

The ball and the wheel turn in opposite senses when the ball has backspin, so their relative rotation is $s + \Omega$. The motor is a linear DC torque–speed line referred to the wheel shaft (gear ratio $G =$ motor revolutions per wheel revolution, $V/V_0 =$ applied voltage over $12\,\mathrm{V}$[C], $\omega_{free} = 2\pi n_{free}/60$):

$$\tau_m(\Omega) = G\,\tau_s \left(\frac{V}{V_0} - \frac{G\,\Omega}{\omega_{free}} \right), \tag{99}$$

and the default is $\tau_m = 0$ (the wheel *coasts* through the contact, as in Brettle's analysis). The roller bearing drag is

$$\tau_b(\Omega_t) = \tau_c \tanh\frac{\Omega_t}{0.5\,\mathrm{rad/s}} + c_v\,\Omega_t. \tag{100}$$

Stick or slip: one step of the solver. Coulomb's law says $|F_w| \le \mu_w N_w$ and $|F_t| \le \mu_t N_t$, with equality while the contact slides. The code advances by $\Delta t = 10^{-5}\,\mathrm{s}$[A] with forces held constant over the step and chooses F_w, F_t so that the slips *at the end* of the step obey Coulomb's law. Differentiate (85) and substitute (90)–(92):

$$\dot\sigma_w = R_w\dot\Omega - \dot v - r_c\dot s = b_w - \left(\tfrac{R_w^2}{I_w} + \tfrac{1}{m} + \tfrac{r_c^2}{I_b}\right)F_w - \left(\tfrac{1}{m} - \tfrac{r_c^2}{I_b}\right)F_t, \tag{101}$$

where $b_w = R_w\dot\Omega_0 - r_c\dot s_0$ collects the force-free parts ($\dot\Omega_0 = (\tau_m + M_w)/I_w$, $\dot s_0 = (M_w + M_t)/I_b$). The same steps for (86) give $\dot\sigma_t = b_t - (\frac{1}{m} - \frac{r_c^2}{I_b})F_w - (\frac{R_t^2}{I_t} + \frac{1}{m} + \frac{r_c^2}{I_b})F_t$ with $b_t = R_t\dot\Omega_{t,0} + r_c\dot s_0$ (the R_t^2/I_t and $R_t\dot\Omega_{t,0}$ terms are zero for a hood). After one step,

$$\begin{aligned} \boldsymbol{\sigma}^+ &= \Delta t\,(\mathbf{q} - \mathbf{A}\mathbf{F}), \qquad \mathbf{q} = \frac{\boldsymbol{\sigma}}{\Delta t} + \mathbf{b}, \\ \mathbf{A} &= \begin{pmatrix} \frac{R_w^2}{I_w} + \frac{1}{m} + \frac{r_c^2}{I_b} & \frac{1}{m} - \frac{r_c^2}{I_b} \\ \frac{1}{m} - \frac{r_c^2}{I_b} & \frac{R_t^2}{I_t} + \frac{1}{m} + \frac{r_c^2}{I_b} \end{pmatrix}. \end{aligned} \tag{102}$$

Each contact i must be in one of three modes:

$$\begin{aligned} &\text{stick:} && \sigma_i^+ = 0, && |F_i| \le \mu_i N_i; \\ &\text{slide forward:} && F_i = +\mu_i N_i, && \sigma_i^+ \ge 0; \\ &\text{slide backward:} && F_i = -\mu_i N_i, && \sigma_i^+ \le 0. \end{aligned} \tag{103}$$

$\mathbf{A}$ is symmetric positive definite, so the friction forces that satisfy these conditions are unique; `solve_friction` enumerates the $3 \times 3 = 9$ mode pairs and keeps the first consistent one. The state is then updated with the constant forces, $v^+ = v + \Delta t(F_w + F_t)/m$ and likewise for s, Ω, Ω_t, and the path position with the mid-step speed, $x^+ = x + \Delta t\,(v + v^+)/2$. The shot ends when $x \ge x_{end}$ (`exited`), or is declared stalled if $v \le 0$ after $0.01\,\mathrm{s}$[A], or stops at $t_{max} = 0.25\,\mathrm{s}$[A]. Gravity along the path is not modelled.

Energy ledger. With constant forces in a step, the work done is exactly force times mid-step velocity, so the code's ledger closes to round-off:

$$\underbrace{\tfrac{1}{2}I_w(\Omega_0^2-\Omega^2)}_{\text{wheel KE lost}}+W_m=\Delta KE_{ball}+\Delta KE_{roller}+E_{f,w}+E_{f,t}+E_{hyst}+E_{bear}, \tag{104}$$

where $W_m=\sum\tau_m\bar{\Omega}\Delta t$, $E_{f,i}=\sum F_i\,\bar{\sigma}_i\,\Delta t\geq 0$ is the sliding loss, $E_{hyst}=-\sum[M_w(\bar{s}+\bar{\Omega})+M_t(\bar{s}-\bar{\Omega}_t)]\Delta t$ and $E_{bear}=\sum\tau_b\bar{\Omega}_t\Delta t$ (bars = mid-step values; all in J).

Worked example (nominal hood, 3000 rev/min). `hood_model(3000)` with the defaults of Table 13: $\rho=0.049+0.0335=0.0825\,\mathrm{m}^{[D]}$, path length $\theta\rho=(\pi/2)(0.0825)=0.1296\,\mathrm{m}^{[D]}$, $N_0=4000\times0.004=16\,\mathrm{N}^{[D]}$. The ball leaves after $24.6\,\mathrm{ms}^{[P]}$ with $v=6.996\,\mathrm{m/s}^{[P]}$, $s=208.8\,\mathrm{rad/s}^{[P]}$ ($1994\,\mathrm{rev/min}^{[P]}$), $S=1.0597^{[P]}$ and both slips zero at exit. The wheel ends at $2726.9\,\mathrm{rev/min}^{[P]}$, and $R_w\Omega_f/2=0.049\times285.56/2=6.996\,\mathrm{m/s}^{[D]}$ matches v, as (89) requires for a rolling exit.

4.5 Why a fixed hood gives backspin, and the band of S

Motivation. The hood does not *create* backspin. The wheel does. The hood mainly decides how much of it survives. Two limiting cases bracket every hood shot.

Limit 1: frictionless top. Let only the wheel push ($F_t=0$, $c_{rr}=0$), with the ball fed at v_{in} without spin. Dividing (91) by (90),

$$\frac{I_b\,\mathrm{d}s}{m\,\mathrm{d}v}=r_c\;\Longrightarrow\;s=\frac{mr_c}{I_b}(v-v_{in})\;\Longrightarrow\;S=\underbrace{\frac{m\,r_cR_b}{I_b}}_{S_\infty}\left(1-\frac{v_{in}}{v}\right), \tag{105}$$

and using (81), $S_\infty=(r_c/R_b)/k_I$. This is `frictionless_top_limit`; the code function returns S_∞, the value for a ball that starts at rest.

Which way does the hood rub? In the same frictionless case the hood-side point moves at

$$v-s\,r_c=v-\frac{mr_c^2}{I_b}(v-v_{in})\approx v\,(1-\beta),\qquad\beta\equiv\frac{mr_c^2}{I_b}=\frac{(r_c/R_b)^2}{k_I}, \tag{106}$$

for $v\gg v_{in}$. For a hollow ball $\beta>1$, so the hood-side point moves *backward*, $\sigma_t=-(v-sr_c)>0$, and a real hood's friction acts *forward* on the ball ($F_t>0$). By (91) a forward F_t *removes* backspin and by (90) it adds speed. The hood therefore pulls S down from S_∞ toward the rolling value.

Limit 2: rolling on the hood. If hood friction is high enough or the wrap long enough, the hood contact sticks and (87) holds, so $S=R_b/r_c$ exactly as in (89). For a slow feed, therefore,

$$\frac{R_b}{r_c}\;\leq\;S\;\leq\;S_\infty\left(1-\frac{v_{in}}{v}\right)\;<\;S_\infty. \tag{107}$$

The unit test `test_hood_S_bounds` checks $1\leq S\,r_c/R_b\leq S_\infty r_c/R_b$ over hood friction and wrap, for a ball fed from rest ($v_{in}=0$) with $c_{rr}=0$.

Worked example. With the nominal values, $\beta = 0.8905/0.63 = 1.4135^{[D]}$ and $S_\infty = 0.9437/0.63 = 1.498^{[D]}$. Running the code with $\mu_t = 0$, $c_{rr} = 0$ at $3000\,\mathrm{rev/min}^{[A]}$: from rest ($v_{in} = 0$) it returns $S = 1.4979^{[P]}$, equal to S_∞; with $v_{in} = 0.5\,\mathrm{m/s}^{[A]}$ it returns $v = 6.142\,\mathrm{m/s}^{[P]}$ and $S = 1.3759^{[P]}$, and (105) gives $1.498\,(1 - 0.5/6.142) = 1.3759^{[D]}$. The rolling value depends only on the squeeze:

δ	$2\,\mathrm{mm}^{[A]}$	$4\,\mathrm{mm}^{[U]}$	$8\,\mathrm{mm}^{[A]}$
$S = R_b/r_c$ ($\lambda = 1^{[A]}$)	$1.029^{[D]}$	$1.060^{[D]}$	$1.127^{[D]}$

The band in the parameter box. `reachable_S_summary` sweeps the fixed hood over $k_I \in [0.55, 0.667]^{[A]}$, $\mu_t \in [0.05, 1.0]^{[A]}$, $\theta \in [45°, 150°]^{[A]}$, $\delta \in [2, 8]\,\mathrm{mm}^{[A]}$ and $k_n \in [1.5, 10]\,\mathrm{N/mm}^{[A]}$ at $3000\,\mathrm{rev/min}^{[A]}$ ($405^{[D]}$ cases). It finds S from $1.029^{[P]}$ to $1.274^{[P]}$ and v from $3.18^{[P]}$ to $7.05\,\mathrm{m/s}^{[P]}$. A grippy or long hood sits at the bottom of this band; only a slick, short hood moves toward the top.

$S < 1$ from a fixed hood. Equation (107) assumed a slow feed. If the ball *enters* already moving with little spin and leaves before hood friction rolls it up, the factor $(1 - v_{in}/v)$ and the unfinished approach to rolling can push S below 1. The stored envelope at $3000\,\mathrm{rev/min}^{[A]}$ gives $S = 0.643^{[P]}$ for $\theta = 45°^{[A]}$, $\mu_t = 0.05^{[A]}$, $v_{in} = 5\,\mathrm{m/s}^{[A]}$. Running the code for $\theta = 20°^{[A]}$, $\mu_t = 0.05^{[A]}$, $v_{in} = 4\,\mathrm{m/s}^{[A]}$ gives $S = 0.575^{[P]}$. This needs a fast pre-accelerating feeder and is sensitive to every friction value, so it is not a design recommendation.

4.6 Rolling and grip-limited regimes

Motivation. Above some wheel speed, turning the wheel faster stops making the ball faster. This is the most important practical limit of a single-wheel hood, and it follows directly from Coulomb friction in (103).

Wheel sliding, hood sticking. Suppose the wheel contact slides forward, $F_w = \mu_w N_w$, while the hood sticks ($v = s r_c$, so $\dot v = r_c \dot s$), with $c_{rr} = 0$. Write (91) as $(I_b/r_c^2)\,\dot v = F_w - F_t$ and add it to (90):

$$\Big(m + \frac{I_b}{r_c^2}\Big)\dot v = 2F_w \quad\Longrightarrow\quad m_{eff}\,\dot v = F_w, \qquad m_{eff} = \frac{1}{2}\Big(m + \frac{I_b}{r_c^2}\Big). \tag{108}$$

Putting $\dot v = F_w/m_{eff}$ back into (90) gives the hood force

$$F_t = \frac{m}{m_{eff}} F_w - F_w = \frac{\beta - 1}{\beta + 1}\,F_w, \tag{109}$$

which is forward for $\beta > 1$, as found above. The hood keeps sticking only while $|F_t| \le \mu_t N_t$.

The speed cap. Along the path $\dot v = v\,\mathrm{d}v/\mathrm{d}x$, so with $w = v^2$ and the hood load (95),

$$\frac{\mathrm{d}w}{\mathrm{d}x} = \frac{2\mu_w}{m_{eff}}\Big(k_n\delta - \frac{m\,w}{2\rho}\Big) = \mathcal{A} - \mathcal{B}w, \qquad \mathcal{A} = \frac{2\mu_w k_n \delta}{m_{eff}}, \quad \mathcal{B} = \frac{\mu_w m}{m_{eff}\,\rho}. \tag{110}$$

This linear equation has the solution, from $w(0) = v_{in}^2$ to $x = \theta\rho$,

$$v_{cap}^2 = w_\infty - \big(w_\infty - v_{in}^2\big)\,e^{-\mathcal{B}\theta\rho}, \qquad w_\infty = \frac{\mathcal{A}}{\mathcal{B}} = \frac{2\rho k_n \delta}{m}, \qquad \mathcal{B}\theta\rho = \mu_w\theta\,\frac{2\beta}{1+\beta}. \tag{111}$$

Nothing in (111) depends on the wheel speed: once the wheel still slips at exit, the exit speed is fixed by μ_w, $k_n\delta$, the wrap and the ball inertia. w_∞ is exactly the speed at which the wheel would lose contact. The rpm independence is what `test_grip_limited_speed_independent_of_rpm` checks.

Where the regimes meet. The wheel catches up with the ball (rolling exit) if $R_w\Omega_f/2 \le v_{cap}$. Using (119) from Section 4.8 for Ω_f, the crossover wheel speed is

$$\Omega_i^* = a\,\frac{2v_{cap}}{R_w} - \frac{R_w m v_{in}}{2I_w}. \tag{112}$$

Worked example. Nominal hood: $I_b/r_c^2 = 0.017687\,\mathrm{kg}^{[D]}$, so $m_{eff} = 0.021343\,\mathrm{kg}^{[D]}$; $w_\infty = 2(0.0825)(16)/0.025 = 105.6\,\mathrm{m^2/s^{2[D]}}$ ($\sqrt{w_\infty} = 10.28\,\mathrm{m/s}^{[D]}$); $\mathcal{B}\theta\rho = 0.8(\pi/2)(1.1713) = 1.472^{[D]}$; $e^{-1.472} = 0.2295^{[D]}$. Then $v_{cap}^2 = 105.6 - (105.6 - 0.25)(0.2295) = 81.42\,\mathrm{m^2/s^{2[D]}}$, so $v_{cap} = 9.02\,\mathrm{m/s}^{[D]}$. The code with $c_{rr} = 0$ at 5000 and 6000 rev/min$^{[A]}$ returns 9.021 m/s$^{[P]}$ for both; the small difference is the short initial hood slip that (111) ignores. With the default $c_{rr} = 0.02^{[A]}$ the stored envelope gives 8.909 m/s$^{[P]}$ from 4000$^{[A]}$ to 6000 rev/min$^{[A]}$. The stick check holds: $(\beta-1)/(\beta+1)\,\mu_w N_w = 0.171 \times 0.8\,N_w = 0.137\,N_w \le 0.3\,N_t^{[D]}$ since $N_t \ge N_w$. Equation (112) gives $\Omega_i^* = 402.7\,\mathrm{rad/s}^{[D]}$ (3845 rev/min$^{[D]}$); the envelope (default c_{rr}) is still rolling at 3500$^{[A]}$ and slipping at 4000 rev/min$^{[A]}$ (predicted regimes). For a softer ball, $k_n = 1500\,\mathrm{N/m}^{[A]}$, the code caps at 5.45 m/s$^{[P]}$.

Roller nip cap. In a nip the load is the parabola (96) with no centripetal term. For a top surface that does not move (a very heavy roller, the same m_{eff}), integrating $\mathrm{d}w/\mathrm{d}x = 2\mu_w N_w/m_{eff}$ over $[-x_h, x_h]$ with $\int_{-x_h}^{x_h}(\delta - \frac{\kappa}{2}x^2)\,\mathrm{d}x = \frac{4}{3}\delta x_h$ gives

$$v_{cap,nip}^2 = v_{in}^2 + \frac{8\,\mu_w k_n\,\delta\,x_h}{3\,m_{eff}}. \tag{113}$$

With $R_t = 0.024\,\mathrm{m}^{[A]}$, $\kappa = 28.64\,1/\mathrm{m}^{[D]}$ and $x_h = 0.01671\,\mathrm{m}^{[D]}$ (a 33.4 mm$^{[D]}$ contact against 129.6 mm$^{[D]}$ for the hood), (113) gives $v_{cap,nip} = 5.19\,\mathrm{m/s}^{[D]}$. The code with a fixed-like roller and $c_{rr} = 0$ gives 5.18 m/s$^{[P]}$; the nominal roller ($I_t = 10^{-5}\,\mathrm{kg\,m^{2[A]}}$) caps at 5.03 m/s$^{[P]}$ from 2500 rev/min$^{[P]}$ upward. The short nip is why a roller is grip-limited much earlier than a 90°$^{[A]}$ hood.

4.7 The ratio v/u, and why $v = u/2$ is not universal

Motivation. Teams set the wheel speed, not the exit speed. The link between them is the speed ratio

$$\eta_v \equiv \frac{v}{u_0} = \frac{v}{R_w\Omega_i}, \tag{114}$$

where u_0 and Ω_i are the surface speed and wheel speed *before* the shot.

Rolling regime. From (89) and (119) (with $c_{rr} = 0$),

$$\eta_v = \frac{\Omega_f}{2\Omega_i} = \frac{1}{2a}\left(1 + \frac{R_w m v_{in}}{2I_w\Omega_i}\right) < \frac{1}{2}, \tag{115}$$

because $a > 1$ and the feed term is small ($R_w m v_{in}/(2I_w) < (a-1)\,\Omega_i$, i.e. above about 12 rad/s$^{[D]}$ here): the wheel has slowed by the time the ball rolls off it.

Grip-limited regime. $v = v_{cap}$ is fixed, so $\eta_v = 60\,v_{cap}/(\pi D n)$ falls as $1/n$.

Other losses. Hysteresis (98) lowers η_v further. Effects that the code does *not* include would also lower it: tread compression (an effective wheel radius below R_w), non-diametral contacts, a hood that converges, and frame kick.

Worked example. With $c_{rr} = 0$ at 3000 rev/min[A], (115) gives $(1+1.156/314.16)/(2\times1.0967) = 0.4576$[D]; the code gives 0.4576[P]. With the defaults the code gives the ratios below; at 6000 rev/min[A] the ratio is barely more than half of the ideal 0.5[D].

n [rev/min]	1000[A]	2000[A]	3000[A]	3500[A]	4000[A]	5000[A]	6000[A]
v [m/s]	2.239[P]	4.640[P]	6.996[P]	8.169[P]	8.909[P]	8.909[P]	8.909[P]
$\eta_v = v/u_0$	0.436[P]	0.452[P]	0.454[P]	0.455[P]	0.434[P]	0.347[P]	0.289[P]
regime		rolling exit			wheel slipping at exit		

Literature band. Because nothing has been measured on POLLEN, the design work uses a band from `research/params_lit.json` (`launcher.v_over_surface_speed`, grade C): low 0.30[A], nominal 0.42[A], high 0.50[A]. These come from slow-motion and forum estimates on other balls and from the rigid ideal; they are literature bounds, not POLLEN measurements. The code adds 0.25[A] (`V_OVER_U` in `src/shooter_spec.py`) from a multi-roller table-tennis machine whose measured ratio, about 0.48[A] of the *mean* roller speed, maps to about 0.24–0.25[D] of u_0 for a hood (`research/EXTRA_SOURCES.md`). The PM-001 model's own nominal hood, 0.436–0.455[P] below the grip limit, sits on the optimistic side of that band.

4.8 Flywheel slow-down and energy per shot

Motivation. A coasting wheel gives up speed to the ball. That drop sets Ω_f in (89), the recovery time between shots, and how much of the motor's work reaches the ball.

Impulse bookkeeping. Integrate (90)–(92) over the contact with $\tau_m = 0$, $c_{rr} = 0$ and define the friction impulses $J_w = \int F_w\,\mathrm{d}t$, $J_t = \int F_t\,\mathrm{d}t$ (N s):

$$m(v - v_{in}) = J_w + J_t, \qquad \frac{I_b s}{r_c} = J_w - J_t, \qquad I_w(\Omega_i - \Omega_f) = R_w J_w. \tag{116}$$

Adding the first two eliminates the unknown hood impulse:

$$m(v - v_{in}) + \frac{I_b s}{r_c} + \frac{2I_w(\Omega_f - \Omega_i)}{R_w} = 0, \tag{117}$$

which `test_generalised_momentum_hood` checks to 10^{-9}.

Generalised Brettle drop. For a rolling exit, $v = s r_c = R_w\Omega_f/2$. Then $J_w = \tfrac{1}{2}[m(v - v_{in}) + I_b v/r_c^2]$, and the wheel equation in (116) becomes

$$I_w\Omega_i - I_w\Omega_f = \frac{R_w}{2}\left[\left(m + \frac{I_b}{r_c^2}\right)\frac{R_w\Omega_f}{2} - m v_{in}\right]. \tag{118}$$

Collecting the Ω_f terms and dividing by I_w:

$$\Omega_f = \frac{\Omega_i + \dfrac{R_w m v_{in}}{2I_w}}{a}, \qquad a = 1 + \frac{R_w^2\,(m + I_b/r_c^2)}{4I_w}, \tag{119}$$

which is `brettle_omega_f` and `brettle_ratio`. For $r_c = R_b$ and $v_{in} = 0$, using (81), $a = 1 + (1 + k_I)\,mR_w^2/(4I_w)$, Brettle's published ratio. The drop per shot is $\Delta n = n_i - n_f$.

Grip-limited drop. If the wheel still slips at exit, the hood still sticks, so $s = v_{cap}/r_c$ and the same J_w applies with $v = v_{cap}$:

$$\Delta\Omega = \frac{R_w}{2I_w}\Big[m(v_{cap} - v_{in}) + \frac{I_b\,v_{cap}}{r_c^2}\Big], \tag{120}$$

which is also independent of wheel speed.

Efficiency. With $v_{in} = 0$, the ball's kinetic energy at a rolling exit is $\frac{1}{2}(m+I_b/r_c^2)v^2 = \frac{1}{2}I_w\Omega_f^2\,(a-1)$ (substitute $v = R_w\Omega_f/2$ and the definition of a). The wheel loses $\frac{1}{2}I_w(\Omega_i^2-\Omega_f^2) = \frac{1}{2}I_w\Omega_f^2(a^2-1)$. Their ratio is

$$\eta = \frac{KE_{ball}}{\Delta KE_{wheel}} = \frac{a-1}{a^2-1} = \frac{1}{a+1} < \frac{1}{2}. \tag{121}$$

The rest, a times the ball's energy, is lost to sliding at the contacts while the ball spins up (in the nominal run below almost all of it at the wheel).

Recovery. After the shot the motor (99) at full voltage refills the wheel: $I_w\dot\Omega = T_0(1 - \Omega/\Omega_{free})$ with $T_0 = G\tau_s V/V_0$ and $\Omega_{free} = (V/V_0)\,\omega_{free}/G$. Separating variables,

$$t_{rec} = \tau_{mw}\ln\frac{\Omega_{free} - \Omega_{start}}{\Omega_{free} - \Omega_{target}}, \qquad \tau_{mw} = \frac{I_w\Omega_{free}}{T_0}, \tag{122}$$

infinite when $\Omega_{target} \geq \Omega_{free}$ (`recovery_time`; no current limit, no controller lag, no rotor inertia).

Worked example. Rigid Brettle case: $a = 1 + 0.049^2(0.025 + 0.01575)/(4 \times 2.65 \times 10^{-4}) = 1.0923^{[D]}$, a drop of $1 - 1/a = 8.45\,\%^{[D]}$, i.e. $253.5\,\mathrm{rev/min}^{[D]}$ at $3000\,\mathrm{rev/min}^{[A]}$ (the `hood_ideal_brettle` row). With $r_c = 0.0335\,\mathrm{m}^{[D]}$, $a = 1.0967^{[D]}$; adding $v_{in} = 0.5\,\mathrm{m/s}^{[A]}$, (119) gives $n_f = 2745.6\,\mathrm{rev/min}^{[D]}$, a drop of $254.4\,\mathrm{rev/min}^{[D]}$, and the code with $c_{rr} = 0$ gives $254.43\,\mathrm{rev/min}^{[P]}$. With hysteresis the nominal drop is $273.1\,\mathrm{rev/min}^{[P]}$. In the grip-limited regime (120) with $v_{cap} = 9.021\,\mathrm{m/s}^{[P]}$ gives $328.9\,\mathrm{rev/min}^{[D]}$ (code, $c_{rr} = 0$: $328.9\,\mathrm{rev/min}^{[P]}$; default: $343.6\,\mathrm{rev/min}^{[P]}$). Equation (121) gives $\eta = 1/2.0967 = 0.477^{[D]}$. The nominal ledger at $3000\,\mathrm{rev/min}$: wheel loses $2.273\,\mathrm{J}^{[P]}$, the ball gains $1.042\,\mathrm{J}^{[P]}$ ($\eta = 0.458^{[P]}$), wheel-side sliding $1.093\,\mathrm{J}^{[P]}$, hood sliding $0.0009\,\mathrm{J}^{[P]}$, hysteresis $0.137\,\mathrm{J}^{[P]}$; the residual is zero to the printed digits. Recovery with the goBILDA 5203 1:1$^{[C]}$ at $12\,\mathrm{V}^{[C]}$: $\Omega_{free} = 628.3\,\mathrm{rad/s}^{[D]}$, $\tau_{mw} = 2.65 \times 10^{-4} \times 628.3/0.144 = 1.156\,\mathrm{s}^{[D]}$, $\Omega_{start} = 314.16 - 28.60 = 285.56\,\mathrm{rad/s}^{[D]}$, so $t_{rec} = 1.156\ln(342.76/314.16) = 0.101\,\mathrm{s}^{[D]}$. Holding full voltage *during* contact instead of coasting gives $7.153\,\mathrm{m/s}^{[P]}$ instead of $6.996\,\mathrm{m/s}^{[P]}$ at $3000\,\mathrm{rev/min}^{[A]}$, and adds nothing in the grip-limited regime.

4.9 The passive upper roller

Motivation. A common idea is that a free top roller "spins up and reduces the spin difference". The model tests that with the roller equation (93).

Rolling state with a moving top surface. Let $u_w = R_w\Omega$ and $u_t = R_t\Omega_t$ be the two surface speeds (forward positive). Rolling at both contacts means $\sigma_w = \sigma_t = 0$: $u_w = v + sr_c$ and $u_t = v - sr_c$. Adding and subtracting,

$$v = \frac{u_w + u_t}{2}, \qquad s = \frac{u_w - u_t}{2r_c}, \qquad S = \frac{R_b}{r_c} \cdot \frac{u_w - u_t}{u_w + u_t}, \tag{123}$$

which is `roller_rolling_state`. A fixed hood is $u_t = 0$. S drops below R_b/r_c only if the top surface moves *forward* ($u_t > 0$).

Which way the passive roller turns. By (106), for a slow feed the ball's top point moves backward, so it drags the roller to $\Omega_t < 0$, i.e. $u_t < 0$, and (123) gives $S > R_b/r_c$. This needs $v \gg v_{in}$: at $1000\,\mathrm{rev/min}^{[A]}$ the $0.5\,\mathrm{m/s}^{[A]}$ feed is a large part of the exit speed, the nominal roller is spun *forward* ($+8.25\,\mathrm{rad/s}^{[P]}$) and $S = 0.974^{[P]}$; the backward drag appears from about $2000\,\mathrm{rev/min}^{[P]}$ upward. The two inertia limits are: a very heavy roller at rest behaves like a fixed hood ($S \to R_b/r_c$), and a massless frictionless roller transmits no force, giving (105) (checked by `test_massless_roller_equals_frictionless_top`).

Between shots. The roller keeps its exit speed and coasts for Δt_{shot} under $I_t\dot{\Omega}_t = -\tau_b(\Omega_t)$ (`roller_coast`, $1\,\mathrm{ms}^{[A]}$ explicit steps, set to zero if the sign would flip).

Worked example. At $2000\,\mathrm{rev/min}^{[A]}$ the nominal roller exits rolling at both contacts with $u_w = 9.375\,\mathrm{m/s}^{[P]}$ and $u_t = -0.166\,\mathrm{m/s}^{[P]}$. Equation (123) gives $v = 4.605\,\mathrm{m/s}^{[D]}$ and $S = 1.0597 \times 9.541/9.210 = 1.098^{[D]}$, matching the simulated $S = 1.0978^{[P]}$. At $3000\,\mathrm{rev/min}$ it is grip-limited: $v = 5.029\,\mathrm{m/s}^{[P]}$, $S = 1.107^{[P]}$, roller at $-89.9\,\mathrm{rev/min}^{[P]}$ ($-9.41\,\mathrm{rad/s}^{[P]}$). Coasting: the bearing torque is about $5 \times 10^{-4} + 10^{-6} \times 9.41 = 5.09 \times 10^{-4}\,\mathrm{N\,m}^{[D]}$, a deceleration of $50.9\,\mathrm{rad/s}^{2[D]}$, so the roller stops in about $0.19\,\mathrm{s}^{[D]} < 0.5\,\mathrm{s}^{[A]}$. Every shot of a burst therefore starts from rest, and the stored `roller_nominal_steady` rows equal `roller_nominal_rest`.

Roller band. Over $I_t \in [10^{-6}, 2 \times 10^{-4}]\,\mathrm{kg\,m}^{2[A]}$, $R_t \in [12.5, 50]\,\mathrm{mm}^{[A]}$, bearing torque 0–$3 \times 10^{-3}\,\mathrm{N\,m}^{[A]}$ and $\delta \in \{4, 8\}\,\mathrm{mm}^{[A]}$ (shots 1–3 of a burst, $216^{[D]}$ cases) the code finds S from $1.060^{[P]}$ to $1.242^{[P]}$ and v from $4.86^{[P]}$ to $7.09\,\mathrm{m/s}^{[P]}$ at $3000\,\mathrm{rev/min}^{[A]}$. With a fast feed the roller is spun forward: a light roller ($I_t = 2 \times 10^{-6}\,\mathrm{kg\,m}^{2[A]}$) gives $S = 0.921^{[P]}$ at $v_{in} = 2\,\mathrm{m/s}^{[A]}$ and $0.674^{[P]}$ at $4\,\mathrm{m/s}^{[A]}$. So the passive roller, fed slowly at a working wheel speed, slightly *raises* S and costs exit speed; it does not reduce spin.

4.10 Inverse problem: wheel speed for a target exit speed

Motivation. The trajectory sections give a target exit speed v. The team needs a wheel speed. Inverting (114) with (79):

$$u_0 = \frac{v}{\eta_v} \implies n = \frac{60\,u_0}{\pi D} = \frac{60\,v}{\eta_v\,\pi D}, \tag{124}$$

which is `wheel_rpm` in `src/shooter_spec.py`. It is rounded to a whole rpm and evaluated at $\eta_v \in \{0.25, 0.30, 0.42, 0.50\}^{[A]}$. The matching ball spin for a design S follows from (80): $n_{ball} = |S|\,v\,60/(2\pi R_b)$ (`spin_rpm`).

Worked example. For the exit-speed band 6.5–9.0 m/s[P] of the shooter specification:

target v	$\eta_v = 0.25$[A]	0.30[A]	0.42[A]	0.50[A]
6.5 m/s[P]	5067 rev/min[D]	4222[D]	3016[D]	2533[D]
7.0 m/s[P]	5457[D]	4547[D]	3248[D]	2728[D]
9.0 m/s[P]	7016[D]	5847[D]	4176[D]	3508[D]

For example $n = 60 \times 9.0/(0.25\,\pi \times 0.098) = 7016$ rev/min[D]. Two checks against the forward model. First, 7016 rev/min[D] is above the 6000 rev/min[C] free speed of a 1:1 motor, so the 0.25 case needs a step-up drive or a different motor. Second, the inverse (124) assumes a constant η_v, which only holds in the rolling regime. The nominal hood of Section 4.6 cannot reach 9.0 m/s[P] at *any* wheel speed (cap 8.91 m/s[P]); a higher cap needs more grip, more squeeze or a stiffer ball, which is what measuring μ_w, k_n and η_v is for. Spin for $S = 1$ at 7.0 m/s[P]: 1883 rev/min[D].

4.11 Assumptions

- Planar, one-dimensional contact path: a hood concentric with the wheel (uniform gap) or a straight pass through a two-cylinder nip. No converging hood, entry ramp, off-axis contact, sidespin or frame kick.
- Linear normal law $N = k_n\delta$ with an unmeasured k_n; a real hollow shell is non-linear and can buckle. The squeeze shortens the rolling radius by $\lambda\delta/2$ with unknown λ; the tread is rigid and R_w is fixed.
- Constant Coulomb coefficients μ_w, μ_t (no dependence on slip speed, heat or wear); none is measured on POLLEN.
- Rolling hysteresis as a moment $c_{rr} r_c N$ smoothed with tanh; c_{rr} is a guess.
- The ball enters with speed v_{in} and no spin; gravity along the path is ignored.
- The wheel coasts during contact unless a motor is given; the motor is a linear torque–speed line with no current limit, controller lag or rotor inertia.
- Ball mass, radius and inertia vary between pieces only through the parameter boxes; k_I is derived from a uniform shell, not measured.
- The speed ratios η_v used in the inverse problem are literature bounds from other balls and launchers, not POLLEN measurements.
- Every simulated value in this section is a prediction of an uncalibrated model. A single test shot would not establish repeatability; the calibration needs repeated shots at several wheel speeds.

5 Contact mechanics: rim and interior bounces

Motivation. A ball that clears the HIVE cleanly is only half the story. On the unfavourable side the CELL mouth faces away from the shooter, so many shots touch something: the lower lip (the *far rim*), the end of the roof (the *near rim*), or the CELL floor just after a clean entry. Each touch is a short, hard collision that can change both the ball's velocity and its spin. Backspin enters the

story here a second time: in the air it bends the path through the Magnus force (an earlier section), and at a contact it changes how hard friction grips. This section derives, one step at a time, the collision law that `src/contact.py` uses (functions `impact` and `impact_batch`), proves that it never creates energy, works out which way backspin pushes the ball at each kind of contact, and explains how `simulate_batch` turns many such collisions into one of six outcomes. It ends with the model's explanation of why backspin can *lower* retention on the unfavourable side.

Every retention number in this section is a model output ([P]). The collision coefficients for POLLEN on the HIVE skin have never been measured; they are swept over assumed ranges.

5.1 Frame, contact normal and contact-point velocity

We work in the 2D *shot frame* shared by `flight.py`, `hive.py` and `contact.py`: x is horizontal toward the target, z is up, and y completes a right-handed set, so $\hat{\boldsymbol{x}} \times \hat{\boldsymbol{z}} = -\hat{\boldsymbol{y}}$. The ball centre is $\boldsymbol{c} = (x, z)$ in m, its velocity is $\boldsymbol{v} = (v_x, v_z)$ in m/s, and its spin is the single scalar $\omega = \omega_y$ in rad/s (the spin vector is $\boldsymbol{\omega} = \omega\,\hat{\boldsymbol{y}}$). The ball has mass m (kg), radius R (m) and moment of inertia I (kg m^2) about its centre.

Which sign is backspin? For any in-plane velocity, the cross product with a y-directed spin is

$$\boldsymbol{\omega} \times \boldsymbol{v} = \omega\,\hat{\boldsymbol{y}} \times (v_x\hat{\boldsymbol{x}} + v_z\hat{\boldsymbol{z}}) = \omega\,(v_z,\, -v_x)\,, \tag{125}$$

using $\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}} = -\hat{\boldsymbol{z}}$ and $\hat{\boldsymbol{y}} \times \hat{\boldsymbol{z}} = \hat{\boldsymbol{x}}$. For a ball moving toward $+x$ ($v_x > 0$, $v_z = 0$) the Magnus direction $\boldsymbol{\omega} \times \boldsymbol{v} = (0, -\omega v_x)$ points up only when $\omega < 0$. So **backspin on a $+x$ ball is $\omega < 0$**, matching `flight.backspin_omega`. The dimensionless spin ratio is

$$S = \frac{|\omega|\,R}{|\boldsymbol{v}|}, \tag{126}$$

and the sweeps build a backspin state from it as $\omega = -S\,|\boldsymbol{v}|/R$ (`launch_states`, `incidence_states`).

Contact normal and tangent. The HIVE section is a list of straight segments (walls, lips, floor, roof). For a ball centre $\boldsymbol{c}$, let $\boldsymbol{q}$ be the closest point on the nearest segment (it may be an end point, that is, a corner or edge). A contact exists when $|\boldsymbol{c} - \boldsymbol{q}| < R$. The code defines

$$\boldsymbol{n} = \frac{\boldsymbol{c} - \boldsymbol{q}}{|\boldsymbol{c} - \boldsymbol{q}|} \qquad \text{(unit normal, pointing from the obstacle to the ball centre)}, \tag{127}$$

so a flat face and a sharp corner are handled by the same formula. The unit tangent is

$$\boldsymbol{t} = \hat{\boldsymbol{y}} \times \boldsymbol{n} = \hat{\boldsymbol{y}} \times (n_x\hat{\boldsymbol{x}} + n_z\hat{\boldsymbol{z}}) = (n_z,\, -n_x)\,, \tag{128}$$

which is $\boldsymbol{n}$ rotated by 90° clockwise in the x–z plane. One useful identity follows directly: $(\boldsymbol{n}\times\boldsymbol{t})_y = n_z t_x - n_x t_z = n_z^2 + n_x^2 = 1$.

Velocity of the contact point. The point of the ball that touches the obstacle sits at $\boldsymbol{r} = -R\,\boldsymbol{n}$ from the centre. Its velocity is the centre velocity plus the rotational part. Using (125) with $\boldsymbol{r}$ in place of $\boldsymbol{v}$,

$$\begin{aligned} \boldsymbol{v}_c &= \boldsymbol{v} + \boldsymbol{\omega} \times \boldsymbol{r} = \boldsymbol{v} + \omega\,(r_z,\, -r_x) = \boldsymbol{v} + \omega\,(-Rn_z,\, Rn_x) \\ &= \boldsymbol{v} - R\,\omega\,\boldsymbol{t}. \end{aligned} \tag{129}$$

Projecting on $\boldsymbol{n}$ and $\boldsymbol{t}$ gives the two numbers the impact law uses:

$$v_{cn} = \boldsymbol{v}\cdot\boldsymbol{n} \qquad \text{normal approach speed, m/s; } v_{cn} < 0 \text{ means approaching,} \tag{130}$$

$$v_{ct} = \boldsymbol{v}\cdot\boldsymbol{t} - R\,\omega \qquad \text{slip velocity of the contact point, m/s.} \tag{131}$$

Spin appears *only* in v_{ct}. This single fact drives everything in Section 5.5.

5.2 Ball inertia: the hollow-shell factor α

The code writes the moment of inertia as

$$I = \alpha\, m R^2, \tag{132}$$

with the dimensionless factor α read from the active parameter file (`ball_props`: key `I_b.alpha_I_over_mR2`). POLLEN is a thin perforated polyethylene shell, not a solid ball. For a uniform spherical shell with outer radius R_o and inner radius R_i (both m), subtracting a solid sphere of radius R_i from one of radius R_o (each with $I = \frac{2}{5}MR^2$ and mass proportional to R^3) gives

$$I_{\text{shell}} = \frac{2}{5}\, m\, \frac{R_o^5 - R_i^5}{R_o^3 - R_i^3} \quad\Longrightarrow\quad \alpha = \frac{2}{5}\,\frac{R_o^5 - R_i^5}{\left(R_o^3 - R_i^3\right) R_o^2}. \tag{133}$$

Two limits check the formula. For $R_i \to 0$ (solid ball) $\alpha \to 2/5$. For $R_i \to R_o$ (very thin wall) both differences vanish; dividing the derivatives, $\frac{2}{5}\cdot 5R_o^4/(3R_o^2\cdot R_o^2) = 2/3$:

$$\lim_{R_i\to 0}\alpha = \tfrac{2}{5}, \qquad \lim_{R_i\to R_o}\alpha = \tfrac{2}{3}. \tag{134}$$

Worked example. With $R_o = 35.5\,\text{mm}^{[C]}$ and a wall of $1.78\,\text{mm}^{[A]}$, (133) gives $\alpha = 0.634^{[D]}$; with a $2.1\,\text{mm}^{[A]}$ wall, $\alpha = 0.629^{[D]}$. The model uses $\alpha = 0.63^{[D]}$ with bounds $0.60^{[A]}$ (thicker or uneven wall, seam) and $0.667^{[D]}$ (thin-shell limit). `ball_props` applies these bounds through α only and keeps m and R nominal: $m = 24.95\,\text{g}^{[A]}$, $R = 35.5\,\text{mm}^{[C]}$, so $I = 1.981\times10^{-5}\,\text{kg m}^{2[D]}$. These numbers are the same in `research/params_lit.json` (the current default) and in the older `data/aero_params.json`, so the impact law below does not depend on which file is active.

5.3 The impact law, step by step

Model. The collision is treated as instantaneous: a finite impulse $\boldsymbol{P} = P_n\boldsymbol{n} + P_t\boldsymbol{t}$ (N s) acts at the contact point during a vanishing time, so gravity, drag and Magnus do nothing during it. The obstacle is rigid and fixed (the HIVE rests on its damper stop). This is the Walton/Brach rigid-body impulse model used by Cross and by Allen for bouncing balls.

Step 1: Newton's laws in impulse form. Linear and angular momentum change by the impulse and its moment about the centre:

$$m\,\Delta\boldsymbol{v} = \boldsymbol{P} = P_n\boldsymbol{n} + P_t\boldsymbol{t}, \tag{135}$$

$$I\,\Delta\omega = (\boldsymbol{r}\times\boldsymbol{P})_y = -R\,(\boldsymbol{n}\times(P_n\boldsymbol{n} + P_t\boldsymbol{t}))_y = -R\,P_t\,(\boldsymbol{n}\times\boldsymbol{t})_y = -R\,P_t. \tag{136}$$

The normal impulse has no lever arm (it passes through the centre), and $(\boldsymbol{n}\times\boldsymbol{t})_y = 1$ from Section 5.1. This is the **torque sign** used in the code (`w2 = w - R*P_t/I`): a positive tangential impulse (along $\boldsymbol{t}$) makes ω more negative.

Step 2: normal impulse from restitution. Taking the $\boldsymbol{n}$ component of (135), and noting that ω does not enter v_{cn},

$$v'_{cn} = v_{cn} + \frac{P_n}{m}. \tag{137}$$

The coefficient of restitution e (dimensionless, $0 \le e \le 1$) is *defined* by $v'_{cn} = -e\, v_{cn}$. Substituting into (137) and solving,

$$P_n = \begin{cases} -(1+e)\, m\, v_{cn}, & v_{cn} < 0 \text{ (approaching)}, \\ 0, & v_{cn} \ge 0 \text{ (separating: no impulse)}. \end{cases} \tag{138}$$

Since $v_{cn} < 0$ whenever an impulse acts, $P_n \ge 0$: the obstacle can only push.

Step 3: how the tangential impulse changes the slip. The $\boldsymbol{t}$ component of (135) and (136) give $\Delta(\boldsymbol{v}\cdot\boldsymbol{t}) = P_t/m$ and $\Delta\omega = -RP_t/I$. Insert both into (131):

$$\Delta v_{ct} = \frac{P_t}{m} - R\left(-\frac{RP_t}{I}\right) = P_t\left(\frac{1}{m} + \frac{R^2}{I}\right) = P_t\left(\frac{1}{m} + \frac{1}{\alpha m}\right) = \frac{P_t}{m_t}, \tag{139}$$

$$m_t = \frac{m\,\alpha}{1+\alpha} \qquad \text{(tangential effective mass, kg)}. \tag{140}$$

The tangential effective mass is smaller than m because a tangential push both slows the centre and spins the ball; the two effects add.

Step 4: the grip ("stick") candidate. If friction is strong enough, the code lets the contact point leave with slip $v'_{ct} = -e_t\, v_{ct}$, where e_t (dimensionless, $0 \le e_t \le 1$) is the tangential restitution. $e_t = 0$ means the ball leaves exactly rolling; $e_t > 0$ means the slip reverses (the ball "grips and springs back"). From (139),

$$P_t^* = -(1+e_t)\, m_t\, v_{ct}. \tag{141}$$

Step 5: Coulomb friction decides stick or slip. Friction cannot exceed μP_n, with μ the Coulomb friction coefficient (dimensionless). The code chooses

$$P_t = \begin{cases} P_t^*, & |P_t^*| \le \mu P_n \quad (\text{"stick"}, \text{ grip regime}), \\ -\mu P_n\, \operatorname{sgn}(v_{ct}), & |P_t^*| > \mu P_n \quad (\text{"slip"}, \text{ Coulomb sliding throughout}). \end{cases} \tag{142}$$

Because $\operatorname{sgn}(P_t^*) = -\operatorname{sgn}(v_{ct})$, the two branches are one saturation ("clip") function of the slip:

$$P_t = -\operatorname{clip}\big((1+e_t)\, m_t\, v_{ct},\ -\mu P_n,\ +\mu P_n\big)\,. \tag{143}$$

Dividing the stick test by $m_t|v_{cn}|$ and using (138) and (140), a contact grips when

$$\frac{|v_{ct}|}{|v_{cn}|} \le \frac{\mu\,(1+e)\,(1+\alpha)}{(1+e_t)\,\alpha}. \tag{144}$$

Step 6: the post-impact state. Putting the pieces together, `impact_batch` returns

$$\boldsymbol{v}' = \boldsymbol{v} + \frac{P_n\,\boldsymbol{n} + P_t\,\boldsymbol{t}}{m}, \tag{145}$$

$$\omega' = \omega - \frac{R\,P_t}{I}, \tag{146}$$

and, from (137) and (139), the contact-point velocities after impact are

$$v'_{cn} = -e\, v_{cn}, \qquad v'_{ct} = \begin{cases} -e_t\, v_{ct}, & \text{stick}, \\ v_{ct} - \mu P_n\, \operatorname{sgn}(v_{ct})/m_t, & \text{slip}. \end{cases} \tag{147}$$

Two checks. (i) With $\mu = 0$ the clip range shrinks to the single value 0, so $P_t = 0$ and $\omega' = \omega$: a frictionless surface cannot change spin. (ii) A ball sliding without spin ($\omega = 0$) onto a flat floor ($\boldsymbol{n} = \hat{\boldsymbol{z}}$, $\boldsymbol{t} = \hat{\boldsymbol{x}}$) with enough friction and $e_t = 0$ gets $P_t = -m_t v_x$, so

$$v_x' = v_x - \frac{m_t}{m} v_x = v_x \left(1 - \frac{\alpha}{1+\alpha}\right) = \frac{v_x}{1+\alpha}, \tag{148}$$

the textbook 5/7 for a solid sphere ($\alpha = 2/5$). Both are checked by the unit tests in the file `tests/test_contact.py`, which also confirm that a ball on an incline accelerates at $g \sin\theta/(1+\alpha)$.

Worked example: the CELL floor. For POLLEN, $m_t/m = \alpha/(1+\alpha) = 0.3865^{[\mathrm{D}]}$ and $m_t = 9.643\,\mathrm{g}^{[\mathrm{D}]}$. Take a ball that has just entered the CELL cleanly and meets the floor at speed $v_i = 5\,\mathrm{m/s}^{[\mathrm{A}]}$ descending at $\varphi = 76^{\circ[\mathrm{A}]}$ below horizontal, so $\boldsymbol{v} = (1.2096,\ -4.8515)\,\mathrm{m/s}^{[\mathrm{D}]}$. The floor rises toward the mouth at the HIVE tilt of $30^{\circ[\mathrm{C}]}$, so its normal is $\boldsymbol{n} = (-0.5,\ 0.866)^{[\mathrm{D}]}$ and $\boldsymbol{t} = (0.866,\ 0.5)^{[\mathrm{D}]}$ (pointing along the floor toward the mouth). Use $e = 0.6^{[\mathrm{A}]}$, $\mu = 0.4^{[\mathrm{A}]}$, $e_t = 0^{[\mathrm{A}]}$. Running `contact.impact` gives Table 14. Backspin $S = 1^{[\mathrm{A}]}$ means $\omega = -140.85\,\mathrm{rad/s}^{[\mathrm{D}]}$. The grip limit (144) is $1.656^{[\mathrm{D}]}$; both cases are below it and stick.

Table 14: One floor impact computed with `contact.impact` (POLLEN, $e = 0.6^{[\mathrm{A}]}$, $\mu = 0.4^{[\mathrm{A}]}$, $e_t = 0^{[\mathrm{A}]}$, $\varphi = 76^{\circ[\mathrm{A}]}$, $v_i = 5\,\mathrm{m/s}^{[\mathrm{A}]}$).

Quantity	$S = 0^{[\mathrm{A}]}$	$S = 1^{[\mathrm{A}]}$
v_{cn} (m/s)	$-4.806^{[\mathrm{D}]}$	$-4.806^{[\mathrm{D}]}$
v_{ct} (m/s)	$-1.378^{[\mathrm{D}]}$	$3.622^{[\mathrm{D}]}$
P_n (mN s)	$191.9^{[\mathrm{P}]}$	$191.9^{[\mathrm{P}]}$
P_t^* (mN s)	$13.3^{[\mathrm{P}]}$	$-34.9^{[\mathrm{P}]}$
μP_n (mN s)	$76.7^{[\mathrm{P}]}$	$76.7^{[\mathrm{P}]}$
regime	stick	stick
$\boldsymbol{v}'$ (m/s)	$(-2.174,\ 2.075)^{[\mathrm{P}]}$	$(-3.848,\ 1.108)^{[\mathrm{P}]}$
$\boldsymbol{v}' \cdot \boldsymbol{t}$ (m/s)	$-0.846^{[\mathrm{P}]}$	$-2.778^{[\mathrm{P}]}$
ω' (rad/s)	$-23.8^{[\mathrm{P}]}$	$-78.3^{[\mathrm{P}]}$
kinetic energy before $\to$ after (mJ)	$311.9 \to 118.3^{[\mathrm{P}]}$	$508.4 \to 260.7^{[\mathrm{P}]}$

Both balls rebound off the floor with the same normal speed $2.884\,\mathrm{m/s}^{[\mathrm{P}]}$. The difference is along the floor: without spin the ball drifts toward the back wall at $0.846\,\mathrm{m/s}^{[\mathrm{P}]}$; with backspin it is driven there at $2.778\,\mathrm{m/s}^{[\mathrm{P}]}$. Section 5.8 shows why that matters.

Worked example: a slipping far-rim contact. At the lower-lip corner the normal depends on where the ball touches. Take $\boldsymbol{n} = (-\sin c, \cos c)$ with $c = 15^{\circ[\mathrm{A}]}$, the same incoming $\boldsymbol{v}$ and $S = 1^{[\mathrm{A}]}$, but $\mu = 0.2^{[\mathrm{A}]}$. Then v_{ct}/v_{cn} has magnitude $0.983^{[\mathrm{D}]}$ against a grip limit of $0.828^{[\mathrm{D}]}$, so the contact *slips*: $P_t = -\mu P_n = -39.9\,\mathrm{mN\,s}^{[\mathrm{P}]}$, $\boldsymbol{v}' = (-2.406,\ 2.461)\,\mathrm{m/s}^{[\mathrm{P}]}$, $\omega' = -69.3\,\mathrm{rad/s}^{[\mathrm{P}]}$, and the contact point still slips at $v_{ct}' = 0.774\,\mathrm{m/s}^{[\mathrm{P}]}$ as it leaves.

5.4 Energy can only be lost

The kinetic energy of the ball (J) is

$$T = \tfrac{1}{2} m |\boldsymbol{v}|^2 + \tfrac{1}{2} I \omega^2 = \tfrac{1}{2} m\,(\boldsymbol{v} \cdot \boldsymbol{n})^2 + \tfrac{1}{2} m\,(\boldsymbol{v} \cdot \boldsymbol{t})^2 + \tfrac{1}{2} I \omega^2, \tag{149}$$

because $\boldsymbol{n}$ and $\boldsymbol{t}$ are orthonormal.

Proposition. For $0 \le e \le 1$, $0 \le e_t \le 1$ and $\mu \ge 0$, every impact of (138)–(146) satisfies $T' \le T$.

Proof. For any quantity, $a'^2 - a^2 = (a' - a)(a' + a)$. Apply this to each term of (149), using $\Delta(\boldsymbol{v} \cdot \boldsymbol{n}) = P_n/m$, $\Delta(\boldsymbol{v} \cdot \boldsymbol{t}) = P_t/m$ and $\Delta\omega = -RP_t/I$:

$$\begin{aligned} T' - T &= \tfrac{1}{2} m \, \frac{P_n}{m} \left(v_{cn} + v'_{cn} \right) + \tfrac{1}{2} m \, \frac{P_t}{m} \left(\boldsymbol{v} \cdot \boldsymbol{t} + \boldsymbol{v}' \cdot \boldsymbol{t} \right) + \tfrac{1}{2} I \left(-\frac{RP_t}{I} \right) \left(\omega + \omega' \right) \\ &= \underbrace{\tfrac{1}{2} P_n \left(v_{cn} + v'_{cn} \right)}_{\Delta T_n} + \underbrace{\tfrac{1}{2} P_t \left(v_{ct} + v'_{ct} \right)}_{\Delta T_t}, \end{aligned} \tag{150}$$

where the last two tangential terms combined into the contact-point slip (131), because $(\boldsymbol{v} \cdot \boldsymbol{t} - R\omega) + (\boldsymbol{v}' \cdot \boldsymbol{t} - R\omega') = v_{ct} + v'_{ct}$. For a sphere the normal and tangential problems therefore decouple.

Normal part. With $v'_{cn} = -e v_{cn}$ and (138),

$$\Delta T_n = \tfrac{1}{2} \left(-(1+e) m v_{cn} \right) (1 - e) \, v_{cn} = -\tfrac{1}{2} \left(1 - e^2 \right) m \, v_{cn}^2 \;\le\; 0. \tag{151}$$

Tangential part, stick. With $P_t = P_t^*$ and $v'_{ct} = -e_t v_{ct}$,

$$\Delta T_t = \tfrac{1}{2} \left(-(1+e_t) m_t v_{ct} \right) (1 - e_t) \, v_{ct} = -\tfrac{1}{2} \left(1 - e_t^2 \right) m_t \, v_{ct}^2 \;\le\; 0. \tag{152}$$

Tangential part, slip. Write $L = \mu P_n \ge 0$ and $s = \operatorname{sgn}(v_{ct})$, so $P_t = -Ls$ and, from (147), $v_{ct} + v'_{ct} = 2v_{ct} - Ls/m_t$. Then

$$\begin{aligned} \Delta T_t &= \tfrac{1}{2}(-Ls) \left(2v_{ct} - \frac{Ls}{m_t} \right) = -L|v_{ct}| + \frac{L^2}{2m_t} = L \left(\frac{L}{2m_t} - |v_{ct}| \right) \\ &\le L \left(\frac{(1+e_t)|v_{ct}|}{2} - |v_{ct}| \right) = -\tfrac{1}{2} \left(1 - e_t \right) L \, |v_{ct}| \;\le\; 0, \end{aligned} \tag{153}$$

where the inequality uses the slip condition $L < |P_t^*| = (1 + e_t) m_t |v_{ct}|$ and $L \ge 0$ (it is an equality when $L = 0$, for example $\mu = 0$). Adding (151) and (152) or (153) gives $T' - T \le 0$. □

Worked check. In Table 14 the floor impact removes $193.6\,\mathrm{mJ}^{[\mathsf{P}]}$ ($S = 0$) and $247.7\,\mathrm{mJ}^{[\mathsf{P}]}$ ($S = 1$). From (151) the normal part alone is $\frac{1}{2}(1 - 0.6^2)\, m\, (4.806)^2 = 184.4\,\mathrm{mJ}^{[\mathsf{P}]}$ in both cases; the rest is friction, which is larger with backspin because there is more slip to remove. The slipping far-rim example loses $313.0\,\mathrm{mJ}^{[\mathsf{P}]}$.

5.5 Which way does backspin push? The kick direction

Compare two balls that reach the same contact with the same $\boldsymbol{v}$ and $\boldsymbol{n}$, one without spin and one with backspin $\omega = -|\omega| < 0$. The normal impulse (138) is identical, and by (131) the only difference is

$$v_{ct}^{\text{spin}} = v_{ct}^{0} + R|\omega|. \tag{154}$$

By (143), P_t is a non-increasing function of v_{ct}, with slope at most $(1 + e_t) m_t$ in magnitude and total range $2\mu P_n$. Hence

$$-\min\{(1 + e_t)\, m_t R|\omega|,\; 2\mu P_n\} \;\le\; \Delta P_t \equiv P_t^{\text{spin}} - P_t^{0} \;\le\; 0, \tag{155}$$

and $\Delta P_t = 0$ exactly when both balls slide the same way (both on the same flat part of the clip). By (145) the difference in outgoing velocity is

$$\boldsymbol{v}'^{\,\text{spin}} - \boldsymbol{v}'^{\,0} = \frac{\Delta P_t}{m} \, \boldsymbol{t} = \frac{|\Delta P_t|}{m} \, \boldsymbol{k}, \qquad \boldsymbol{k} = -\boldsymbol{t} = (-n_z,\; n_x)\,, \tag{156}$$

which is `spin_kick_direction`: $\boldsymbol{n}$ rotated by $+90°$. Its size is between 0 and $\min\{(1 + e_t) m_t R|\omega|,\, 2\mu P_n\}/m$.

Far rim (lower lip). When the ball's lower-front surface catches the lower-lip corner on the way in, the centre is up and to the left of the corner: $\boldsymbol{n} = (-\sin c,\, \cos c)$, with c in the corner's cone of normals. Then

$$\boldsymbol{k} = (-\cos c,\; -\sin c)\,, \tag{157}$$

which points back toward the shooter and down. The CELL's inward axis is $-\boldsymbol{n}_{\text{out}} = (-\cos 30°, -\sin 30°)$, so at $c = 30°$[D] the kick is exactly into the CELL: the "check" effect. Repeating the far-rim worked example ($c = 15°$[A]) with $\mu = 0.4$[A] instead of 0.2[A], both runs stick, and the extra velocity is $1.93\,\text{m/s}$[P] along $\boldsymbol{k} = (-0.966, -0.259)$[D]. The component of $\boldsymbol{v}'$ along $\boldsymbol{n}_{\text{out}}$ changes from $+0.725\,\text{m/s}$[P] (moving away from the mouth) to $-1.142\,\text{m/s}$[P] (moving into it).

CELL floor. The floor normal $(-\sin 30°, \cos 30°)$ is the $c = 30°$ case of (157), so the backspin kick on the floor is along the inward axis, toward the back wall. In Table 14 it adds $1.93\,\text{m/s}$[P] toward the back wall, which equals the stick-case value $m_t R|\omega|/m = 0.3865 \times 5 = 1.93\,\text{m/s}$[D].

Near rim (roof end, upper lip). A ball that strikes the roof-end corner from above or behind (a shot slightly short) has $\boldsymbol{n} \approx (-\sin a, \cos a)$ and receives the same backward-and-down kick. Geometrically that is backward *around the corner*, over the roof, *away* from the mouth, because the mouth lies on the $+x$ side of that corner.

5.6 From one impact to a trajectory

`simulate_batch` advances many balls at once with a fixed-step fourth-order Runge–Kutta integrator (step $\Delta t = 1\,\text{ms}$[A]) using the same drag, Magnus and spin-decay model as `flight.py`. After each step it looks for a contact:

1. **Detection.** If the nearest-segment distance $d = |\boldsymbol{c} - \boldsymbol{q}|$ at the end of the step satisfies $d < R$, the ball has penetrated.

2. **Locating the contact.** The code bisects the straight chord between the step's start state $\boldsymbol{Y}_0$ and end state $\boldsymbol{Y}_1$ (22[A] halvings) for the largest fraction f that does not penetrate,

$$f = \max\{f \in [0,1] : d(\boldsymbol{c}_0 + f(\boldsymbol{c}_1 - \boldsymbol{c}_0)) \geq R\}\,, \qquad \boldsymbol{Y}_c = \boldsymbol{Y}_0 + f\,(\boldsymbol{Y}_1 - \boldsymbol{Y}_0), \tag{158}$$

   where $\boldsymbol{Y} = (x, z, v_x, v_z, \omega)$ and the velocity and spin are interpolated linearly too.

3. **Impact.** (138)–(146) are applied at $\boldsymbol{Y}_c$ with

$$e_{\text{eff}} = \begin{cases} 0, & -v_{cn} < V_{\text{rest}}, \\ e, & \text{otherwise}, \end{cases} \qquad V_{\text{rest}} = 0.05\ \text{m/s}^{[\text{A}]}, \tag{159}$$

   and the rest of the step, $(1 - f)$ of what remained, is re-integrated (up to 8[A] contacts per step).

4. **Slow contacts.** If the approach is slower than V_{rest}, the code skips the bisection: it moves the centre out to distance R along $\boldsymbol{n}$ and applies a plastic ($e = 0$, $e_t = 0$) impulse with Coulomb friction. Repeated micro-impulses of this kind reproduce rolling and sliding on a slope.

Contacts approaching faster than $V_{\text{bounce}} = 0.10\,\text{m/s}$[A] are counted as "bounces" in the statistics; this does not change the physics.

5.7 Outcome classes and the trapping test

Entry. A ball has *entered* the CELL when its centre lies inside the CELL polygon: the cavity behind the mouth plane, bounded by the walls moved inward by 1 mm[A]. (The code builds it by growing `hive.py`'s centre region, which keeps R from every wall, outward by $R - 1$ mm.) On the unfavourable side the mouth's outward normal is $\boldsymbol{n}_{\text{out}} = (\cos 30^\circ, \sin 30^\circ) = (0.866,\ 0.5)$[D]. A ball moving toward $+x$ at speed $|\boldsymbol{v}|$ and descent angle φ has $\boldsymbol{v} = |\boldsymbol{v}|(\cos\varphi, -\sin\varphi)$, and it can cross the mouth inward only if

$$\begin{aligned} \boldsymbol{v}\cdot\boldsymbol{n}_{\text{out}} < 0 &\iff \cos\varphi\cos 30^\circ < \sin\varphi\sin 30^\circ \\ &\iff \tan\varphi > \cot 30^\circ \iff \varphi > 60^\circ. \end{aligned} \tag{160}$$

So on this side every ball that enters straight from flight (still moving toward $+x$) is descending at more than 60°[D]; only a ball turned back by a contact can enter at a shallower angle.

Trapped. Define the energy height (m), the total mechanical energy divided by mg:

$$H = z + \frac{|\boldsymbol{v}|^2}{2g} + \frac{I\omega^2}{2mg} = z + \frac{|\boldsymbol{v}|^2 + \alpha R^2\omega^2}{2g}, \tag{161}$$

with $g = 9.80665\ \text{m/s}^2$[C]. Between impacts,

$$\frac{d}{dt}(mgH) = \boldsymbol{v}\cdot\boldsymbol{F}_D + \boldsymbol{v}\cdot\boldsymbol{F}_M - k\,I\omega^2 = -\tfrac{1}{2}\rho C_d A|\boldsymbol{v}|^3 + 0 - k\,I\omega^2 \ \le\ 0, \tag{162}$$

because drag $\boldsymbol{F}_D$ opposes $\boldsymbol{v}$, the Magnus force $\boldsymbol{F}_M$ is along $\boldsymbol{\omega}\times\boldsymbol{v} = \omega(v_z, -v_x)$, which is perpendicular to $\boldsymbol{v}$, and spin decays as $\dot\omega = -k\omega$. (Here ρ is air density in kg/m^3, C_d the drag coefficient, $A = \pi R^2$ in m^2, and k the spin-decay rate in 1/s.) The impacts never add energy (Section 5.4). So H never increases. To leave, the centre must cross the mouth plane somewhere on the gate, and the lowest point of the gate is $z_{\text{bar}} = 1.387$ m[D] (the ball resting on the floor at the lower lip; the lip crest itself is 1.392 m[D]). The code therefore declares a ball **trapped** when

$$\boldsymbol{c} \in \text{CELL} \quad \text{and} \quad H < z_{\text{bar}} - 10^{-4}\ \text{m}^{[\text{A}]}. \tag{163}$$

Counting the spin energy as if it could lift the ball makes this test strict (conservative): a few slow, spinning balls counted as "not yet trapped" would in reality settle.

Worked example. A ball inside the CELL at $z = 1.30$ m[A] moving at 1 m/s[A] without spin has $H = 1.30 + 0.051 = 1.351$ m[D] < 1.387 m[D]: trapped. The same ball spinning at 50 rad/s[A] adds $\alpha R^2\omega^2/(2g) = 0.101$ m[D], so $H = 1.452$ m[D] and the test waits.

The six outcomes. A ball that is outside the CELL with $H < z_{\text{bar}} - 10^{-4}$ m can never enter (the same argument mirrored) and is resolved at once. A ball also stops when it falls below $z = 0.70$ m[A] while descending, leaves $|x| < 3$ m[A], drops below $z = 0$, or reaches the time limit $t_{\max}$ (3.5 s[A] by default; 2.0 s[A] in the incidence sweep). Table 15 lists the classes as the code assigns them.

5.8 Why backspin can reduce retention on this side

Section 5.5 shows that backspin has two opposite effects at a contact:

- at the **far rim** it kicks a ball that is clipping the lower lip into the CELL (helps);

Table 15: Outcome classes of `simulate_batch` (`OUTCOMES`).

Class	Rule in the code
`retained_clean`	First entry had no earlier contact, and the ball is trapped (163). Later contacts inside the CELL are allowed.
`retained_contact`	At least one contact before the first entry, then trapped.
`bounced_out`	Entered at some time, then stopped (fell, left the region, or was locked out) without being trapped.
`blocked`	Touched the HIVE, never entered. Also a ball that starts inside an obstacle, and a ball that touched the HIVE without entering and is still unresolved at $t_{\max}$ (for example resting on the roof).
`miss`	Never touched the HIVE, never entered.
`timeout`	Entered, but neither trapped nor gone at $t_{\max}$; counted as not retained.

- on the **CELL floor**, right after a clean entry, it kicks the ball along the floor toward the back wall (Table 14: $2.78\,\mathrm{m/s}^{[P]}$ versus $0.85\,\mathrm{m/s}^{[P]}$). The ball then hits the back-wall/roof corner with more speed and can rebound out through the mouth (hurts).

On this side clean entries are more common than far-rim clips, so the second effect can win.

Worked example (incidence sweep, one angle). The current default aero file, `research/params_lit.json`, was used with two functions of `contact.py`: `incidence_states` builds the states and `simulate_batch` runs them. Inputs: crossing speed $v_i = 5\,\mathrm{m/s}^{[A]}$, descent $\varphi = 76°^{[A]}$, $e = 0.6^{[A]}$, $\mu = 0.4^{[A]}$, $e_t = 0^{[A]}$, and gate coordinate s from $-0.10\,\mathrm{m}^{[A]}$ to $0.45\,\mathrm{m}^{[A]}$ in $5\,\mathrm{mm}^{[A]}$ steps ($s = 0$ is the far rim; the gate is $0.286\,\mathrm{m}^{[D]}$ long). The crossing state is identical for every S, so the table isolates the collision effect of spin. Widths are the total length of s giving each outcome.

Table 16: Outcome widths (mm) along the gate, $\varphi = 76°^{[A]}$, $v_i = 5\,\mathrm{m/s}^{[A]}$, $e = 0.6^{[A]}$, $\mu = 0.4^{[A]}$.

S	retained clean	retained contact	bounced out	of which, clean entry	blocked
$0^{[A]}$	$190^{[P]}$	$40^{[P]}$	$0^{[P]}$	$0^{[P]}$	$325^{[P]}$
$0.5^{[A]}$	$190^{[P]}$	$55^{[P]}$	$0^{[P]}$	$0^{[P]}$	$310^{[P]}$
$1^{[A]}$	$0^{[P]}$	$70^{[P]}$	$195^{[P]}$	$190^{[P]}$	$290^{[P]}$
$2^{[A]}$	$0^{[P]}$	$40^{[P]}$	$235^{[P]}$	$190^{[P]}$	$280^{[P]}$

Both effects are visible. Contact-assisted retention grows from $40\,\mathrm{mm}^{[P]}$ at $S = 0$ to $70\,\mathrm{mm}^{[P]}$ at $S = 1$ (the far-rim check), but every clean entry at this angle bounces out at $S = 1$ (the floor kick), so total retention drops from $230\,\mathrm{mm}^{[P]}$ to $70\,\mathrm{mm}^{[P]}$. $S = 0.5$ is neutral to slightly positive here, and $S = 2$ is worse than $S = 1$. A single ball crossing at $s = 0.15\,\mathrm{m}^{[A]}$ shows the mechanism: with $S = 0$ it hits the floor (approach speed $5.03\,\mathrm{m/s}^{[P]}$), makes $5^{[P]}$ contacts and is retained cleanly; with $S = 1$ it makes $3^{[P]}$ contacts and bounces out.

What the full sweeps add. The published sweeps in `docs/05_contact.md` were run with the older `data/aero_params.json` (the contact parameters m, R, α are identical in both files; the aero sets differ). Averaged over descent angles 64–86°[A] at $e = 0.6$[A] and $\mu = 0.4$[A], the best retention probability for an assumed 20 mm[A] Gaussian aim scatter falls from 0.89[P] ($S = 0$) to 0.86[P] ($S = 0.5$), 0.72[P] ($S = 1$) and 0.57[P] ($S = 2$). In controlled first-contact pairs, backspin $S = 1$ raises the retained fraction of far-rim clips from 0.36[P] to 0.60[P] but lowers that of floor contacts after a clean entry from 1.00[P] to 0.086[P]. Two cautions keep this from becoming a slogan:

- **More spin is not always better, and not always worse.** The penalty depends strongly on e: in the full launch sweep (launch height 0.737 m[A], distance 0.90 m[A]) at $e = 0.3$[A] backspin $S = 1$ only ties no spin (tight-scatter probability 0.85[P] versus 0.81[P], within noise), while at $e = 0.6$[A] it falls to 0.48[P] versus 0.79[P]. In the air, too, the lift coefficient saturates in this model ($C_l = \min(c_1 S, C_{l,\max})$), so extra spin stops adding curvature.
- **These are model probabilities**, not measured hit rates. One successful shot on a real HIVE would say nothing about repeatability; many shots per setting are needed.

5.9 What is assumed

- **Contact coefficients.** e, μ and e_t for POLLEN on the HIVE skin are unmeasured. The sweeps use e from 0.3[A] to 0.8[A], μ from 0.2[A] to 0.8[A], and $e_t = 0$[A] with a spot check at 0.3[A]. Pickleball and smooth-sphere literature values only set these ranges; they are not POLLEN measurements. e is the most important unknown.
- **Constant coefficients.** e does not depend on impact speed (real plastic balls usually lose restitution at higher speed).
- **Rigid sphere, rigid obstacle.** The ball is a smooth rigid sphere with $I = \alpha m R^2$, $\alpha = 0.63$[D]; perforations catching on edges, shell deformation, panel compliance and HIVE motion are not modelled. The HIVE rests on its damper stop.
- **Instantaneous impact.** Gravity and aerodynamics act only between impacts.
- **2D centre plane.** One section through the HIVE; no side walls, off-centre entries, sidespin or tilted spin axes.
- **Empty CELL.** The preloaded NECTAR against the back skin is not modelled; it could absorb back-wall rebounds and soften the floor-kick penalty.
- **Numerics.** Step 1 ms[A], 22[A] bisection halvings, $V_{\text{rest}} = 0.05$ m/s[A], trap margin 10^{-4} m[A].
- **Geometry.** The HIVE section comes from the official CAD with the 30°[C] tilt; landmarks such as z_{bar} are derived from it.
- **Unfavourable side only.** The favourable side was not swept for contact.

6 From windows to requirements: robustness, Monte Carlo and the shooter specification

The earlier sections give, for one release state, a yes-or-no answer: does the ball enter the upward-facing CELL of the HIVE cleanly or not? A shooter, though, is not a single release state. Its exit

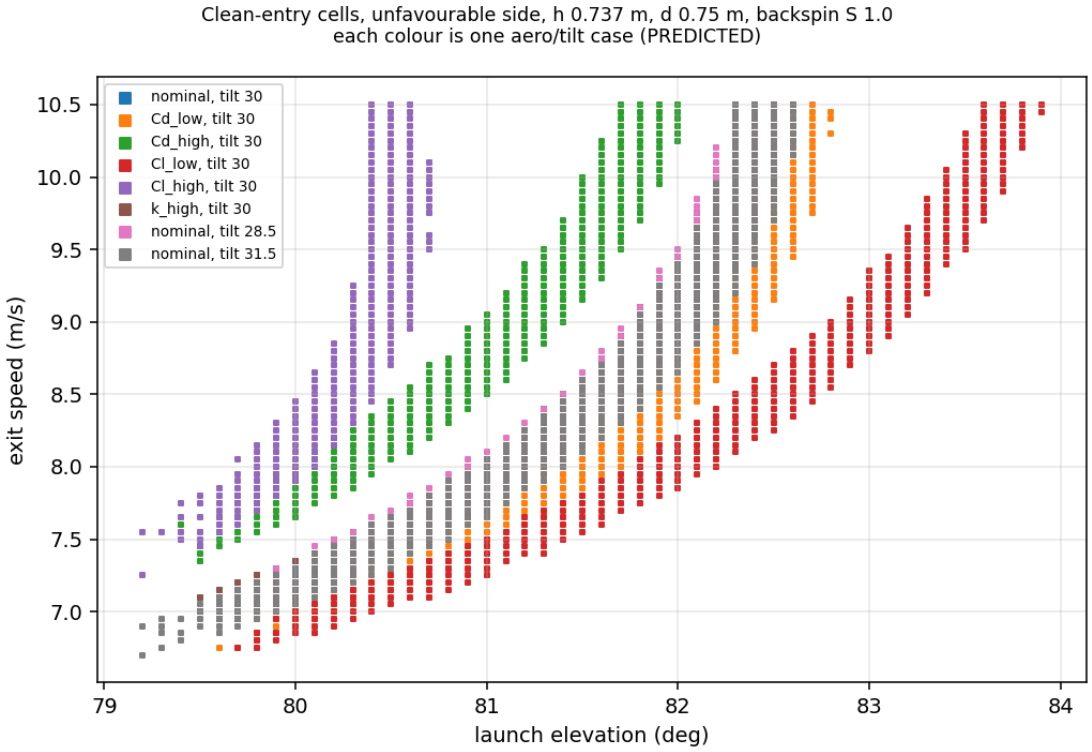


Figure 2: PREDICTED clean-entry cells in the (launch elevation, exit speed) plane for an unfavourable-side shot from release height $h = 0.737\,\mathrm{m}^{[A]}$ and distance $d = 0.75\,\mathrm{m}^{[A]}$ with backspin $S = 1.0^{[A]}$. Each colour is one aerodynamic or tilt condition from `research/params_lit.json` (nominal, C_d and lift bounds, fast spin decay) or the tilt bounds $28.5^{\circ[D]}$ and $31.5^{\circ[D]}$. The clean bands of the different conditions do not overlap in one common cell (Eq. (175)), which is why the specification relies on tuning the wheel speed on the field (Section 6.3). All cells are model output, not measurements.

speed and its angle scatter from shot to shot, the ball's mass and its aerodynamic coefficients are only known to within literature bounds, and the HIVE itself sits on the field with a tolerance. This section turns the yes-or-no windows into numbers a CAD designer can build to. It goes in five steps: (i) sample the clean window on a grid; (ii) find the largest clean square in that grid and read it as a $\pm$ tolerance; (iii) ask which hood angle stays usable when the aerodynamics change and the team re-tunes only the wheel speed; (iv) smooth the window with Gaussian release errors, and run a full three-dimensional Monte Carlo with per-shot physics, wind and HIVE pose, scored with a Wilson interval; (v) convert the result into wheel and ball rpm and a CAD card.

All targets at the end of this section are model outputs, so they carry [P]. They are design targets to be checked on the field, not measured performance.

Throughout, the frame is the one of Section 3: x is horizontal toward the HIVE, y is horizontal and sideways, and z points up. The shot comes from the unfavourable side: the ball has to pass over the roof of the HIVE and then drop into the CELL, whose mouth faces outward and upward at the tilt angle $\beta = 30^\circ$[C] (Competition Manual figure and official CAD).

6.1 The clean window on a grid

Motivation. The flight solver answers "clean or not" for one release state. To talk about tolerances we need that answer on a regular grid of release states, so that neighbouring cells can be compared.

Release state. A shot is released from horizontal distance d (m) before the HIVE pivot plane and at height h (m) above the tiles, with exit speed v (m/s), elevation angle θ (deg, above horizontal) and spin parameter S (dimensionless, $S = \omega R/|\boldsymbol{v}|$ with ball radius R in m and spin rate ω in rad/s). In two dimensions there is no wind, so the forces depend only on the velocity and the spin, not on the position. The solver `integrate_batch` therefore integrates each path once from the origin, giving $\boldsymbol{r}_0(t; v, \theta, S)$, and translates it:

$$\boldsymbol{r}(t; h, d, v, \theta, S) = \boldsymbol{r}_0(t; v, \theta, S) + (-d,\ h)^{\mathsf{T}}. \tag{164}$$

This is exact because the right-hand side of the equations of motion contains no position (it depends on the velocity and, through the spin decay, on the time since release only). One integration serves every (h, d) pair, which is what makes a dense grid affordable.

Clean indicator. Let $\boldsymbol{n}_{\text{out}}$ be the outward unit normal of the CELL mouth (dimensionless) and let $c(t)$ (m) be the raster clearance between the ball surface and the nearest HIVE part. Following `evaluate_gate`, a path is clean when it crosses the mouth edge (the gate segment between the two lips) at a time t_g (s), moving inward, without touching anything on the way in:

$$M(h, d, v, \theta) = \begin{cases} 1, & \text{if } \exists\, t_g :\ \boldsymbol{r}(t_g) \in \text{gate},\quad \boldsymbol{v}(t_g) \cdot \boldsymbol{n}_{\text{out}} < 0,\quad c_{\min}(t_g) \geq 0, \\ 0, & \text{otherwise,} \end{cases} \tag{165}$$

where $c_{\min}(t_g) = \min_{t_{\text{box}} \leq t \leq t_g} c(t)$ (m) and t_{box} (s) is the time the ball enters the raster box around the HIVE.

Grid. The specification grid of `shooter_spec.grid` is

$$v_i = v_{\min} + i\,\Delta v, \qquad \theta_j = \theta_{\min} + j\,\Delta\theta, \tag{166}$$

with $v_{\min} = 5.0\,\mathrm{m/s}^{[A]}$, $\Delta v = 0.05\,\mathrm{m/s}^{[A]}$ and $111^{[D]}$ speeds up to $10.5\,\mathrm{m/s}^{[A]}$; $\theta_{\min} = 62°^{[A]}$, $\Delta\theta = 0.1°^{[A]}$ and $261^{[D]}$ elevations up to $88°^{[A]}$. The release heights are 0.20, 0.35, 0.50, 0.65 and $0.737\,\mathrm{m}^{[A]}$, and distances run from $0.60\,\mathrm{m}^{[A]}$ to $1.75\,\mathrm{m}^{[A]}$ in steps of $0.05\,\mathrm{m}^{[A]}$ ($24^{[D]}$ values). The spin values are $S \in \{0, 0.5, 1.0, 1.5, -1.0\}^{[A]}$, where negative S is topspin.

The flight solver returns candidate distances d_k (m) at which a path hits the gate. `grid_from_candidates` marks a grid distance d clean only if two candidates at consecutive time samples k, $k+1$ of the same path are both clean and bracket it:

$$M(d) = 1 \iff \exists k:\ M(d_k) = M(d_{k+1}) = 1,\quad \min(d_k, d_{k+1}) \le d \le \max(d_k, d_{k+1}). \tag{167}$$

This is conservative: a clean point that is isolated between two dirty neighbours is not counted.

Worked example. Table 17 lists the number of clean (v, θ) cells $N(h, d) = \sum_{i,j} M(h, d, v_i, \theta_j)$ for the nominal aerodynamic set of `research/params_lit.json`. At $d \le 0.70\,\mathrm{m}^{[A]}$ no cell is clean at the heights $h = 0.50$ to $0.737\,\mathrm{m}^{[A]}$: the ball cannot both clear the roof and descend steeply enough into the mouth from that close. The count peaks near $d = 0.80\,\mathrm{m}^{[P]}$ and then falls slowly with distance. At $d = 0.75\,\mathrm{m}^{[A]}$ the window shrinks with height: $369^{[P]}$ cells at $h = 0.737\,\mathrm{m}^{[A]}$, $295^{[P]}$ at $0.65\,\mathrm{m}^{[A]}$ and $170^{[P]}$ at $0.50\,\mathrm{m}^{[A]}$ (no spin). For $S = 1$ at $h = 0.50\,\mathrm{m}^{[A]}$ the cell count at $d = 0.75\,\mathrm{m}^{[A]}$ is already $0^{[P]}$.

Table 17: Clean cells on the 111×261 speed–elevation grid at $h = 0.737\,\mathrm{m}$, nominal aerodynamics (`research/params_lit.json`).

d (m)	0.75	0.80	0.85	0.90	1.15	1.50	1.75
N, $S = 0$	$369^{[P]}$	$457^{[P]}$	$445^{[P]}$	$429^{[P]}$	$355^{[P]}$	$288^{[P]}$	$244^{[P]}$
N, $S = 1$	$303^{[P]}$	$418^{[P]}$	$410^{[P]}$	$398^{[P]}$	$321^{[P]}$	$249^{[P]}$	$214^{[P]}$

Backspin does not make the window larger here: $S = 1$ has fewer clean cells than $S = 0$ at every distance in the table. The case $S = 1.5$ gives exactly the same window as $S = 1.0$, because the lift coefficient has already saturated (Section 3): with the nominal set, $C_L = \min(c_1 S, C_{L,\max})$ stops growing at $S_{\mathrm{sat}} = C_{L,\max}/c_1 = 0.25/0.35 = 0.714^{[D]}$. More spin above that adds nothing.

Assumptions.

- The 2D model has no wind and no sideways motion; azimuth is added in Section 6.5.
- A grid cell is a point sample; nothing is claimed about the space between grid points.
- The aerodynamic coefficients are literature bounds from pickleball and smooth-sphere data, not POLLEN measurements.
- The HIVE tilt is the nominal $30°^{[C]}$ unless a variant says otherwise.

6.2 The largest clean square and the ± tolerance

Motivation. A shooter set to (v_c, θ_c) that actually fires anywhere in $[v_c - \delta_v, v_c + \delta_v] \times [\theta_c - \delta_\theta, \theta_c + \delta_\theta]$ is clean every time only if that whole box is clean. The largest all-clean $k \times k$ block of cells is therefore the best repeatability target the window allows. `best_square` finds it in $O(N_v N_\theta)$ work per size using a prefix sum.

Prefix sum. Let $M_{ij} \in \{0,1\}$ be the mask on the $N_v \times N_\theta$ grid for fixed (h, d, S). Define the two-dimensional prefix sum (a count, dimensionless)

$$P_{ij} = \sum_{i'=0}^{i-1} \sum_{j'=0}^{j-1} M_{i'j'}, \qquad P_{0j} = P_{i0} = 0. \tag{168}$$

Block sum, step by step. The number of clean cells in the $k \times k$ block whose lower corner is (i_0, j_0) is $B_k(i_0, j_0) = \sum_{i=i_0}^{i_0+k-1} \sum_{j=j_0}^{j_0+k-1} M_{ij}$. The sum $P_{i_0+k,\, j_0+k}$ covers the rectangle $[0, i_0 + k) \times [0, j_0 + k)$. Subtracting $P_{i_0,\, j_0+k}$ removes the rows below i_0, and subtracting $P_{i_0+k,\, j_0}$ removes the columns left of j_0. The corner $[0, i_0) \times [0, j_0)$ has now been removed twice, so it is added back once (inclusion–exclusion):

$$B_k(i_0, j_0) = P_{i_0+k,\, j_0+k} - P_{i_0,\, j_0+k} - P_{i_0+k,\, j_0} + P_{i_0,\, j_0}. \tag{169}$$

Because each $M_{ij} \le 1$, the block is all clean exactly when its count is the largest possible:

$$\text{block clean} \iff B_k(i_0, j_0) = k^2. \tag{170}$$

Monotone search. Every $k \times k$ clean block contains a $(k-1) \times (k-1)$ clean block, so if no block of size k exists, none of a larger size exists either. The code increases k from 1 and stops at the first size with no hit:

$$k^* = \max\{k : \ \exists (i_0, j_0),\ B_k(i_0, j_0) = k^2\}. \tag{171}$$

When several blocks of size k^* exist, the code keeps the one whose lower corner is closest (in index units) to the centroid $(\bar{\imath}, \bar{\jmath})$ of the lower corners of all hits (the same as comparing block centres, since every centre is the corner shifted by $(k^* - 1)/2$), which keeps the choice away from the edge of a cluster:

$$(i_0, j_0) = \arg\min_{\text{hits}} \Big[\big(i_0 - \bar{\imath}\big)^2 + \big(j_0 - \bar{\jmath}\big)^2 \Big]. \tag{172}$$

Centre and tolerance. The k cell centres of the block span $k - 1$ grid steps. The set point is the middle of that span, and the half-span is the tolerance:

$$v_c = v_{\min} + \Big(i_0 + \frac{k^* - 1}{2}\Big)\Delta v, \qquad \theta_c = \theta_{\min} + \Big(j_0 + \frac{k^* - 1}{2}\Big)\Delta\theta, \tag{173}$$

$$\delta_v = \frac{k^* - 1}{2}\,\Delta v, \qquad \delta_\theta = \frac{k^* - 1}{2}\,\Delta\theta. \tag{174}$$

Here v_c and δ_v are in m/s, θ_c and δ_θ in degrees. A single clean cell ($k^* = 1$) gives a zero tolerance: the set point works only if the shooter is perfect.

Worked example. At $h = 0.737\,\mathrm{m}^{[\mathsf{A}]}$, $d = 0.75\,\mathrm{m}^{[\mathsf{A}]}$ and $S = 0^{[\mathsf{A}]}$, nominal aerodynamics, the search ends at $k^* = 4^{[\mathsf{P}]}$ with $(i_0, j_0) = (78, 223)^{[\mathsf{P}]}$. Then $v_c = 5.0 + (78 + 1.5)(0.05) = 8.975\,\mathrm{m/s}^{[\mathsf{P}]}$ and $\theta_c = 62 + (223 + 1.5)(0.1) = 84.45^{\circ[\mathsf{P}]}$, with $\delta_v = 1.5 \times 0.05 = 0.075\,\mathrm{m/s}^{[\mathsf{D}]}$ and $\delta_\theta = 1.5 \times 0.1 = 0.15^{\circ[\mathsf{D}]}$. The smallest raster clearance inside the block is only $0.2\,\mathrm{mm}^{[\mathsf{P}]}$. For $S = 0.5^{[\mathsf{A}]}$ the square is also $k^* = 4^{[\mathsf{P}]}$ at $(9.275\,\mathrm{m/s}^{[\mathsf{P}]}, 82.45^{\circ[\mathsf{P}]})$; for $S = 1.0^{[\mathsf{A}]}$ it is $k^* = 4^{[\mathsf{P}]}$ at $(9.025\,\mathrm{m/s}^{[\mathsf{P}]}, 81.95^{\circ[\mathsf{P}]})$. Over every unfavourable-side row of the sweep with $S \in \{0, 0.5, 1\}$, the square sizes are $k^* = 4$ in $16^{[\mathsf{P}]}$ rows, $k^* = 3$ in $251^{[\mathsf{P}]}$, $k^* = 2$ in $40^{[\mathsf{P}]}$, $k^* = 1$ in $1^{[\mathsf{P}]}$ and empty in $52^{[\mathsf{P}]}$. No row reaches $k^* = 5$. So $\pm 0.075\,\mathrm{m/s}^{[\mathsf{P}]}$ and $\pm 0.15^{\circ[\mathsf{P}]}$ is the best repeatability the grid admits anywhere, and it holds for one set of aerodynamic coefficients only.

Assumptions.

- A square block in index space is used; with $\Delta v = 0.05\,\mathrm{m/s}$[A] and $\Delta\theta = 0.1^\circ$[A] this fixes the aspect ratio of the tolerance box.
- Cleanliness is checked only at the cell centres of the block.
- One aerodynamic set, one tilt and one (h, d) at a time.

6.3 Many conditions: the robust intersection and the field-tuned hood

Motivation. The square above is for the nominal coefficients. On the field the drag and lift of a real POLLEN, and the HIVE tilt, can be anywhere inside their bounds. A fixed hood fixes the elevation angle; the wheel speed is the one thing the team can tune at the field. This subsection asks which hood angle leaves a usable speed under every condition.

Conditions. A condition c is one aerodynamic variant combined with one HIVE tilt. `shooter_spec.variants` gives the nominal set and the drag bounds (Cd_low, Cd_high); for $S \neq 0$ it adds the lift bounds (Cl_low, Cl_high) and fast spin decay (k_high). The tilts are 28.5°[D], 30°[C] and 31.5°[D] (the field acceptance lip-height tolerance). That gives $3 \times 3 = 9$[D] conditions for $S = 0$ and $6 \times 3 = 18$[D] for $S \neq 0$. Each condition has its own mask $M_c(v_i, \theta_j)$.

Robust intersection. The strictest requirement is a single (v, θ) that is clean in every condition:

$$M_{\mathrm{rob}}(v_i, \theta_j) = \prod_c M_c(v_i, \theta_j). \tag{175}$$

On the unfavourable side this mask is empty in every one of the 600[P] rows of `data/spec_robust.csv`: no single speed and angle works for all drag, lift and tilt bounds at once. This is why the specification must rely on field tuning.

Elevation band. A real hood angle is set to within $\pm\tau$ (deg). With $n_b = \mathrm{round}(\tau/\Delta\theta)$ (cells), a speed cell is usable at hood index j only if it is clean at every elevation in the band:

$$G_c(v_i, j) = \prod_{|j'-j| \le n_b} M_c(v_i, \theta_{j'}), \qquad n_b \le j < N_\theta - n_b. \tag{176}$$

Longest run. For fixed j, `longest_run` scans $G_c(\cdot, j)$ along the speed axis. With the run length $L_i = (L_{i-1} + 1)\, G_c(v_i, j)$, $L_{-1} = 0$, the longest run and its centre (a speed index, possibly a half integer) are

$$n_c(j) = \max_i L_i, \qquad \hat{\imath}_c(j) = \arg\max_i L_i - \frac{n_c(j) - 1}{2}. \tag{177}$$

The nominal-only fixed-hood value in `describe` uses only the nominal condition and picks the j with the longest run (ties broken by the larger worst-case clearance), with set speed $(v_{\mathrm{lo}} + v_{\mathrm{hi}})/2$ and tolerance $(v_{\mathrm{hi}} - v_{\mathrm{lo}})/2$ over that run.

Field-tuned hood. `tunable_rows` lets each condition use its own speed. The hood is feasible at index j if every condition has at least one usable speed run. The guaranteed run length is the worst one,

$$w(j) = \min_c n_c(j), \qquad \text{feasible} \iff w(j) > 0, \tag{178}$$

and the spread of the run centres measures how far the team may have to re-tune, $\Sigma(j) = \max_c \hat{\imath}_c(j) - \min_c \hat{\imath}_c(j)$ (cells; $\Delta v\,\Sigma(j)$ in m/s). The code ranks hood indices by a lexicographic score (run length first, small spread second; the spread is at most $N_v - 1 = 110$ cells, so the factor 1000 keeps the run length dominant):

$$j^* = \arg\max_j \Big[1000\, w(j) - \Sigma(j)\Big]. \tag{179}$$

The reported tuning range and the speed tolerance that every condition keeps at the chosen hood are

$$v_{\text{tune}} \in \Big[v_{\min} + \Delta v \min_c \hat{\imath}_c(j^*),\ v_{\min} + \Delta v \max_c \hat{\imath}_c(j^*)\Big], \qquad \delta_{v,\text{tune}} = \frac{w(j^*) - 1}{2}\,\Delta v. \tag{180}$$

Note that the masks are not intersected here: Eq. (178) needs a run to exist in each condition separately, and the runs may sit at different speeds.

Worked example. At $h = 0.737\,\text{m}^{[\text{A}]}$ and $d = 0.75\,\text{m}^{[\text{A}]}$ with $\tau = 0.2^{\circ[\text{A}]}$ ($n_b = 2^{[\text{D}]}$):

- No spin: the hood is $\theta_{\text{hood}} = 80.8^{\circ[\text{P}]}$. The nine run centres lie between $6.65\,\text{m/s}^{[\text{P}]}$ (Cd_low) and $7.35\,\text{m/s}^{[\text{P}]}$ (Cd_high); the nominal-drag centres are 6.825–$6.9\,\text{m/s}^{[\text{P}]}$. The worst run is $w = 1^{[\text{P}]}$ cell, so $\delta_{v,\text{tune}} = 0^{[\text{D}]}$: each condition has exactly one tuned speed that works.
- Backspin $S \in \{0.5, 1.0, 1.5\}$: no hood index is feasible at $\tau = 0.2^{\circ[\text{A}]}$. With $\tau = 0.1^{\circ[\text{A}]}$ ($n_b = 1^{[\text{D}]}$) the hood is feasible at $80.3^{\circ[\text{P}]}$ ($S = 0.5$) and $80.4^{\circ[\text{P}]}$ ($S = 1.0$) with $w = 2^{[\text{P}]}$, i.e. $\delta_{v,\text{tune}} = 0.025\,\text{m/s}^{[\text{D}]}$.
- With $\tau = 0.5^{\circ[\text{A}]}$ no row of the sweep is feasible for any spin.

Without the band ($\tau = 0$), every condition has clean speeds at the backspin hood $80.5^{\circ[\text{P}]}$ (conditions of $S = 0.5$ and $S = 1$ together); the lowest clean speed per condition ranges from $6.9\,\text{m/s}^{[\text{P}]}$ to $8.9\,\text{m/s}^{[\text{P}]}$ (the high-lift bound needs the most speed). For no spin at $80.8^{\circ[\text{P}]}$ the union of the nine clean speed sets is 6.6–$7.45\,\text{m/s}^{[\text{P}]}$. An exit speed range of about 6.5–$9.0\,\text{m/s}^{[\text{P}]}$ therefore contains a clean speed for every condition at the chosen hood. The nominal-only fixed hood of `describe` at $\tau = 0.2^{\circ[\text{A}]}$ is $82.8^{\circ[\text{P}]}$ for no spin (7.75–$7.8\,\text{m/s}^{[\text{P}]}$) and $80.5^{\circ[\text{P}]}$ for $S = 0.5$ and $S = 1$; at $\tau = 0.5^{\circ[\text{A}]}$ it finds none.

Assumptions.

- The team can re-tune wheel speed at the field but not the hood angle.
- The conditions are the corners of the literature bounds, not a measured distribution.
- Topspin ($S = -1^{[\text{A}]}$) is feasible at $\tau = 0.2^{\circ[\text{A}]}$ in the sweep (for example $81.6^{\circ[\text{P}]}$, 6.4–$7.05\,\text{m/s}^{[\text{P}]}$) but is not the chosen configuration.

6.4 Release errors as a Gaussian blur

Motivation. The square and the band treat the tolerance as a hard box. Real release errors are random, so the right question is the probability of a clean shot when the shooter aims at a cell. For independent Gaussian errors this probability is the clean mask blurred by a Gaussian.

Expectation as convolution. Let the actual distance, speed and elevation be $d+\delta_d$, $v(1+\varepsilon_v)$ and $\theta+\delta_\theta$, with $\delta_d \sim \mathcal{N}(0,\sigma_d^2)$ (m), $\varepsilon_v \sim \mathcal{N}(0,\sigma_{v,\mathrm{rel}}^2)$ (dimensionless) and $\delta_\theta \sim \mathcal{N}(0,\sigma_\theta^2)$ (deg), all independent. The probability of a clean shot is

$$p(d,v,\theta) = \mathbb{E}\big[M(d+\delta_d,\ v(1+\varepsilon_v),\ \theta+\delta_\theta)\big]. \tag{181}$$

Near a reference speed v_{ref} (m/s) the speed error is $v\varepsilon_v \approx v_{\mathrm{ref}}\varepsilon_v$, an additive error of standard deviation $\sigma_v = \sigma_{v,\mathrm{rel}} v_{\mathrm{ref}}$ (m/s). Writing the expectation as an integral against the product density ϕ gives a convolution:

$$\begin{aligned} p(d,v,\theta) &= \iiint M(d',v',\theta')\,\phi_{\sigma_d}(d'-d)\,\phi_{\sigma_v}(v'-v)\,\phi_{\sigma_\theta}(\theta'-\theta)\,\mathrm{d}d'\,\mathrm{d}v'\,\mathrm{d}\theta' \\ &= (M * \phi)(d,v,\theta). \end{aligned} \tag{182}$$

On the grid, `smooth_prob` evaluates this with `scipy.ndimage.gaussian_filter`, whose widths are in cells:

$$s_d = \frac{\sigma_d}{\Delta d}, \qquad s_v = \frac{\sigma_{v,\mathrm{rel}}\, v_{\mathrm{ref}}}{\Delta v}, \qquad s_\theta = \frac{\sigma_\theta}{\Delta\theta}. \tag{183}$$

Cells outside the grid count as dirty (`mode="constant"`, value 0) and the kernel is cut at 3σ. The speed width is applied piecewise: the mask is filtered three times and each speed row keeps the result for its nearest reference speed,

$$v_{\mathrm{ref}}(v_i) = \arg\min_{r\in\{3.5,\,6.0,\,8.5\}} |v_i - r|. \tag{184}$$

The reference speeds 3.5, 6.0 and 8.5 m/s[A] are code constants. The best aim at each distance, for a launcher that cannot exceed v_{cap} (m/s), is

$$P_{\mathrm{kin}}(d) = \max_{v_i \le v_{\mathrm{cap}},\ \theta_j} p(d, v_i, \theta_j). \tag{185}$$

Error sets. The release error sets of `ERR_SETS` are design assumptions, not measurements. Listed as $\sigma_{v,\mathrm{rel}}$, σ_θ, σ_d, they are: tight 0.005[A], 0.15°[A], 0.010 m[A]; nominal 0.015[A], 0.40°[A], 0.025 m[A]; loose 0.030[A], 0.80°[A], 0.050 m[A]. The azimuth and lateral spreads σ_ψ, σ_y are 0.25°[A], 0.010 m[A] (tight), 0.50°[A], 0.025 m[A] (nominal) and 1.00°[A], 0.050 m[A] (loose).

Worked example. On the spec grid ($\Delta d = 0.05$ m[A]) with the nominal errors, $s_d = 0.025/0.05 = 0.5$[D] cells, $s_\theta = 0.40/0.1 = 4.0$[D] cells and $s_v = 0.015 \times 6.0/0.05 = 1.8$[D] cells near 6 m/s (1.05[D] near 3.5 and 2.55[D] near 8.5 m/s). At $h = 0.737$ m[A] the best smoothed probability with no spin is 0.419[P] at $d = 0.75$ m[A] (aim 7.1 m/s[P], 81.4°[P]), 0.426[P] at $d = 0.80$ m[A] (7.15 m/s[P], 81.2°[P]) and 0.364[P] at $d = 1.15$ m[A]. With $S = 1$ it is 0.423[P] at $d = 0.75$ m[A] (7.2 m/s[P], 79.9°[P]) and 0.418[P] at $d = 0.80$ m[A] (7.25 m/s[P], 79.7°[P]). Two points stand out. First, the smoothed optimum sits near 80–81°, not at the 84.45° square of Section 6.2: a blur of 4 cells in elevation rewards the part of the window that is deep in elevation, while the square rewards the part that is wide in both directions. Second, even the best aim is clean only about four times in ten with nominal release errors. On the robustness grid of `robustness.py` ($\Delta d = 0.01$ m[A]) the same errors give $s_d = 2.5$[D] cells.

Assumptions.

- Errors in d, v and θ are Gaussian, independent and the same for every shot.
- The relative speed error is approximated by three fixed additive widths.
- The mask is for one aerodynamic condition; physical spread is added in Section 6.6.

6.5 Azimuth: why the in-plane distance scales with $\cos\psi$

Motivation. A robot that is not square to the HIVE shoots at an azimuth ψ (deg or rad) away from the x axis. Re-running the 3D solver for every azimuth would be expensive. `clean_masks` instead reuses the 2D paths and scales their horizontal coordinate and velocity by $\cos\psi$. This subsection shows why that is exact in still air.

Rotation about the vertical. Let

$$\mathsf{R}_\psi = \begin{pmatrix} \cos\psi & -\sin\psi & 0 \\ \sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{186}$$

be the rotation by ψ about $\hat{\boldsymbol{z}}$. It preserves lengths, $|\mathsf{R}_\psi \boldsymbol{a}| = |\boldsymbol{a}|$, it commutes with the cross product, $(\mathsf{R}_\psi \boldsymbol{a}) \times (\mathsf{R}_\psi \boldsymbol{b}) = \mathsf{R}_\psi(\boldsymbol{a} \times \boldsymbol{b})$ (true for every proper rotation), and it leaves the vertical unchanged, $\mathsf{R}_\psi \hat{\boldsymbol{z}} = \hat{\boldsymbol{z}}$.

The acceleration is equivariant. In still air the 3D acceleration of `integrate3d` is

$$\boldsymbol{a}(\boldsymbol{v}, \boldsymbol{\omega}) = -q\, C_D\, |\boldsymbol{v}|\, \boldsymbol{v} + q\, C_L(S)\, |\boldsymbol{v}|^2\, \frac{\boldsymbol{\omega} \times \boldsymbol{v}}{|\boldsymbol{\omega} \times \boldsymbol{v}|} - g\, \hat{\boldsymbol{z}}, \qquad q = \frac{\rho \pi R^2}{2m}, \tag{187}$$

with air density ρ (kg/m^3), ball mass m (kg), drag and lift coefficients C_D, C_L (dimensionless), q in 1/m and g in m/s^2. (The code regularises the two denominators with $\varepsilon = 10^{-6}$[A]; this does not affect the argument.) The spin parameter $S = |\boldsymbol{\omega}|R/|\boldsymbol{v}|$ is a ratio of lengths, so it is unchanged by the rotation. Applying the three properties term by term:

$$\boldsymbol{a}(\mathsf{R}_\psi \boldsymbol{v}, \mathsf{R}_\psi \boldsymbol{\omega}) = -qC_D|\boldsymbol{v}|\, \mathsf{R}_\psi \boldsymbol{v} + qC_L|\boldsymbol{v}|^2\, \frac{\mathsf{R}_\psi(\boldsymbol{\omega} \times \boldsymbol{v})}{|\boldsymbol{\omega} \times \boldsymbol{v}|} - g\, \mathsf{R}_\psi \hat{\boldsymbol{z}} = \mathsf{R}_\psi\, \boldsymbol{a}(\boldsymbol{v}, \boldsymbol{\omega}). \tag{188}$$

The spin decays as $\boldsymbol{\omega}(t) = \boldsymbol{\omega}_0 e^{-kt}$ about a fixed axis, with decay rate k (1/s), so the rotated spin stays the rotated spin at all times.

The backspin axis rotates with the shot. The launch direction is $\hat{\boldsymbol{v}}_0 = (\cos\theta\cos\psi,\ \cos\theta\sin\psi,\ \sin\theta)$ and `backspin_axis` takes the unit vector along $\hat{\boldsymbol{v}}_0 \times \hat{\boldsymbol{z}}$:

$$\begin{aligned} \hat{\boldsymbol{v}}_0 \times \hat{\boldsymbol{z}} &= (\cos\theta\sin\psi,\ -\cos\theta\cos\psi,\ 0), \\ \hat{\boldsymbol{e}}_\omega &= (\sin\psi,\ -\cos\psi,\ 0) = \mathsf{R}_\psi(0, -1, 0)^{\mathsf{T}}. \end{aligned} \tag{189}$$

At $\psi = 0$ this is $-\hat{\boldsymbol{y}}$, and $(-\hat{\boldsymbol{y}}) \times (v_x, 0, v_z) = (-v_z, 0, v_x)$, which is the 2D Magnus direction of Section 3.

The path rotates rigidly. Let $\boldsymbol{r}_2(t) = (x(t), 0, z(t))$ be the 2D path launched at $\psi = 0$. By Eqs. (188)–(189), $\mathsf{R}_\psi \boldsymbol{r}_2(t)$ satisfies the same equation of motion as the 3D path launched at ψ, from the rotated initial state. Solutions of an ODE with a Lipschitz right-hand side are unique, so

$$\boldsymbol{r}_3(t) = \mathsf{R}_\psi\, \boldsymbol{r}_2(t) = \big(x(t)\cos\psi,\ x(t)\sin\psi,\ z(t)\big). \tag{190}$$

The HIVE cross-section does not change along the pivot axis y (inside the CELL width), so only the x–z projection matters for the clean test. That projection keeps z and scales the horizontal coordinate and velocity:

$$X(t) = x(t)\cos\psi, \qquad V_X(t) = v_x(t)\cos\psi, \qquad Z(t) = z(t), \qquad V_Z(t) = v_z(t), \tag{191}$$

which is exactly what `clean_masks` applies before the gate test. Wind breaks the argument: with $\boldsymbol{v}_r = \boldsymbol{v} - \boldsymbol{w}$ and a fixed wind $\boldsymbol{w}$ (m/s), $\mathsf{R}_\psi \boldsymbol{w} \neq \boldsymbol{w}$ in general, so the 3D Monte Carlo integrates with wind directly.

Azimuth from the lateral offset. A shot aimed at lateral offset y (m) from the CELL centre reaches the gate plane at horizontal distance $d \pm x_g$, where $x_g = 0.401\,\mathrm{m}^{[\mathsf{D}]}$ is the gate position in the HIVE frame. The robustness maps use

$$\psi = \arctan\frac{|y|}{d \pm x_g}, \qquad s_{\mathrm{eff}} = \min(0.3\tan\psi,\ 0.2)\ \mathrm{m}, \tag{192}$$

with azimuth bins 0° to $60^{\circ\,[\mathsf{A}]}$ in $10^{\circ\,[\mathsf{A}]}$ steps; s_{eff} is an assumed effective sideways shift of the mouth seen at an angle (coefficients $0.3^{[\mathsf{A}]}$ and $0.2\,\mathrm{m}^{[\mathsf{A}]}$).

Lateral factor. The sideways miss at the gate is e_y (m). An azimuth error $\delta_\psi \sim \mathcal{N}(0, \sigma_\psi^2)$ (rad) over a lever arm $L = \sqrt{(d \pm x_g)^2 + y^2}$ (m, the horizontal release-to-gate distance used by `position_prob`; $L = d + x_g$ for an unfavourable shot with $y = 0$) moves the ball by $L\tan\delta_\psi \approx L\delta_\psi$. Adding a lateral release error σ_y (m) and a wind drift σ_{dr} (m), all independent and Gaussian, the variances add:

$$\sigma_{\mathrm{lat}}^2 = (L\sigma_\psi)^2 + \sigma_y^2 + \sigma_{\mathrm{dr}}^2. \tag{193}$$

The ball fits if $|e_y| < w_{\mathrm{lat}}$, with half-width w_{lat} (m). For $e_y \sim \mathcal{N}(0, \sigma_{\mathrm{lat}}^2)$, substituting $u = e/(\sqrt{2}\sigma_{\mathrm{lat}})$,

$$\begin{aligned}\Pr(|e_y| < w_{\mathrm{lat}}) &= \int_{-w_{\mathrm{lat}}}^{w_{\mathrm{lat}}} \frac{e^{-e^2/(2\sigma_{\mathrm{lat}}^2)}}{\sqrt{2\pi}\,\sigma_{\mathrm{lat}}}\,\mathrm{d}e = \frac{2}{\sqrt{\pi}}\int_0^{w_{\mathrm{lat}}/(\sqrt{2}\sigma_{\mathrm{lat}})} e^{-u^2}\mathrm{d}u \\ &= \operatorname{erf}\Big(\frac{w_{\mathrm{lat}}}{\sqrt{2}\,\sigma_{\mathrm{lat}}}\Big).\end{aligned} \tag{194}$$

`position_prob` combines it with the kinematic probability, treating the two as independent:

$$P_{\mathrm{pos}}(d, \psi) = P_{\mathrm{kin}}(d, \psi)\ \operatorname{erf}\Big(\frac{w_{\mathrm{lat}}}{\sqrt{2}\,\sigma_{\mathrm{lat}}}\Big). \tag{195}$$

The code uses $w_{\mathrm{lat}} = 0.10\,\mathrm{m}^{[\mathsf{A}]}$ (low), $0.15\,\mathrm{m}^{[\mathsf{A}]}$ (primary) and $0.2185\,\mathrm{m}^{[\mathsf{D}]}$ (high, the CELL half-width $0.254\,\mathrm{m}^{[\mathsf{C}]}$ minus R).

Worked example. For the 3D path with params_lit nominal aerodynamics, $v = 7.25\,\mathrm{m/s}^{[A]}$, $\theta = 79.7^{\circ\,[A]}$, $S = 1^{[A]}$ and $\psi = 30^{\circ\,[A]}$, the backspin axis is $(0.5, -0.866, 0)^{[D]}$. Comparing `integrate3d` with the 2D solution rotated by Eq. (190), the largest differences are $1.1 \times 10^{-7}\,\mathrm{m}^{[P]}$ in x, $1.1 \times 10^{-16}\,\mathrm{m}^{[P]}$ in y and $0^{[P]}$ in z: the projection is exact to round-off. On this path the apex is $2.339\,\mathrm{m}^{[P]}$ above the release point; there $v_z = 0$ but the ball still moves horizontally at $v_x = 0.903\,\mathrm{m/s}^{[P]}$, so it is not at rest at the apex.

For the lateral factor take $d = 0.81\,\mathrm{m}^{[A]}$, so $L = 0.81 + 0.401 = 1.211\,\mathrm{m}^{[D]}$, and the nominal $\sigma_\psi = 0.5^\circ = 0.00873\,\mathrm{rad}^{[A]}$ and $\sigma_y = 0.025\,\mathrm{m}^{[A]}$. Then $L\sigma_\psi = 0.0106\,\mathrm{m}^{[D]}$ and, without wind, $\sigma_{\mathrm{lat}} = \sqrt{0.0106^2 + 0.025^2} = 0.0271\,\mathrm{m}^{[D]}$; the factor is $1.000^{[D]}$ for $w_{\mathrm{lat}} = 0.15\,\mathrm{m}^{[A]}$. With the wind drift $\sigma_{\mathrm{dr}} = 0.0615\,\mathrm{m}^{[D]}$ from `wind_drift_sd` (module-default coefficients `data/aero_params.json`; $0.0634\,\mathrm{m}^{[D]}$ with `research/params_lit.json`), $\sigma_{\mathrm{lat}} = 0.0672\,\mathrm{m}^{[D]}$ and the factor drops to $\mathrm{erf}(0.15/(\sqrt{2}\cdot 0.0672)) = 0.974^{[D]}$, or $0.863^{[D]}$ for $w_{\mathrm{lat}} = 0.10\,\mathrm{m}^{[A]}$. Sideways aim is a small loss next to the in-plane window.

Assumptions.

- Still air for the projection; wind enters only through σ_{dr} and the 3D Monte Carlo.
- The HIVE section is uniform along the pivot axis across the CELL width.
- Lateral and in-plane outcomes are independent in Eq. (195).
- Azimuth bins stop at $60^{\circ\,[A]}$ (`PSI_MAX`).

6.6 Three-dimensional Monte Carlo

Motivation. The blur of Section 6.4 covers release errors in one aerodynamic condition. A real match mixes everything at once: release errors, ball-to-ball spread, unknown coefficients, air movement and the HIVE pose. `mc_shots` draws all of these per shot and flies each shot in 3D.

Per-shot sampling. Each shot n draws independent standard normals $\varepsilon_1, \varepsilon_2, \cdots \sim \mathcal{N}(0,1)$. The release state is

$$v = v_{\mathrm{aim}}(1 + \sigma_{v,\mathrm{rel}}\varepsilon_1)(1 + b) - \frac{\partial v}{\partial n_w}\,\Delta n_w\, f, \tag{196}$$

$$\theta = \theta_{\mathrm{aim}} + \sigma_\theta \varepsilon_2, \qquad \psi = \sigma_\psi \varepsilon_3, \tag{197}$$

$$\boldsymbol{r}_0 = \big(-(d + \sigma_d\varepsilon_4),\ \sigma_y\varepsilon_5,\ h + \sigma_h\varepsilon_6\big), \tag{198}$$

where b (dimensionless) is a uniform speed bias used by the battery-sag scenarios, $\partial v/\partial n_w$ (m/s per rpm) is the launcher-curve slope, Δn_w (rpm) the wheel-speed drop after a shot and f a uniform recovery-deficit fraction for the rapid-fire scenarios; in the nominal scenario $b = f = 0$. The release-height error is $\sigma_h = 0.005\,\mathrm{m}^{[A]}$. The ball and the air are drawn as

$$\begin{aligned} m &= \mathrm{clip}(\bar{m} + \sigma_m\varepsilon_7,\ m_{\mathrm{lo}}, m_{\mathrm{hi}}), & R &= \mathrm{clip}(\bar{R} + \sigma_R\varepsilon_8,\ R_{\mathrm{lo}}, R_{\mathrm{hi}}), \\ C_D &= \bar{C}_D(1 + 0.05\,\varepsilon_9), & (c_1, C_{L,\max}) &= (\bar{c}_1, \bar{C}_{L,\max})(1 + 0.10\,\varepsilon_{10}), \\ \boldsymbol{w} &= \sigma_w(\varepsilon_{11}, \varepsilon_{12}, 0), & S &\sim \mathcal{U}(S_{\mathrm{lo}}, S_{\mathrm{hi}}), \end{aligned} \tag{199}$$

with $\sigma_m = 0.5\,\mathrm{g}^{[A]}$, $\sigma_R = 0.25\,\mathrm{mm}^{[A]}$, the coefficient spreads $5\,\%^{[A]}$ and $10\,\%^{[A]}$ (one factor shared by c_1 and $C_{L,\max}$, so S_{sat} does not move), a wind of $\sigma_w = 0.30\,\mathrm{m/s}^{[A]}$ per horizontal component held constant during the shot, and a spin parameter uniform on $[1.03, 1.27]^{[P]}$ (the contact model's

hood band); the release-error-only scenario fixes $S = 1.0597^{[P]}$. The HIVE is drawn per shot with tilt $\beta = \mathrm{clip}(30^\circ + 0.75^\circ\varepsilon_{13},\ 28.5^\circ, 31.5^\circ)$ (spread $0.75^{\circ[A]}$, clipped to the derived tilt bounds), pivot height $1.11633\,\mathrm{m}^{[C]}$ with spread $0.005\,\mathrm{m}^{[A]}$ clipped to 1.09093–1.14173 $\mathrm{m}^{[C]}$ (the manual's ± 1 in field tolerance), and an x offset of spread $0.01\,\mathrm{m}^{[A]}$. The initial spin is backspin about the rotated axis of Eq. (189):

$$\boldsymbol{\omega}_0 = \hat{\boldsymbol{e}}_\omega(\psi)\,\frac{S\,v}{R}. \tag{200}$$

The path is integrated with Eq. (187) using the relative velocity $\boldsymbol{v}_r = \boldsymbol{v} - \boldsymbol{w}$.

Scoring one shot. Where the path (with x measured from the drawn HIVE position) last crosses the gate plane $x = x_g$ between samples k and $k+1$, the lateral position is interpolated linearly:

$$y_g = y_k + (y_{k+1} - y_k)\,\frac{x_g - x_k}{x_{k+1} - x_k}. \tag{201}$$

If $|y_g| > w_{\mathrm{cell}} - R$, with CELL half-width $w_{\mathrm{cell}} = 0.254\,\mathrm{m}^{[C]}$, the shot hits a side wall (or misses when it never crosses the plane). Otherwise a HIVE with the drawn tilt, pivot height and ball radius, offset sideways by $|y_g|$, runs `classify_path` on the in-plane path. The shot counts as clean only if the outcome is `clean`. Out of n shots, k are clean, and the point estimate is

$$\hat{p} = \frac{k}{n}. \tag{202}$$

Worked example. The stored run `data/robust_mc.csv` aims at $d = 0.81\,\mathrm{m}^{[P]}$, $v = 7.25\,\mathrm{m/s}^{[P]}$, $\theta = 79.7^{\circ[P]}$ with backspin and $n = 1500^{[A]}$ shots per scenario. Its nominal scenario has $k = 455^{[P]}$, $\hat{p} = 0.303^{[P]}$, Wilson interval $[0.281, 0.327]^{[D]}$, and a gate-lateral spread of $0.049\,\mathrm{m}^{[P]}$. The other scenarios (all at $n = 1500^{[A]}$):

- release errors only (fixed physics, no wind): $\hat{p} = 0.362^{[P]}$;
- tight release errors: $0.411^{[P]}$; loose: $0.157^{[P]}$;
- wind $\sigma_w = 1.0\,\mathrm{m/s}^{[A]}$: $0.111^{[P]}$;
- true lift at the low or high bound while the aim was tuned for nominal lift: $0.097^{[P]}$ and $0.054^{[P]}$; re-tuned for low lift ($\theta = 80.7^{\circ[P]}$): $0.325^{[P]}$;
- the no-spin twin aimed at $81.4^{\circ[P]}$: $0.317^{[P]}$ $[0.294, 0.341]^{[D]}$;
- the favourable side for comparison: $0.996^{[P]}$.

The spin and no-spin intervals overlap, so at this aim backspin gives no measurable gain. Using the wrong lift coefficient costs far more than the release scatter, which is why the specification relies on tuning at the field. A fresh run of $400^{[A]}$ shots with seed $1000^{[A]}$ and the same code gives $k = 110^{[P]}$, $\hat{p} = 0.275^{[P]}$, Wilson $[0.234, 0.321]^{[D]}$; it overlaps the stored result. The two most common failures there are the outer roof ($147^{[P]}$ shots) and the floor ($62^{[P]}$). These Monte Carlo results were computed with the older first-pass coefficient file `data/aero_params.json`; a rerun with `research/params_lit.json` has not been made.

Assumptions.

- All spreads in Eqs. (196)–(199) are assumed, independent and Gaussian or uniform; none is measured on the robot.
- The wind is constant during one shot and horizontal.
- The aerodynamic coefficients are literature bounds, not POLLEN measurements.
- Shots are independent trials, which Section 6.7 needs.

6.7 How sure is a success rate? The Wilson interval

Motivation. A rate of k clean shots out of n is itself random. The simple interval $\hat{p} \pm z\sqrt{\hat{p}(1-\hat{p})/n}$ fails badly for small n or for $\hat{p}$ near 0 or 1: for one clean shot out of one it gives $[1, 1]$. `wilson` uses the score interval instead, which is the right tool for field tests.

Derivation. For n independent shots with true probability p, k is binomial, and for moderate n

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \approx \mathcal{N}(0, 1). \tag{203}$$

The Wilson interval is the set of p for which this score is within $\pm z$, where $z = \Phi^{-1}(0.975) = 1.959964^{[D]}$ for 95 %. Note that the variance uses the unknown p, not $\hat{p}$. The boundary satisfies

$$(\hat{p} - p)^2 = \frac{z^2}{n}\, p(1-p). \tag{204}$$

Expanding both sides and collecting powers of p gives a quadratic:

$$\Big(1 + \frac{z^2}{n}\Big)p^2 - \Big(2\hat{p} + \frac{z^2}{n}\Big)p + \hat{p}^2 = 0. \tag{205}$$

Its discriminant simplifies as

$$\Big(2\hat{p} + \frac{z^2}{n}\Big)^2 - 4\Big(1 + \frac{z^2}{n}\Big)\hat{p}^2 = \frac{4z^2}{n}\hat{p}(1-\hat{p}) + \frac{z^4}{n^2} = 4z^2\Big(\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}\Big), \tag{206}$$

and the two roots, divided top and bottom by 2, are the interval:

$$p_{\pm} = \frac{\hat{p} + \dfrac{z^2}{2n} \pm z\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}. \tag{207}$$

The code clips both ends to $[0, 1]$. The centre $(\hat{p} + z^2/2n)/(1 + z^2/n)$ is pulled toward $1/2$, and the width never collapses to zero, even at $\hat{p} = 0$ or 1.

Worked example. For $k = 455^{[P]}$ of $n = 1500^{[A]}$: $\hat{p} = 0.3033^{[D]}$, $z^2/n = 0.00256^{[D]}$, and Eq. (207) gives $[0.2806, 0.3271]^{[D]}$. For $\hat{p} = 0.5$ the half-width is $0.263^{[D]}$ at $n = 10$, $0.201^{[D]}$ at $n = 20$, $0.096^{[D]}$ at $n = 100$ and $0.025^{[D]}$ at $n = 1500$. One clean shot out of one gives $[0.207, 1]^{[D]}$: a single shot proves only that the clean rate is probably above about one in five, and says nothing about repeatability. Zero clean shots out of 20 still allows a rate up to $0.161^{[D]}$. A field test that should separate a 30 % shooter from a 50 % one therefore needs on the order of 100 shots per setting, not a handful.

Assumptions.

- Shots are independent with a common probability (no warm-up drift).
- The normal approximation in Eq. (203); the Wilson form keeps good coverage down to small n.

6.8 From exit speed to wheel and ball rpm

Motivation. The CAD designer chooses a motor and gear ratio, so the speed requirement must be stated as a wheel rpm, and the spin requirement as a ball rpm.

Wheel rpm. A wheel of diameter D (m) turning at n_w (rpm) has surface speed $u = \pi D\, n_w/60$ (m/s). The exit speed of the ball is a fraction $r = v/u$ (dimensionless) of the surface speed. Solving for the wheel speed:

$$n_w = \frac{60\, v}{r\, \pi D}. \tag{208}$$

This is `wheel_rpm`. The ratio r is not universal. With a single powered wheel against a fixed hood the ball rolls on the hood, and the simple rolling picture gives $r = 1/2$; slip, compression and a moving upper roller change it. The code tabulates $r \in \{0.25, 0.30, 0.42, 0.50\}^{[A]}$ (grade C), and the contact model of the previous section is the place where r is predicted.

Ball rpm. From $S = \omega R/v$, the spin rate is $\omega = Sv/R$ (rad/s); converting with $1\,\text{rev} = 2\pi\,\text{rad}$ and $1\,\text{min} = 60\,\text{s}$,

$$n_b = \frac{S\, v}{R} \cdot \frac{60}{2\pi}. \tag{209}$$

This is `spin_rpm`. There is no point in asking for spin far above S_{sat}: the lift coefficient saturates and $S = 1.5$ gives the same window as $S = 1.0$ (Section 6.1).

Worked example. With $D = 0.098\,\text{m}^{[U]}$ and $R = 0.0355\,\text{m}^{[C]}$ (the manual's ball diameter of about 7.1 cm):

- $v = 6.5\,\text{m/s}^{[P]}$: $n_w = 60 \times 6.5/(0.25\,\pi\,0.098) = 5067^{[D]}$ rpm at $r = 0.25^{[A]}$, and $4222^{[D]}$, $3016^{[D]}$, $2533^{[D]}$ rpm at $r = 0.30$, 0.42, 0.50.
- $v = 9.0\,\text{m/s}^{[P]}$: $7016^{[D]}$, $5847^{[D]}$, $4176^{[D]}$, $3508^{[D]}$ rpm.
- $v = 7.0\,\text{m/s}^{[P]}$: $5457^{[D]}$, $4547^{[D]}$, $3248^{[D]}$, $2728^{[D]}$ rpm.
- Ball spin: $S = 0.5^{[A]}$ at 6.5 m/s is $n_b = 0.5 \times 6.5/0.0355 \times 60/(2\pi) = 874^{[D]}$ rpm; $S = 1^{[A]}$ at 7.0 m/s is $1883^{[D]}$ rpm and at 9.0 m/s is $2421^{[D]}$ rpm.

The full exit-speed band 6.5–9.0 m/s therefore asks for anything from about 2500 to 7000 rpm at the wheel depending on r; measuring r on the first prototype narrows this to one gear choice.

Assumptions.

- r is constant over the speed range for a given shooter build.
- The wheel speed does not drop during the shot (the drop is a Monte Carlo scenario, Eq. (196)).

6.9 The shooter specification card

Motivation. The final step collects the results above into the few numbers the CAD designer needs. Each target traces back to an equation and a worked example of this section. All of them are model outputs.

Table 18: CAD-card targets for the unfavourable-side shot (single powered 98 mm wheel, fixed hood). All targets are model predictions to be confirmed on the field.

Quantity	Target	Where it comes from
Release height h	0.65–0.737 m[P]	Table 17; window shrinks below 0.65 m[P]
Distance d	0.75–0.85 m[P]	no clean cell at $d \leq 0.70$ m[P]; peak near 0.80 m[P]
Hood angle, no spin	about 81°[P]	Eq. (179): 80.8°[P] at $\tau = 0.2°$[A]
Hood angle, backspin	80–80.5°[P]	Eq. (179) at $\tau = 0.1°$[A]; nominal band hood
Exit speed range	6.5–9.0 m/s[P]	every condition has a clean speed (Sec. 6.3)
Speed repeatability	±0.075 m/s[P]	Eq. (174), $k^* = 4$[P]
Angle repeatability	±0.15°[P]	Eq. (174), $k^* = 4$[P]
Wheel speed	about 2500–7000 rpm[P]	Eq. (208), depends on r

How to read the card.

- *Height and distance.* The window opens at $d \approx 0.75$ m[P] and is largest near 0.80 m[P]; beyond 0.85 m it shrinks slowly. Lower release heights shrink it quickly (Table 17 and the h comparison in Section 6.1).
- *Hood angle.* The hood is chosen so that, for every drag, lift and tilt corner, some tuned wheel speed is clean (Eq. (178)). The backspin hood is slightly flatter than the no-spin hood, but backspin does not enlarge the window and its hood needs a tighter angle tolerance (±0.1° instead of ±0.2°).
- *Exit speed.* The range is a tuning range, not a tolerance: the team sets one speed at the field and keeps it.
- *Repeatability.* The ±0.075 m/s and ±0.15° figures are the largest clean square anywhere in the sweep (Section 6.2) for one set of coefficients. They are a target for shot-to-shot scatter, roughly the tight release set of the Monte Carlo ($\sigma_\theta = 0.15°$[A]), which raises the modelled clean rate from 0.30[P] to 0.41[P]. They are not a guarantee of clean entry under every condition.

Assumptions.

- A single powered wheel with a fixed hood (an optional passive upper roller changes r and the spin, not the geometry of the window).
- The coefficient ranges of `research/params_lit.json` for the spec grid; the Monte Carlo rates were computed with `data/aero_params.json`.

- No claim here is based on a single test shot; field confirmation needs a Wilson interval from many shots (Section 6.7).

7 Whole-field model, live solver and missing effects

Sections 2–6 treat one shot straight down the HIVE tipping plane. This section describes how the same flight model is used for a release point anywhere on the field (the explorer's Robot view, `viz/explorer/solver.js`, with a Python copy in `src/`), which effects were added on top of it, and which effects are still missing. Every window here is model output[P] with the literature aerodynamics[A]; none of it is a hit rate.

7.1 Whole-field oblique model

The release point is (x, y) in the field frame of the HIVE section: x along the shot axis with the pivot plane at $x = 0$, the far (unfavourable) side at $x < 0$, and the up-cell gate centre at $(x_g, 0)$ with $x_g = 0.401\,\mathrm{m}$[D]. The shot is aimed at the gate centre. With $d = |x|$ the distance to the pivot plane,

$$L_h = \begin{cases} d + x_g & \text{far side,} \\ \max(d - x_g,\, 10^{-3}\,\mathrm{m}) & \text{near side,} \end{cases} \qquad \psi = \arctan\frac{|y|}{L_h}, \tag{210}$$

where ψ is the bearing off the HIVE centre line[D]. The ball flies in the vertical plane of that bearing with its spin axis perpendicular to it, so there is no sideways Magnus force. Its in-plane path is the planar path of Section 3 with the horizontal coordinate and velocity projected on the shot axis,

$$X_{\mathrm{axis}}(t) = X(t)\cos\psi, \qquad V_{X,\mathrm{axis}}(t) = V_X(t)\cos\psi, \tag{211}$$

and the lateral offset of the ball centre from the cell centre line is $s(t) = (x_g - x_{\mathrm{axis}}(t))\tan\psi$. The projection is exact for walls extruded along y and still air. Bearings above 60°[A] are flagged as extrapolated.

7.2 The 3D cell: side walls and pentagon roof

The precomputed maps use the centre section at one assumed lateral offset per bearing and do not model the side walls. The live solver instead tests the ball-centre path (x, y, z) against the cell as 3D faces inflated by $R = 0.0355\,\mathrm{m}$[C]: the centre section extruded across the cell, side walls at $\pm 0.254\,\mathrm{m}$[C] with inner face, rim and outer face, wall thickness $0.0508\,\mathrm{m}$[A], and a pentagon roof whose ridge drops linearly towards the side walls (from the opening profile $0.3556 \to 0.19329\,\mathrm{m}$ over $0.254\,\mathrm{m}$[D]). The clearance is the exact segment-to-polygon distance minus R. A shot is clean when the centre enters the cell interior before any face comes closer than R.

At $\psi = 0$ with the ridge off the 3D test reproduces the section test exactly (2500 shots per side). With the ridge on it is slightly stricter on the centre line, because the roof is lower off it. Off the centre line the 3D cell is usually more permissive than the old single-offset section: at $d = 1.0\,\mathrm{m}$, $h = 0.5\,\mathrm{m}$, $S = 1.0$ the far-side clean grid cells go from 306 to 301 at $\psi = 0$, from 212 to 309 at 20° and from 164 to 362 at 40°[P] (no-spin twin $348 \to 346$, $253 \to 347$, $186 \to 401$[P]). These are counts on a coarse search grid, not probabilities.

7.3 Live solver: knife-edge and tolerant minima

For a continuous release point and height the solver flies a coarse grid (exit speed 2–16 m/s in 0.1 m/s steps, hood angle in 0.1° far-side or 0.25° near-side steps) and then refines near the minima. It reports two kinds of minimum:

- **Knife edge** (`minRpm`, `minAngle`): the lowest clean exit speed or hood angle the search found. These are grazing shots. Their clearance is typically 0–0.5 mm[P], and the value moves with the resolution because the window edge carries about 0.1° of path-sampling noise at the 2 ms step. They are shown, but a robot must not be set there.
- **Tolerant** (`minRpmTol`, `minAngleTol`): the lowest speed whose continuous clean hood-angle run is at least 0.2°[A] wide, and the lowest hood angle whose continuous clean speed run is at least 0.1 m/s[A] wide, with clearance $\geq$ 2 mm[A] at every sample. The most forgiving shot is the centre of the largest all-clean square on the search grid.

Every result is solved twice: with the chosen backspin and as the no-spin twin ($S = 0$, no Magnus term), with the same air and drag settings. The fused integrate-and-classify fast path is checked against the reference integrator and classifier for equal class and clearance to 10^{-12} m.

At the explorer's default robot spot ($x = -1.0$, $y = 0.2$ m[U], $h = 0.737$ m[U], $\psi = 8.1°$[D], $S = 1.0$[A], $v/u = 0.42$[A], nominal aerodynamics) the solver gives

	backspin $S = 1.0$	no-spin twin $S = 0$
knife-edge min. rpm	2677 rpm at 71.74°[P]	2682 rpm at 72.86°[P]
tolerant min. rpm	2784 rpm at 73.30°[P]	2784 rpm at 74.47°[P]
tolerant min. hood angle	75.80° at 3023 rpm[P]	77.30° at 3034 rpm[P]
most forgiving	4269 rpm at 80.8°[P]	4872 rpm at 84.7°[P]

The rpm values are derived from the exit speed with the assumed grip and a 98 mm[U] wheel. The knife-edge clearance at the minimum rpm is 0.63 mm[P]. At this spot backspin does not lower the tolerant minimum wheel speed; it lowers the hood angle at that speed by about 1.2°[P]. More spin is not always better: the lift saturates (Section 3).

7.4 Robot and turret view

The Robot view places a 0.457 m[C] robot anywhere on the field at any heading. The release point is the robot centre at the chosen height[A], and a turret aims at the cell centre exactly[A]. The page reports the turret yaw relative to the robot front and, if the chosen turret travel (360°, ±180°, ±90°, ±45° or fixed) cannot reach that yaw, how far to turn the robot. Poses inside the A-frame base are illegal; the second HIVE is only flagged as an obstacle. The recommended setting is the most forgiving shot, unless it needs more exit speed than the grip limit below. In that case the lowest tolerant wheel speed is recommended. At the default spot the most forgiving shot (4269 rpm, about 9.2 m/s[P]) is above the limit, so the page recommends 2784 rpm at 73.3°[P] (twin 2784 rpm at 74.5°[P]).

7.5 Venue air, drag options, drafts and the grip limit

These were added from the ranked list in `research/MISSING_EFFECTS.md` because they need no new measurement. Python (`src/air.py`, `src/feasibility.py`) and JavaScript (`physics.js`, `solver.js`) carry the same formulas. Tests compare trajectories from the two at 2×10^{-6} relative, which is the float32 storage limit.

Air density. Simplified CIPM-2007 with a Magnus-form vapour pressure and a US Standard Atmosphere pressure from altitude,

$$\rho = \frac{p - 0.378\,e}{R_d\,T}, \quad e = \phi\,611.2 \exp\frac{17.62\,t}{243.12 + t}, \quad p = 101325\,(1 - 2.25577 \times 10^{-5}\,z)^{5.25588}, \quad (212)$$

with $R_d = 287.06\,\mathrm{J/(kg\,K)}$, t in °C, $T = t + 273.15$, humidity ϕ and altitude z in m (user inputs). Viscosity follows Sutherland's law. Still air at 20°C, 101.325 kPa, dry gives $\rho = 1.2041\,\mathrm{kg/m^3}$[D] and $\nu = 1.506 \times 10^{-5}\,\mathrm{m^2/s}$[D]. This is now the live solver's default instead of the rounded 1.20[A] used by the precomputed maps. At the default spot the change moves the tolerant minimum from 2793 rpm at 73.43° to 2784 rpm at 73.30°[P]. The twin stays at 2784 rpm at 74.47°[P] to the precision shown.

Drag options. C_d stays the constant literature value by default. Two alternatives are available. The first is the smooth-sphere correlation $C_d(\mathrm{Re})$ of Morrison (2013), a reference curve for a ball without holes and not a POLLEN fact. The second is a what-if partial drag crisis that falls from the constant C_d to 0.33[A] with a smoothstep between $\mathrm{Re} = 3 \times 10^4$ and 5×10^4[A]. Backspin can add drag:

$$C_d \leftarrow C_d + \Delta C_d\,\min(S/0.6,\,1), \qquad \Delta C_d \in \{0,\,0.08,\,0.18\}\text{[A]}. \quad (213)$$

The no-spin twin has $S = 0$ and never gets this term.

Draft. A steady draft $\boldsymbol{w} = (w_\parallel, w_z)$ in the shot plane (user input) enters through the relative velocity $\boldsymbol{v}_{\mathrm{rel}} = \boldsymbol{v} - \boldsymbol{w}$. Drag, Magnus force and S all use it. A sideways part is an option on the 3D path (Section 7.6).

Grip limit. An answer that needs an exit speed above 8.9 m/s[P] (the hood launcher model of Section 4) is flagged, not removed. Above that speed, v/u and S are uncertain.

Effect at the default spot (tolerant minimum rpm and its hood angle, twin in brackets; all [P]):

option	backspin $S = 1.0$	twin $S = 0$
still air (default)	2784 rpm, 73.30°	2784 rpm, 74.47°
Morrison $C_d(\mathrm{Re})$	2775 rpm, 73.64°	2765 rpm
what-if drag drop	as still air ($\mathrm{Re} < 3 \times 10^4$ here)	as still air
spin drag $\Delta C_d = 0.18$	2821 rpm, 72.98°	2784 rpm (no term)
tail draft 0.3 m/s	2803 rpm, 73.84°	2784 rpm, 74.8°
updraft 0.3 m/s	2793 rpm, 73.84°	2765 rpm
altitude 1600 m ($\rho = 0.9925$[D])	2775 rpm, 73.77°	2765 rpm, 74.61°

No tolerant minimum at this spot is grip-limited. Each option moves the hood angle by up to about 0.5°[P], which is wider than the 0.2°[A] tolerance. A venue therefore needs its own aim check.

7.6 Shooting on the move, hit chance and lift options

These options come from the same ranked list (`research/MISSING_EFFECTS.md`, build-next 2a–2f). All are off by default, so the default answer above is unchanged. Python (`src/robustness.py`, `src/air.py`, `src/flight.py`) and JavaScript (`solver.js`, `physics.js`) agree on launch states to 10^{-12}, on 3D paths to 10^{-9} relative and on every Monte Carlo count (`tests/test_move.py`, `tests/js_move_check.js`).

Moving robot. The robot's ground velocity at the exit point is $\boldsymbol{v}_r = a\,\hat{\boldsymbol{u}} + c\,\hat{\boldsymbol{n}} + \omega_z \hat{\boldsymbol{z}} \times \boldsymbol{r}_{\text{exit}}$, with $\hat{\boldsymbol{u}}$ the horizontal aim direction, $\hat{\boldsymbol{n}} = \hat{\boldsymbol{z}} \times \hat{\boldsymbol{u}}$, and the speeds a, c and turn rate ω_z user inputs (field-frame v_x, v_y are also accepted). The ball leaves with

$$\boldsymbol{v}_0 = \boldsymbol{v}_{\text{shooter}}(v,\ \theta_e) + \boldsymbol{v}_r, \qquad \theta_e = \theta_{\text{hood}} + \delta_0, \tag{214}$$

where δ_0 is an exit-angle offset ([A], default 0). With lead on (default) the shooter turns so the sideways part of $\boldsymbol{v}_0$ cancels, which needs $|\boldsymbol{v}_r \cdot \hat{\boldsymbol{n}}| \le v \cos\theta_e$. The backspin axis is horizontal and normal to the shooter direction, plus ω_z about the vertical; the no-spin twin has no backspin. The flight is then 3D in the 3D cell, with a sideways draft allowed. The solver also reports how far the centre of the clean run moves per 0.1 m/s toward the goal and sideways.

Hit chance. At one setting the solver fires $n = 1000$ shots per aero set (lift low, nominal and high) with ASSUMED Gaussian launch scatter: speed 1.5 %, elevation 0.40°, heading 0.50° (1σ, the nominal set of Section 6.6). All sets and the twin use the same seeded draws. The result is k/n with a Wilson 95 % interval (Section 6.7). It is a model probability under assumed scatter, not a measured repeatability. At the default spot and the recommended 2784 rpm, 73.3°: nominal 0.124 [0.105, 0.146], lift low 0.092, lift high 0.074; the twin at its own setting (2784 rpm, 74.47°) about 0.13 (all [P]). Backspin does not raise the chance here.

Lift options. Gyro-correct spin ratio $S_\perp = |\boldsymbol{\omega} \times \boldsymbol{v}_{\text{rel}}| R / |\boldsymbol{v}_{\text{rel}}|^2$ on the 3D path (equal to S for pure backspin, tested). A user table $C_l(S)$, linear between points and flat after the last. Reverse Magnus: when $\text{Re} < 4.6 \times 10^4$ and $S < 0.5$ the lift blends (smoothstep near both bounds) to $-C_{l,\text{rev}} \min(S/0.1, 1)$, $C_{l,\text{rev}} \in [0, 0.1]$[A] (smooth-sphere data of Kim et al. 2014 and Kray et al. 2012, not POLLEN).

Effective mass. $\varepsilon = (0.5 + \phi)\rho V/m$ with $\phi \in [0, 1]$[A] the share of the air inside the ball that moves with it, and $\dot{\boldsymbol{v}} = (\boldsymbol{F}_{\text{aero}}/m + \boldsymbol{g})/(1+\varepsilon)$. For POLLEN $\varepsilon = 0.0045$ to 0.0136[D]. Default $\varepsilon = 0$.

Effect at the default spot (tolerant minimum rpm and hood angle; [P]):

option	backspin $S = 1.0$	no-spin twin
none (default)	2784 rpm, 73.30°	2784 rpm, 74.47°
moving, all speeds 0	same	same
0.5 m/s toward the goal	2738 rpm, 78.09°	2719 rpm, 79.01°
0.3 m/s sideways, lead	2812 rpm, 73.41°	2784 rpm, 74.15°
0.3 m/s sideways, no lead	no clean shot	no clean shot
turning 90°/s	2793 rpm, 73.42°	as default
$\delta_0 = +1°$	2784 rpm, 72.30°	2784 rpm, 73.47°
sideways draft 0.3 m/s	2803 rpm, 73.54°	2793 rpm, 74.61°
C_l table $(0.5, 0.2), (1.5, 0.35)$	2803 rpm, 73.25°	as default
$\phi = 1$	2765 rpm, 73.34°	2756 rpm, 74.36°

Driving toward the goal moves the hood angle by about 1° per 0.1 m/s[P], several times the ASSUMED aim tolerance. Sideways motion with lead halves the backspin window (315 to 158 clean pairs, twin 364 to 347), because the spin axis follows the turned shooter. Reverse Magnus does not act here ($S = 1$).

7.7 Hit-chance guidance and the measurement protocol

Best-chance setting. The lowest-rpm setting above sits near the edge of the clean window, so the page now recommends the clean setting with the highest model chance to go in instead (`solver.js bestChance`, replayed in `src/robustness.py best_chance`). The candidates are clean cells inside the PREDICTED grip limit (8.9 m/s), ranked by their share of clean neighbours under a Gaussian with the scatter as σ and picked at least one σ apart (up to 24), plus the tolerant minima. Stage 1 scores each with 64 shots per air set; at most 5 whose pooled Wilson upper bound reaches the best lower bound go to stage 2 with 400 shots. All candidates in a stage see the same seeded draws (common random numbers). The winner is scored again on 1000 fresh shots per air set, so the reported chance is not biased by the selection. The score is the mean over the three air sets (lift low, nominal, high), equal weight[A]. The no-spin twin is searched on its own grid, and one scatter source at a time names the one that matters most.

Default spot ($x = -1.0$, $y = 0.2$ m, $h = 0.737$ m[U], $S = 1.0$[U], grip 0.42[A], scatter 1.5 %, 0.40°, 0.50°[A]; all chances [P], Wilson 95 % ranges):

setting	lift low	nominal	lift high	mean
best chance, 3062 rpm[D], 76.2°	0.207	0.295 [0.268, 0.324]	0.064	0.189
lowest rpm, 2784 rpm[D], 73.3°	0.092	0.124 [0.105, 0.146]	0.074	0.097
no-spin twin at its best, 3851 rpm[D], 82.1°	0.375	[0.346, 0.405] (no lift: one set)		0.375
no-spin twin at its lowest rpm, 2784 rpm, 74.47°	0.132	[0.112, 0.154]		0.132

Here the hood angle is the scatter source that matters most for both shots: with only one source on, the nominal chance is 0.46 (speed only), 0.40 (angle only) and 1.00 (aim only)[P]; for the twin 0.74, 0.44 and 1.00[P]. At 3 % speed scatter[U] the best setting moves to 3341 rpm, 78.0° (nominal 0.258), speed matters most, and the twin's best is 3990 rpm, 82.6° (0.290) (all [P]). At this spot the no-spin twin has the higher chance: more spin is not always better. Over the lift range the backspin chance moves by a factor of almost five (0.064 to 0.295), so the measured C_l decides whether backspin helps here at all.

What the chance is not. It is a model probability under ASSUMED scatter and ASSUMED air sets, not a measured repeatability. It becomes a prediction a team can test only after the scatter itself is measured (the standard deviation of exit speed and elevation over about 20 shots at one setting, logged with wheel speed and battery voltage) and the hit rate is counted with a Wilson interval from many shots (Section 6.7).

Measurement protocol. `docs/MEASURE.md` and the browser kit `viz/measure/measure.html` (linked from the explorer) turn the two largest unknowns into numbers with one phone camera at 240 fps or faster. The ball centre is clicked frame by frame and the track is fitted with the same RK4 flight model as `physics.js` (Python `src/fit_aero.py` agrees with the page to 5×10^{-9}).

- **Drop test** (no launcher, no spin): C_d at zero spin. With one 4 m drop and 4 mm click noise the 95 % error is about ±0.008, and about ±0.085 from 1.5 m[D].
- **Twin test:** shot A with no spin gives v_0, the launch angle and C_d; shot B, the same setting with backspin, holds A's C_d and gives C_l beside its no-spin twin. On synthetic `physics.js` tracks (truth ASSUMED, $C_d = 0.50$, $C_l = 0.25$) one pair gives sd 0.008 in C_d and 0.010 in C_l at 4 mm noise and 0.7 s tracked[D]. These are synthetic checks, not POLLEN data.

- **Length scale.** The scale bar is the largest error: +1 % in scale moves C_l by about 19 %[D]. Shot A also fits the scale against gravity (sd about 0.3 %[D]), and the fit can use either scale.
- **Exit speed and spin:** wheel rpm with the video gives the grip v/u; counting turns of a striped ball gives S at the first tracked frame.

The results go into the explorer's Advanced settings (nearest air preset and grip), and the best-chance search then runs on measured rather than assumed inputs. Until then every chance above stays a model prediction.

7.8 Missing effects and how to measure them

`research/MISSING_EFFECTS.md` ranks 32 effects by how far they move the answer. The largest ones that are still not in the model, or are in it only as bracket sets, need measurements rather than code:

effect	how to measure
Spin lift level $C_l(S)$ (slope and cap)	side video at 240–480 fps, wheel above vs below at equal rpm, taped-hole ball (T1)
Drag level C_d	the same flight video and a drop test (T1)
Grip-limited exit speed, slip v/u, ball squeeze	exit video with a wheel encoder (T2); mass, diameter, squeeze (T0/T4)
Hood angle $\neq$ exit angle, exit scatter	exit video over a hood-angle sweep (T2)
Knuckle force on a low-spin holed ball	shot scatter with and without spin (T1)
Exit skew and spin-axis tilt (sideways error)	axle alignment check; spin axis from video
Balls and NECTAR in the cell, restitution, TIP mid-volley	drop tests on skin, rib and NECTAR; cell tests (T5/T6)
Effective mass (an ASSUMED option only)	absorbed by tuning to real shots; mass and volume (T0)
Flywheel droop, battery; real scatter while moving	burst tests with encoder and voltage logs; shots on the move
Venue drafts and air state	air speed at robot height, temperature, humidity, station pressure

Until those numbers exist, the model's job is to say which measurement changes the answer most. It does not predict a scoring rate.

8 Limitations and what to measure

Every result above is a prediction of a model whose most important inputs have never been measured on POLLEN. The model is useful for deciding *what to build and what to measure first*; it is not evidence that the shot works. The prototype test plan in `docs/07_test_plan.md` turns the list below into concrete tests with pass and fail lines.

- **Shot scatter.** The clean window on the unfavourable side is narrow: the largest clean square found is about $\pm 0.075\,\mathrm{m/s}$[P] by $\pm 0.15^\circ$[P]. Whether a real shooter repeats that well is the

deciding question for hypothesis (A). It must be measured as the standard deviation of exit speed and elevation over many shots at one setting, with the wheel-speed sag from rapid fire and battery voltage logged alongside.

- **The ratio v/u.** The exit speed as a fraction of the wheel surface speed depends on the hood friction, the compression and the ball; the model does not treat $v = u/2$ as universal. It sets the wheel speed needed for a target exit speed and must be measured on the built shooter.
- **Exit spin parameter S.** The spin the hood or roller actually puts on the ball sets where the clean band sits. It can be read from high-speed video of a striped ball.
- **Drag C_d and spin lift.** The literature allows C_d and the lift law to vary over a range that moves the clean band by more than its width. Because POLLEN is perforated, its spin lift may be well below the smooth-sphere value. Full-flight video tracking at two spin levels pins both. In this model the lift coefficient saturates, so more spin past saturation does not add curvature.
- **Contact restitution e and friction μ.** Retention after a rim or floor contact (hypothesis B) depends mostly on e, and backspin can raise or lower it. Drop and spin tests on a sample of the HIVE wall material give e and μ.
- **Real HIVE tilt.** The $30°$[C] tilt comes from the official CAD and the manual figure, and the field acceptance tolerance gives 28.5–$31.5°$[D]. The tilt of the real field element, resting on its damper stop, should be measured, because it moves the minimum descent angle one-for-one.
- **Geometry and sideways effects.** The precomputed maps use one section through the HIVE centre plane. The live solver flies the oblique shot against the 3D cell with side walls and the pentagon roof (Section 7.2). Sideways drafts, robot motion and a spin axis tilted by turning are options (Section 7.6), but launcher sidespin and exit skew are still not modelled (Section 7.8).
- **One shot is not a rate.** A success rate must be reported with a Wilson interval from many shots (Section 6.7). A single good shot on a real HIVE says nothing about repeatability.

When these measurements come back, the model is re-run with the measured v/u, S, C_d, spin lift, e and μ, and the predicted scoring rate is compared with the measured one.

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